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Value-at-Risk

– An Empirical Analysis of the Market-Price-Risk Estimation –

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Abstract

Given the increased regulatory and societal pressures, there has been a growing need for managing and predicting financial risk. In this study we evaluate and compare the daily returns of both the *Dow Jones Industrial Average (DowJones)* and the *Deutscher Aktienindex (DAX)* from over 30 years of performance. The researched Value-at-Risk models are Historical Simulation, Variance- Covariance Simulation, and Monte-Carlo Simulation. These methods were selected because of their prevalence and popularity in both industry and academic settings. The daily Value-at-Risk for each of these indexes was computed using all three models at both the 95 and 99 percent confidence intervals. The results were evaluated using *Christoffersen's* backtesting methods to allow for comparability and evaluation. Each of the models was found to be inadequate in terms of losses not exceeding the Value-at-Risk at the rate specified in the confidence interval.

However, the findings resulted in Historical Simulation faring the best under our test conditions, followed by Variance-Covariance – Moving-Average, Variance-Covariance – Exponential Weighted Moving-Average, and Monte-Carlo respectively. The analysis also introduces Basel II methods for evaluating the success of Value-at-Risk models, which is often used for regulatory requirements. This model involved categorizing the success of models into three separate categories of ‘green,’ ‘yellow,’ and ‘red,’ ranging from good to poor respectively. Using this framework, our models appear to be in a more acceptable range. As an area of future research, it would an area of intrigue to analyze if some of the other more sophisticated varieties of the Variance-Covariance and Monte-Carlo method could exceed the performance of the Historical Simulation method.

Keywords: Empirical finance, Value-at-Risk, Historical Simulation, Variance-Covariance, Monte-Carlo Simulation

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1. Introduction

The role of a functional risk management has become more important in companies during the past years. Substantial occurrences, such as the financial crisis in 2008 or increased volatility in the second half of 2011, reinforce the importance of a risk management. The new parameters as well as globalization and the growing dependence of all financial markets creates the need for companies to reevaluate their risk management systems and their risk calculation models. The optimization should help to provide a more realistic impression of the probability of extreme events.

Due to the fact that banks are the larger protagonists within financial markets, they need to face the new requirements of transparency, being flexible, and having up to date risk management systems. In addition, banks need to fulfill all requirements under the strict control of regulators to guarantee the stability of the financial system. The underlying policy to follow exists out of three pillars and was formed by the Basel Committee on Banking Supervision, named “Basel II”. The *first pillar* describes the minimum capital requirements, the *second* the supervisory review, and the *third* the market discipline. With specific attention given to the minimum capital requirements, which is the amount of equity which is retained as a security for potential losses, depends on the credit risk, market risk and operational risk (Basel Committee on Banking Supervision, 2006). The accuracy to quantify the risk is essential for a bank. On the one side, it should be avoided to retain too much equity for potential losses and on the other side, to guarantee the stability of business.

Moreover, banks received admission on calculating the equity ratio for their risk positions with an internal risk model, called *internal rating based (IRB)* (Basel Committee on Banking Supervision, 2001). As a consequence, the Value-at-Risk factor became more important. Most of the risk models are based on the Value-at-Risk because it was an improvement compared to former models, which were based mainly on rough estimations. The Value-at-Risk model is, however, more precise in quantifying the risk. Based on these facts, the Value-at-Risk is still today a popular method for estimating and valuating the market risk (Jorion, 2007).

Besides banks, other companies are interested in the Value-at-Risk model and use it for their own calculation. Mainly companies which are trading at the market and facing financial transactions during their daily business operations are using the Value-at-Risk measure. Due to their participation in the financial market, they are vulnerable to market risks, such as interest risk, foreign exchange risk or market price changes (Bonke, 2007).

The goal of this Master Thesis is the research and investigation of some of the most known calculation models for the Value-at-Risk (Historical Simulation, Variance-Covariance and Monte-Carlo simulation) to quantify the market risk. However, the focus of our investigation will be more on the market price risk. The investigated calculation models will be filled with historic, empirical data from a chosen portfolio. Thereby, we will investigate further the predictive power of risk by using a backtesting-analysis. Finally, it will be possible to observe the deviation of the market price risk and conclude whether the underlying model was precise in its prediction.

1.1. Statement of the Problem

The definition of the Value-at-Risk as a risk measure is the amount of loss of a financial asset with an estimated probability, within a certain period (Hull, 2011). As a consequence, the Value-at-Risk enables the stability to determine specific probabilities of the occurrence of a calculation loss. However, it is essential to determine the risk factors to create a probability distribution. The process of defining these risk factors is hardly dependent on the choice of the underlying calculation model. The most popular and frequently used models, being the Historical Simulation, the Variance-Covariance and the Monte-Carlo simulation, are selected as the basis of our research. The Historical Simulation is based on historical market trends, which serves as a fundament for estimating the Value-at-Risk. The Variance-Covariance Simulation is based on distributive assumptions of the market price changes to compute the Value-at-Risk. The Monte-Carlo simulation shows characteristics of the two other simulations. The Monte-Carlo method is additionally using randomly generated market price changes, on the basis of a specified distribution function (Jorion P. , 2007).

The problem of the Value-at-Risk is in the mathematical assumptions, which need to be decided and inferred for different calculation models and distribution functions. These variations result in estimation errors and inaccuracy and thus, a negative impact on the risk quantification (Engle & Manganelli, 2004). Moreover, results for the Value-at-Risk will distinguish due to the differences within the models. The choice historical based data or the estimation of a parameter will have different influence on the distribution function. Hence, the risk prediction power will vary regarding which calculation model is used. This proposes the question whether one calculation model might be more precise than another in its risk prediction. Regarding this matter Prof. Robert F. Engle, Nobel laureate in 2003 for economics, stated: “Despite Value-at-Risk’s conceptual simplicity, its measurement is a very challenging statistical problem, and none of the methodologies developed so far gives satisfactory solutions.” (Engle & Manganelli, 2004, p. 367)

In addition, this thesis is elaborating the question of the risk prediction power of the three evaluated calculation models with respect to the Value-at-Risk. The research will also focus on whether the different models are able to compute the market price risk in a substantial way, so that the Value-at-Risk is not exceeded by realized losses in excess of the stated confidence interval. These results forge the fundament to determine the accuracy and reliability of the calculation models. In case of deviating from predicted losses, it can be concluded that the model might lacking in its reliability.

1.2 Motivation

Value-at-Risk is widely used because it can be delivered as a quantifiable amount that is easy to understand. The reason to investigate and compare these three different Value-at-Risk models is because these three are the most commonly used in practice. They are also the basis for many more variants of Value-at-Risk, (Jorion P. , 2007) yet there is little literature or academic studies comparing these directly. The opportunity for this analysis to provide practical value to businesses, particularly those that have many resources allocated to financial markets, makes this type of research particularly worthwhile. By conducting this empirical analysis, we hope to be able to add more credibility to one of the models, and give a reason to use one model over the others rather than a single attribute such as ease of use or understandability. We also found it valuable to have a control for our analysis, which is why we are looking at a portfolio index on the U.S. exchange, and also a portfolio index on the German exchange. We wanted to be able to analyze if one model would have more precision in a different environment. Conclusively, the inspiration for our research was to add more analysis and research into a measure, Value-at-Risk, that is widely used, yet it still relatively new, so it does not have the years of academic research as some other models in finance do.

1.3 Research Question and Sub-question

Based on the above, this thesis will constitute an empirical analysis of Value-at-Risk's as a risk prediction metric. This leads to the following research question:

“With many variations of the Value-at-Risk measure existing, does one measure perform superior when evaluating based on historic data from major market indexes?”

We believe that this research will provide valuable insights to firms which are in the field of financial risk, and additionally to any firm which uses financial instrument. This basis creates a large breadth

of users who could find value in our findings. Having evidence backing the use of a particular model may serve as guidance to inexperienced users of the Value-at-Risk model.

Additionally, a sub research question was also deemed necessary based on the manner of our analysis. This sub question can be framed as:

“How is a Value-at-Risk measure defined in terms of accuracy, and also evaluated for categorizing the success of a measure?”

This sub question goes hand in hand with finding the answer to the main research question of this thesis. To be able to properly rank the success of a Value-at-Risk measure, we must be able to effectively evaluate the measures, and also have criteria to conduct our evaluation upon. In addition, this sub question will also frame the methods we use for conducting our analysis, thereby further stating the evaluation aspects of this research importance to this report.

1.4 Delimitations

It should be fully emphasized that the purpose of this thesis is to investigate the area of financial risk models, namely Value-at-Risk. Therefore, other considerations which may be pertinent during our evaluation will not be covered in this paper to keep the focus on the methods of using this model to measure financial risk. The choice to delimit the scope of this thesis to this is to keep the paper focused and relevant to users, and avoid scope creep in the form of addressing areas of finance and risk which in the frame of this thesis do not impact our analysis or results. We will however acknowledge some of the shortcomings of Value-at-Risk, both by academics and researchers, and from our own findings. Furthermore, as this research is on the performance of this specific instance of a risk valuation measure, we will *not* focus on how the measure can be used outside of financial risk measurement, and if it is applicable to other aspects of industry.

Additionally, to add further validity and accuracy to our analysis, and because of the resource heavy nature of many of the calculations, we chose to use R programming to perform much of our analysis. Like many individuals in finance, much of our experience for analyzing data figures is done with *Microsoft Excel*; however, because of the iterative qualities of many of the models, and also the need to replicate results and run code scripts, we found it necessary to use a program which provided more functionality in this realm. R also provides the advantage of ‘packages’ (R Core, 2013) which are

programs that contain some of the complex Value-at-Risk functions, and allow us to cross check and conduct certain aspects of the analysis by utilizing the expertise of other individuals in this field.

1.5 Thesis Outline

As a guide to the reader, the structure of the thesis will be provided in the figure below to help ensure an understandable and cohesive reading of the paper. An overview of the content of each chapter is also provided below the figure.

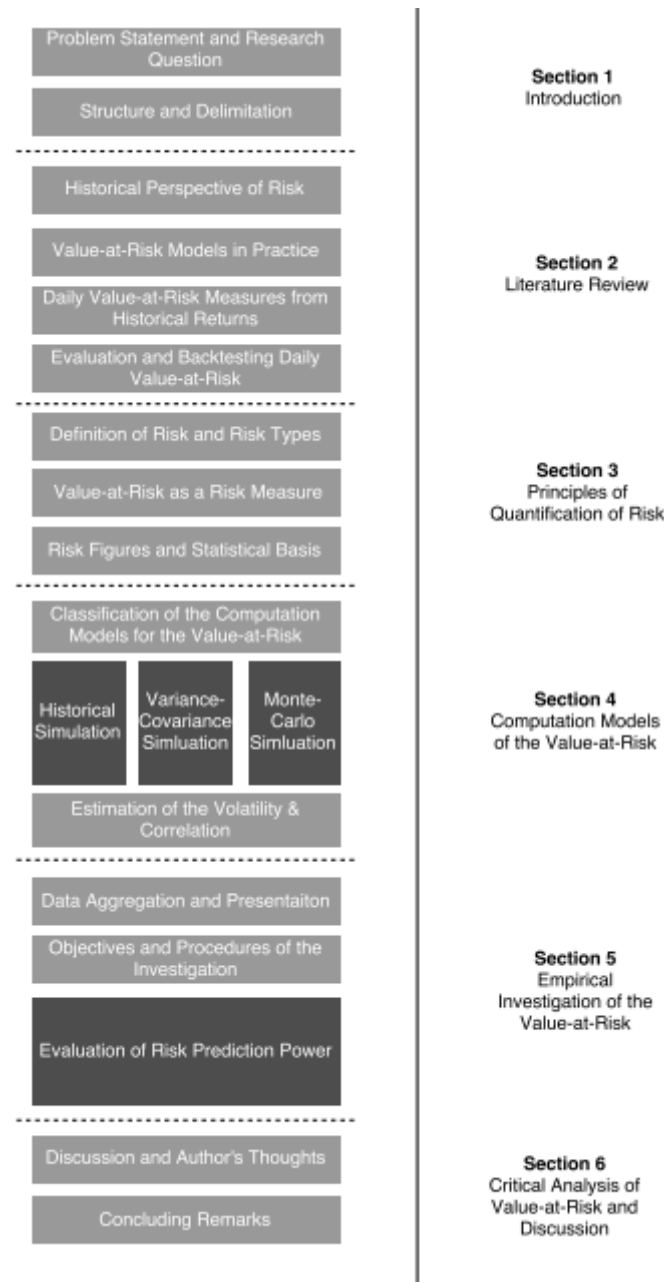


Figure 1: Thesis Structure

Section 1 has provided an introduction to our research topic, stating the research problem and background of the Value-at-Risk, and the delimitation for how we concluded on our research project. Additionally, this section also provides the structure that our research will be provided in.

Section 2 is a thorough literature review which will showcase the underlying theory and academic articles which our research was motivated by and based upon. This section will delve into the specific methods used to investigate in our field, and also include some of our reflections and thoughts to how our methodological choices took place and may influence the findings, while also creating certain limitations. Additionally, this section gives an overview to the empirical nature of our research to provide additional context to the reader.

Section 3 In this section, we will provide an overview of risk principles which build the framework for the Value-at-Risk model. The types of risk will also be differentiated, and provide the opportunity to highlight the risk most important to this research, market risk. Lastly, we will explain some of the background of Value-at-Risk, providing the types of Value-at-Risk models, and also Value-at-Risk's use in practice today.

Section 4 describes and explains each of the three models being analyzed, Historical Simulation, Variance-Covariance, and Monte-Carlo method. Explanation and methods for calculating the Value-at-Risk inputs and parameters will also be expanded upon in this section. This section will contain all of the relevant formulas used in models, along with a description and purpose. In addition to the prior, we will also give an overview of some of the variations which have their basis in these aforementioned models.

Section 5 will showcase the presentation of results showing the performance of the models and how they perform. Discussion of each model in isolation, and in addition context of the relative performance and strengths and weaknesses of the Value-at-Risk models will be provided. Results are displayed through the use of tables and visuals to help ease understandability, with the aid of published works to guide our framing of results.

Section 6 provides discussion of the results from section 5. This section will contain the key thoughts of the authors and also what areas could be expanded upon for future research. Discussion will also range to the application of our findings, and impacts for practice.

2. Literature Review

The literature review takes point of departure in the research area of Value-at-Risk history, development, evaluation and the theory that is at the backbone of Value-at-Risk measurement. When beginning our literature review, criteria was agreed upon to conduct an effective and appropriate method for selecting theories and models for our research. This means that the selection of journals and articles were based upon a certain level of academia being used and focusing on literature which has been referenced by other respected publications and that appear in well regarded finance journals. The chosen articles were usually identified using scholarly databases such as the CBS library, and online articles through Google Scholar. Through our research, we were able to utilize some of the most recognized researchers and scholars within our area of emphasis. By doing so, we were able to examine many of the most important academic and industry articles related to Value-at-Risk and create a well-researched base for our further analysis into Value-at-Risk. Using this point as the emphasis of our literature aggregation, the most relevant areas in regard to Value-at-Risk theories and models will be outlined in the following sections.

2.1 Historical Perspective on Risk and Risk Models

The idea of risk and more specifically, people engaging in the practice risk analysis and risk management can be traced back centuries. Since the practice can be simplified as identifying important dimensions of a problem (Covello & Mumpower, 1985), and then identifying alternative actions and what the outcome would be, or more specifically whether an outcome is positive or negative. With this in mind, we can make the claim that risks and how to analyze and manage them have always been a part of people's lives. The focus of the research and literature of this paper will be on more modern emergences of risk, but it is worth noting the beginnings and iterations with which the concept of risk has gone through.

Risk is usually quantified and 'measured' as a way to show the probability (McGoun, 1995) of an event to take place. Yet, the methods of creating or calculating this measurement is something that business industry and academics alike have continued to attempt to improve since the early modern iterations were created around 1900. In this era, probability was applied to historical cash flows which were used to estimate what level of cash reserves would be necessary for banks (Edgeworth, 1888). It is obvious to say now that the methods used now are far more sophisticated than what Edgeworth introduced back in 1888. However, some of the same principles, such as the reliance on

using historical data to predict future performance, are still commonly used today. A key difference is the lack of sole reliance on only one type of measure, and instead assuming other characteristics for future performance. The emphasis of the following section will be on following the progression of these risk valuation methods, and how these iterations differ and are similar to what is used today.

Early ambassadors of risk management were insurers, (Pfeffer, 1974) whose livelihood depended on ability to predict and measure the riskiness of any activity, ranging from financial investments to farmer's crops. In this case the appearance of insurers was particularly prevalent after the occurrence of a large tragedy, such as the London fire of 1666, which led to huge demand for fire insurance (Covello & Mumpower, 1985). This provides perspective on early risk management methods, where a risk was evaluated only after an occurrence so impactful took place that people had no choice but to take notice and react.

Yet, many more modern risk management techniques began taking place in the beginning of the 20th century due to societal, natural, and academic changes in the world. A notable reason for this increased expertise in this area is the increase in the number of researchers and analysts whose focus was on identifying and managing risks (Covello V. a., 1984), and has allowed risk analysis to emerge as its own identifiable area of practice. This attribute is even more pronounced today, and as showcased by this type of research paper, is so prevalent in society that companies and societies are willing to spend large sums of resources to appropriately analyze and manage risk. Additionally, this type of emphasis on risk is what has led to the evolution of risk measurement evolving from simple probability of future payment based on prior payment history, to the measure that is the focus of this paper, Value-at-Risk.

An early methodology that predated Value-at-Risk was Asset Liability Management (ALM). With this method, earnings would be forecasted based on predetermined price/rate market conditions. (J.P. Morgan, 1996, pp. 22-24). However, the changes in market rates may occur slowly compared to the earnings which are reported on an accrual accounting basis, the forecasts or simulations need to be done continuously until the transactions on the books come to maturity. The risks of this methodology are that it requires the projection of market rates into the future (J.P. Morgan/Reuters, 1996) and gives the perspective that the gains and losses occurred at the time that they impact the P/L of a firm because of the matching principle aspect of accrual accounting. This provides the opportunity for Value-at-Risk.

The most notable difference between the new and old approach is the ability to continually mark-to-market with Value-at-Risk, and decreases the forecasted horizon (J.P. Morgan/Reuters, 1996). As a general rule, shorter forecasts can be considered more accurate than longer forecasts, so this serves as a way to reduce the risk inherent with forecasting. There are also challenges with the constant mark-to-market nature of Value-at-Risk, most notably the difficulty of finding a market which accurately gauge a price for every asset a company holds, particularly if it is not frequently traded or not liquid. However, the improvement Value-at-Risk provides over the older model of ALM and its adoption by not only the finance sector, but all industries has shown its usefulness and applicability over older models.

2.2 Value-at-Risk models in practice

Though the scope of our research focuses on three Value-at-Risk models, there are many variations of this model used in practice for just about any investment in which a manager would be interested in knowing the amount of loss that can be expected with a specified level of confidence. As the variety and specificity of a model increases, the Value-at-Risk measure can become quite niche. An example of this is Value-at-Risk for derivatives, which is not even touched in the research of this paper, provides additional methods for measuring Value-at-Risk, such as the Delta and Gamma approaches (Duffie, 1997). This provides an example of the breadth and depth of the Value-at-Risk measure and how many ways it can and is used in practice.

It is also important to highlight some of the limits of Value-at-Risk and its use in industry. An important note not only for the scope of Value-at-Risk measures in this paper, but for Value-at-Risk measures as a risk measurement tool is the fact that the measurement does not consider ‘worst case scenario’ type of losses in which the loss far exceeds the calculated Value-at-Risk (Ghadoui, 2003). The Value-at-Risk simply states an amount which an investment can lose a percent of the time, instead of the common thought that it provides that absolute loss which can occur. In addition, another aspect of Value-at-Risk to be mindful of when using the model is that the measurement works as more of a passive risk control than an active tool (Berkowitz, Christoffersen, & Pelletier, 2001) used for decision making. More specifically, a trader will not make specific trading decisions in response to Value-at-Risk limits or how management should select the optimal Value-at-Risk limits for their traders. These prior critiques showcase some of the shortfalls of Value-at-Risk, and lead to a more important point. Value-at-Risk can be an extremely valuable tool for use in regulatory reporting and for a company’s investments. However, the use of Value-at-Risk on its own can lead to a distorted

measure, (Jorion, 2007) meaning that a user could misinterpret, or come to an incorrect conclusion by using Value-at-Risk on its own without any other types of risk measures. The Value-at-Risk should be accompanied with other theories, such as Extreme Value Theory, (McNeil, 1997) which provides a measurement of the ‘tails’ for a return distribution, which can measure the larger losses that take place which common Value-at-Risk models do not account for.

With the provided context of how vast the scope of Value-at-Risk as a measurement can be, the focus will be put on what we analyzed during our research for this paper. Much of our analysis is grounded in the findings and research done by JP Morgan, when they created “*RiskMetrics*” (J.P. Morgan/Reuters, 1996) which is considered to be one of the first iterations of Value-at-Risk methodology. The measures being researched Historical Simulation, Variance-Covariance, and Monte-Carlo methods will be provided with more details in section 4.1 about these measures, as they are the basis of our research.

After the introduction of Value-at-Risk by JP Morgan, the next most influential figure in this field is Philippe Jorion, whose literature and work will often be cited throughout this entire research. Jorion’s text book, titled simply “*Value at Risk-The New Benchmark for Managing Financial Risk*” (Jorion P., 2007) provides much of the development and adaptability which took place after JP Morgan initially developed the measure. Jorion thought of the measure within the scope of financial institutions, and being used to manage financial risks, but did not have to be limited to only this risk, but the events which occur from nature or governments, or any other scenario which could impact financial markets. Jorion himself refers to Value-at-Risk as “the latest step in the evolution of risk-tools” (Jorion P., 2007, p. 25) The reason for this belief is because everything which the Value-at-Risk measure can encompass. Value-at-Risk is able to include multiple risks besides single attributes such as the risk only of changing interest rates, and can also include foreign exchange and equity risk. Jorion’s work with Value-at-Risk also had an emphasis on properly estimating the Value-at-Risk for a multi-asset portfolio, while also providing a basis for more basic forms of the model, such as the Value-at-Risk of a single asset. These two aspects of Jorion’s work provide a useful basis for the research in this paper.

2.3 Analysis of daily Value-at-Risk measure based on historical daily returns

With a broad scope of directions available to analyze Value-at-Risk, we chose to keep our focus on using historic data and a way to measure daily Value-at-Risk. A key factor was to be able to have a large enough subject matter that research would be feasible, while also keeping the scope limitation in mind so the final result would be presentable and useful to the reader. It was also valuable that an amount of research had been done on this topic, but not identical to ours. This helped to reinforce the emphasis that there was a need for this type of analysis, while also being enough academic work available to base methods on rather than needing to create new methods or making assumptions which may not be accurate.

An article which served as a reference and inspiration for our research is authored by Darryll Hendricks, titled “*Evaluation of Value-at-Risk Models Using Historical Data*” (Hendricks, 1996). The research done by Hendricks also focuses on comparing the performance of risk models by using historical currency exchange rates as the historic data set, and tests the Value-at-Risk models simulating from this data. This paper also includes evaluations at both the 99th and 95th confidence intervals, which we also included in our analysis as they are the most commonly used (Pearson, 2011) in the Value-at-Risk model. This paper also uses the concept of evaluating Value-at-Risk based on varying number of past trade days in the data set that is being used. This meaning more specifically that not every past closing price is included in the calculation of a Value-at-Risk measure, and instead a fixed number of past data points are included in each Value-at-Risk measure. This idea will be further explored and termed in the next used article. Lastly, with this article exploring the use of Historical Simulation and both the Moving-Average and exponentially weighted Moving-Average varieties of Variance-Covariance, there was strong overlap with the selection of models being evaluated. However, the Monte-Carlo method was left out in this article which serves as a key differing attribute. But utilizing the articles presentation of methods and results for use in our own analysis was invaluable to our research.

An additional article which was used as a constant frame of reference was “*Value-at-Risk Prediction: A Comparison of Alternative Strategies*” (Kuester, Mittnik, & Paolella, 2006) which appeared in the Journal of Financial Econometrics. This article provided the inspiration to use an index portfolio as the chosen data set in our research as well of the chosen time horizon of approximately 30 years. Knowing that a market index portfolio was an option for our analysis intrigued us since it felt more tangible as it is a figure which we are aware of and follow in our daily lives as opposed to, for example,

foreign exchanges currency rates. Along with this article also being an empirical analysis of Value-at-Risk, this paper described the idea of ‘moving windows’ which became a vital framework for our analysis, and was touched upon in the previous paragraph. The window works in that each Value-at-Risk measure has a unique set of trade days associated with its calculation. However, this paper used a much longer trade day window of 1000, compared to our longest window being 250 (J.P. Morgan/Reuters, 1996) as was done by JP Morgan during their development of Value-at-Risk. Using this moving window type of method is how Value-at-Risk is used in practice, and a key input into the measurement of Value-at-Risk being the window length. For this reason, we chose to change this variable to see what type of impact this has on the end measure, and if we could gain insight into the pros and cons of different window lengths. With window lengths of 1000, 500, and 250 evaluated in this article, it also served as further inspiration to evaluate based on different windows. Lastly, the conclusions provided from this research also gave us an ability to confirm and cross check our results, and also assure that our findings did have a basis in other academic research.

[2.4 Evaluation and backtesting daily Value-at-Risk](#)

An early challenge after developing our research question was how to go about evaluating the results of our Value-at-Risk findings. It did not seem sufficient to compare the Value-at-Risk measurement of different models without any context. This has clearly also been a question by other individuals who work and research in this area. This need for a measurement methodology has led to evaluation of the accuracy of Value-at-Risk measures through ‘back-testing’ (J.P. Morgan/Reuters, 1996). This process appears already in the “*RiskMetrics*” paper, with Kupiec (Techniques for verifying the accuracy of risk measurement models, 1995) appearing to perform an earlier iteration of the method for measuring the accuracy of risk models. However, JP Morgan is able to provide back-testing samples with simple calculations for evaluating Value-at-Risk accuracy.

The focus in this paper is on the “Realized percentages of Value-at-Risk violations” (J.P. Morgan/Reuters, 1996, p. 220) where a violation is an occurrence where a realized loss exceeds the Value-at-Risk measurement for a given date. There are of course a specified number of violations which are expected to take place using Value-at-Risk. For example, at the 95% confidence interval, the expectation is that once every twenty trade days (5%) a realized loss will exceed the calculated Value-at-Risk. However, if the realized loss exceedances deviate from the prescribed amount based on the confidence interval, the model can be stated as not performing well. The ease of

understandability and practicality of this measure makes it one of the most common back-testing methods, and is used in our analysis.

Lastly, our processes used for backtesting results were also formed by the work of Peter Christoffersen, with a framework provided by “*Evaluating Interval Forecasts*” (Christoffersen P. F., 1998) and his later article building on many of the ideas of the former, “*Backtesting Value-at-Risk: A Duration Based Approach*” (Christoffersen & Pelletier, 2004). In the 1998 article, Christoffersen establishes a framework to evaluate interval forecasts, using Value-at-Risk as the model to demonstrate his framework on. From this text, we learned of “The Likelihood Ratio Test for Independence” (Christoffersen P. F., 1998, p. 845). This test can be described as evaluating the distribution of exceedances if the likelihood of an individual exceedance is independent of all other exceedances. This can be particularly valuable to a user of Value-at-Risk, since clustering of exceedances of Value-at-Risk could have large consequences by having losses on consecutive days. Additionally, in Christoffersen’s 2004 article the Likelihood Ratio Test is used on portfolio data used in calculating Value-at-Risk. The similarity of this evaluation is what led us to choose to replicate Christoffersen’s process from this article in order to assure consistent results.

2.5 Theoretical Framework

The above reviewed literature constitutes the theoretical framework, on which our data methodology is built upon, as well as the subsequent analysis, discussion, and conclusions. The below illustration 2 highlights the most important theoretical aspects within each reviewed area of emphasis regarding our investigation of comparing different methods of Value-at-Risk measurement.

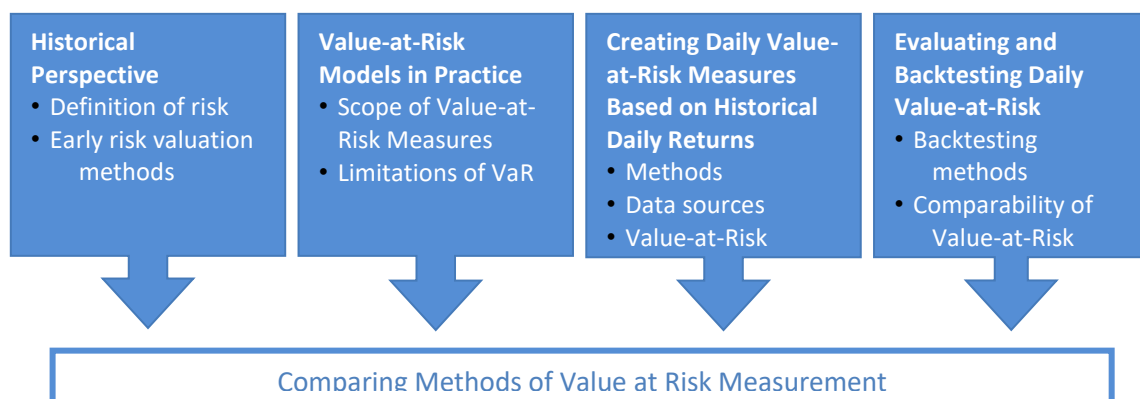


Figure 2: Framework for Literature Review

3. Principles for Quantification of Risk

This chapter will define risk and the usual types of risk as well as their delimitation with respect to this paper. Moreover, the Value-at-Risk is introduced and explained as a risk measure, followed by further fundamental explanations of computation functions.

3.1. Definition of Risk and Risk Types

Generally, risk is defined as a deviation from an expected value, which might occur at some point in the future (Jorion, 2007). Thereby, it is differentiated between a symmetric and an asymmetric risk. The symmetric risk is mainly found in capital market theory and became more known through the publication of Markowitz's portfolio theory. The risk is defined as dispersion of negative and positive deviations from the expected values, also known as volatility (Markowitz, 1952). The asymmetric risk, however, defines the risk exclusively as the negative deviations from the expected value (Eisele, 2004). The decrease of a stock at the stock market is an example for asymmetric risk.

Companies are facing many different risks in their daily business, which need to be considered and evaluated. Because of the diversity of risk classification in the present literature, this paper is based on an own categorization of risk, which is congruent with numerous characteristics in the literature.

The German Federal Institute for Financial Service Supervision ("*BaFin*") published "*Minimum Requirements for the Risk Management*" ("*MaRisk*") for financial institutions, which contains the risk types: *counterparty risks*, *market price risks*, *liquidity risks* and *operational risks*. Moreover, it also contains the regulations for qualitative requirements for the risk precaution (Bundesanstalt für Finanzdienstleistungsaufsicht (BaFin), 2010). The "*Bank for International Settlements*" ("*BIS*") classifies the market risks as interest rates, exchanges rates, liquidity prices and commodity prices (Basel Committee on Banking Supervision, 2006). This paper uses a similar differentiation and is pictured in the following graphic.

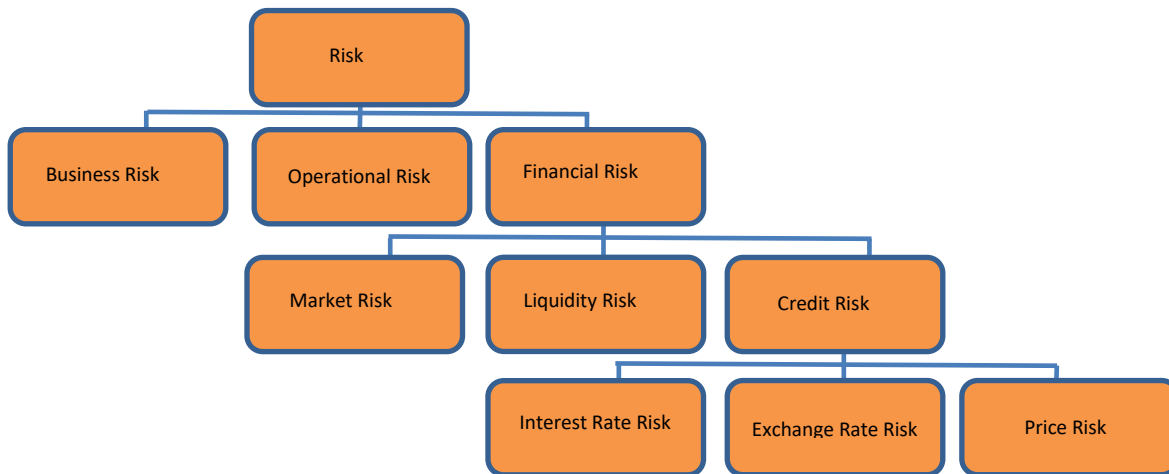


Figure 3: Risk Types¹

The business risk describes the risk of strategic mistakes. For instance, the development of products, which are not going to be sold (*shelf warmer*) (Jorion, 2007, p. 4), unprofitable investment decisions as well as the wrong choice of site selection can be related to the business risk.

The operational risk describes the risk in the own company, which means mistakes in internal processes and resulting eventually in e.g. stop of production systems. In addition, vague distribution of responsibilities in the organization or fraud, caused through a firm's own employees, is also characterized by operational risk (J.P. Morgan/Reuters, 1996).

Besides the two mentioned risks, the financial risk encompasses a major part of the overall company risk. Even though the focus of financial institutions is emphasized on the financial risk because this is the industry in which they act in, and thus face more risk to financial fluctuations. The financial instruments are still used by, in larger companies so the financial risk is still inherent. Therefore, the financial risk has an immense importance and should not be neglected for any type of firm (Scherpereel, 2005).

The financial risk involves the risks and potential losses in the company, which can be caused by any kind of financial transaction. The financial risk can be further classified into *credit risk*, *liquidity risk* and *market risk* (e.g. Choudhry & Wong, 2013; J.P. Morgan/Reuters, 1996; Jorion, 2007). The potential loss of a claim towards a creditor would be condensed as credit risk (Deutsch, 2008). The liquidity risk describes the risk that a company would not be able to settle its expenses in the future

¹ Own Illustration; referring to: Dowd (1998); J.P. Morgan/Reuters (1996); Jorion (2007)

as well as the risk of not being able to sell its traded financial instruments due to a lack of demand at the market (Jorion, 2007).

Last but not least, the financial risk describes the market risk, which means the risk of a prospective loss due to traded financial instruments in the company's portfolio. It can be caused through fluctuations or changes regarding the interests and exchange rates (J.P. Morgan/Reuters, 1996).

There are different risk measurements to measure the named risk types and quantify these risks. Besides the volatility as a symmetrical risk measure, which states the negative and positive dispersion of an expected value, the Value-at-Risk represents a well-known asymmetrical risk measure because it deals exclusively with the potential loss and therefore, expresses the negative deviation of an investment (Koller, 2005). This is the reason why it is also known as the "*Downside*"- risk measure (Schroeck, 2002, p. 168). In this academic paper, with respect to the Value-at-Risk as an asymmetric risk measure, the risk is exclusively defined as a negative deviation, or a potential loss.

The published literature contains the application of the Value-at-Risk regarding several risk types and describes therewith, its functionality in practice. It can be used as a risk measure for operational risks but also for the subcategories of the financial risk, pictured in figure 3. The common utilization of the Value-at-Risk is, however, the determination of the quantification of the market risk (Jorion, 2007). Based on this fact, this paper is considering only the market risks, especially the risk of price changes of financial instruments (market price risk). The theoretical foundation of how to aggregate the quantification of the market price risk in a portfolio is further explained in the following chapter 3.2. Afterwards, chapter 5 is investigating the problem statement whether the computation models can deliver reliable risk quantifications of the market price risk and consequently how significant the risk prediction power is.

3.2 Definition Value-at-Risk as a Risk Measure

A market index portfolio faces several risks. The challenge is to determine and quantify these risks to consider them in a risk analysis and valuation. The Value-at-Risk is in theory as well as in practice an appropriate risk measure, which serves as 'state of the art' when quantifying market risks. It aggregates risks of various variables with respect to existing correlations. Eventually, it transforms them into a final risk figure (Eisele, 2004). Consequently, it is possible to consider known types of risk into the risk management and present a single number to the management to express the total

business risk. Due to that the Value-at-Risk is mostly presented in one currency (often EUR, USD or domestic currency); it is intuitive, easy and fast to understand.

In 1990, the Value-at-Risk as a figure was developed by J.P. Morgan (Holton, 2003) to gain a daily and quick overview of its own risk. Since the Value-at-Risk is not used from banks exclusively, but also from numerous companies. The dissemination was pushed from a software release (including databank references), called *RiskMetrics* (J.P. Morgan/Reuters, 1996), which is used as a key element in the Value-at-Risk concept.

The purpose of the Value-at-Risk is to serve as a risk measure to quantify the risk for a single financial derivative or portfolio. It determines the worst possible loss in the portfolio over a target horizon and a defined (low) probability that will not be exceeded at a given level of confidence. In other words, the portfolio position is not losing more of its value than the actual Value-at-Risk factor defines (Jorion P. , 2007). The two most influential factors for the Value-at-Risk are therefore, the probability and the period of time.

In general, the calculation of the Value-at-Risk is based on yield's probability distribution of the underlying financial derivative or portfolio. More formally, the Value-at-Risk defines the quantile of the projected distribution of gains and losses over the defined period. If c is the selected confidence interval, the Value-at-Risk corresponds to the $1-c$ lower tail level and is mostly expressed as a positive number (Jorion, 2007). Thus, the quantile stands for the threshold, which will not be exceeded with respect to the probability of the confidence level. Stated in a formula, the quantile is defined as α of the probability Pr , which is exceeded by the Value-at-Risk. However, the difference between the actual portfolio value P and the initial portfolio value P_0 describes the portfolio loss (Saita, 2007, p. 25).

$$\Pr(P - P_0 < -VaR_c) = \alpha$$

Equation 1: Definition of the Value-at-Risk

For instance, a Value-at-Risk of € 20,000.00 within a period of one day and a confidence interval of 99% means that the portfolio will not decrease more than € 20,000.00 until the next day with a 99% probability, or the portfolio will decrease more than € 20,000.00 until the next day with a probability smaller than 1%.

Formally, the computation of the Value-at-Risk for each underlying probability distribution for returns $f(u)$ can be determined for the quantile α (and hence the Value-at-Risk, by using the inverse function $f^{-1}(\alpha)$), for which the confidence interval is expressed through the sum of probabilities until this quantile (e.g. Dowd, 2002, p. 45; Jorion, 2007, p. 109; Schroeck, 2002, p. 168):

$$c = \int_{-\infty}^{\alpha} f(u) du$$

$$VaR_c = f^{-1}(\alpha)$$

Equation 2: Computation of Value-at-Risk for random distribution of returns

3.3. Risk Figures & Statistical Basics in Financial Economy

The key figures, which are mainly used for estimating the parameters and eventually computing the Value-at-Risk, have their roots in the statistics and financial environment. Therefore, a real and deeper understanding is essential to comprehend a reasonable computation. One of the key figures is the return, which is portrayed as the variance and standard deviation (volatility) in statistics or the covariance and the correlation coefficient.

In the most cases, the interesting key figure for investors is the return. It describes the relative change of the price P between two points in time, $t-1$ and t . It can be distinguished between a discrete and continuous return u_t in a specific point of time t . The discrete return means the absolute change in price between the two points in time regarding the prices P_{t-1} of the prior point in time $t-1$. The discrete return is computed as followed (e.g. J.P. Morgan/Reuters, 1996, p. 46; Jorion, 2007, p. 93; Tsay, 2002, p. 2):

$$u_{t,discrete} = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Equation 3: Discrete Return

While the discrete return enables the ability to observe a price change to the respective points in time, the continuous return expects a steady price development. It reflects the logarithmic price change and is computed as followed (e.g. J.P. Morgan/Reuters, 1996, p. 46; Jorion, 2007, p. 94):

$$u_{t,continuous} = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Equation 4: Continuous Return

The benefit of the continuous return is the simple application, so that returns of single points in time can be added and conclusively result in the total return. The discrete return does not give this opportunity due to the multiplicative correlation between the single returns, which exacerbate a fast computation. From a mathematical perspective, it can be shown that the discrete as well as the continuous return are approximately equal regarding short observation periods (Eisele, 2004).

Besides the return and the resulting profitability, the correlated risk is another interesting key figure for investors. The risk is fundamentally dependent and calculated from the price variations, which are depict by the statistical key figures variance/volatility. Based on an expected value, it determines the fluctuation range of the price. Ordinarily, the higher the variance/volatility of an instrument, the higher the risk associated with the financial instrument (e.g. Choudhry & Wong, 2013; Dowd, 2002).

The computation of the variance σ^2 is based on the chosen target horizon T , in which the observed returns u are evaluated and eventually a deviation from the return mean is ascertained $\bar{u}$ (e.g. Choudhry & Wong, 2013, p. 14; Hull, 2009, p. 586; Jorion, 2007, p. 95).

$$\sigma^2 = \frac{1}{T-1} * \sum_{i=1}^T (u_i - \bar{u})^2$$

Equation 5: Variance

As a result of the variance you receive the volatility, which is also known as the standard deviation, if the underlying distribution is a normal distribution (Hull, 2011, p. 214).

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{T-1} * \sum_{i=1}^T (u_i - \bar{u})^2}$$

Equation 6: Volatility / Standard Deviation

The variance as well as the volatility can be determined for a single financial instrument but also for an entire portfolio. But computing the key figures for an entire portfolio brings up additional facts to consider, which do not apply to single asset calculations. Besides diversifications between the financial instruments, dependencies as well need to be incorporated in the computation. For instance, the volatility for an entire portfolio is not the sum of the single volatilities of the financial instruments within the portfolio. Due to the risk dispersion to several financial instruments, the total risk decreases and therefore, the volatility is respectively lower (e.g. Alexander, 1998; Markowitz, 1952).

A sample portfolio out of two financial instruments X and Y has the portfolio variance σ_{X+Y}^2 , formed out of the two single variances σ_X^2 and σ_Y^2 , the weightings w_X and w_Y in the portfolio² and the correlation coefficient $p_{X,Y}$, which describes the dependency between both values (Hull, 2011, p. 321).

$$\sigma_{X+Y}^2 = w_X^2 * \sigma_X^2 + w_Y^2 * \sigma_Y^2 + 2 * w_X * w_Y * p_{X,Y} * \sigma_X * \sigma_Y$$

Equation 7: Portfolio Variance for two Financial Instruments X and Y

In general, the portfolio variance for a portfolio out of N financial instruments is computed as followed (Jorion, 2007, p. 161; Markowitz, 1952, p. 81):

$$\sigma_P^2 = \sum_{i=1}^N \sum_{j=1}^N w_i * w_j * p_{i,j} * \sigma_i * \sigma_j$$

Equation 8: Portfolio Variance for N Financial Instruments

² The weighting of a financial instrument results out of the invested amount in this financial instrument divided by the total value of the portfolio

The computation of the dependency between two portfolio values X and Y is also based on a deviation from an expected value $\bar{u}$ (cf. computation of variance) and further a product of both dependent values is ascertained as follows (Jorion, 2007, p. 96).

$$cov_{X,Y} = \frac{1}{T-1} * \sum_{i=1}^T (u_{i,X} - \bar{u}_X) * (u_{i,Y} - \bar{u}_Y)$$

Equation 9: Covariance for two Financial Instruments X and Y

Consequently, the covariance determines the dependency between two portfolio positions. However, in case of more than two portfolio positions $X_1 \dots X_N$ it is plausible, due to simplicity, using the covariance matrix to express the computation (Jorion, 2007, p. 162):

$$cov_{X_1 \dots X_N} = \begin{bmatrix} \sigma_{X_1}^2 & cov_{X_1, X_2} & \dots & cov_{X_1, X_N} \\ cov_{X_2, X_1} & \sigma_{X_2}^2 & \dots & cov_{X_2, X_N} \\ cov_{X_3, X_1} & cov_{X_3, X_2} & \sigma_{X_3}^2 & cov_{X_3, X_N} \\ \dots & \dots & \dots & \dots \\ cov_{X_N, X_1} & cov_{X_N, X_2} & \dots & \sigma_{X_N}^2 \end{bmatrix}$$

Equation 10: Covariance Matrix for N Financial Instruments $X_1 \dots X_N$

The more intuitive correlation coefficient ρ can be derived from the covariance of two portfolio positions (Jorion, 2007, p. 163).

$$\rho_{X,Y} = \frac{cov_{X,Y}}{\sigma_X * \sigma_Y}$$

Equation 11: Correlation Coefficient between X and Y

The correlation coefficient lies between -1 and 1. A negative correlation coefficient means an opposed development of both positions. An increase of position X would bring a decrease in position Y . However, a positive correlation coefficient means that both positions go in the same direction (Jorion, 2007). The higher the correlation coefficient, the stronger is the dependency between both positions. A correlation coefficient of zero means that no dependency exists.

The fundamentals for the computation of important key figures in the financial economy will be used on a repetitive basis in this paper. Moreover, these basics serve as the backbone for further computations regarding the concept of the Value-at-Risk.

4. Computation Models for the Value-at-Risk

This chapter introduces the most common computation models for the Value-at-Risk known in literature. Moreover, the goal is to explain the underlying computation theories as well as to discuss the existing problem. Based on these computation theories, the subsequent empirical research is conducted with respect to compare the models' risk prediction power.

This section begins with a classification of the computational risk models. Following the introduction, we will detail and examine the Historical Simulation, Variance-Covariance Simulation and the Monte-Carlo simulation and ending eventually with the estimation procedures (Moving-Average, EWMA and GARCH) to determine the parameters such as volatility and correlation.

4.1. Classification of the Computation Models for the Value-at-Risk

There are different procedures in the theory for a detailed determination of the Value-at-Risk, which mostly include the three well-known and often used models: Historical Simulation, Variance-Covariance Simulation and Monte-Carlo simulation.

The Value-at-Risk contains a prediction, which is reached with a certain probability. To form such a prediction the computation models need to be provided with information about probable, prospective developments. The common factor for all three models is that all models are based on observations and analyses from the past as well as necessary assumptions for preferable realistic estimations to predict future developments. However, the investigations of the computation models focus on the computation procedure, the inquiry of the underlying data (e.g. the underlying distribution of returns) and the assumptions, which facilitate the depiction of the reality in the theoretical computation. Attributed to some necessary assumptions as well as to several different processes with respect to the Value-at-Risk different values are yielded for the quantification of the market price risk (Manganelli & Engle, 2001). Thus, there is seen an inevitable differentiated force of expression among the different classifications of the computation models. The assumptions will be explained later in more detail.

The investigation method for the Value-at-Risk can be split into two types of classifications. On the one hand, it can be differentiated between non-parametric and parametric. On the other hand, it can be differentiated between simulative and analytical approaches (Manganelli & Engle, 2001). The

following table shows the integration of all three computation models into these investigation methods:

	Non-Parametric Approach	Parametric Approach
Simulative Approach	Historical Simulation	Monte-Carlo Simulation
Analytical Approach	-	Variance-Covariance Simulation

Table 1: Classification of Computation Models for Value-at-Risk (Author Creation)

While the computations in the parametric approaches requests a choice of suitable parameters for the underlying distribution function, the non-parametric approaches base on empirical, historical data, which simulates numerous scenarios for future changes in the value of the portfolio (e.g. Jorion, 2007; Manganello & Engle, 2001). Hence, the crucial difference is that the parametric approaches lean onto the assumption that the market-price-risk-determined risk factors are formed through a prior defined distribution function and is with those statistically delineated (Jorion, 2007). The distribution function is most often based on the normal distribution, due to the fact that it is completely defined with only a few parameters (e.g. the mean, standard deviation and correlation) (Bonke, 2007). This required assumption of a normal distribution is an enormous simplification with respect to the computation of the Value-at-Risk. However, it can be also problematic if the actual distribution function of risk factors deviates from the normal distribution. Thus, often the models are working with leptokurtic distributions, which are characterized by clear margins (*“fat tails”*) and more frequently appearing means (*“thin waists”*). Respectively, there are a larger number of extreme events as predicted in the normal distribution and the frequency of events close to the mean is higher than actually described by a normal distribution (Eisele, 2004).

Due to the independence of predefined distribution assumptions, the non-parametric approaches are not facing this problem. The distribution of returns, used for the computation of the Value-at-Risk, is exclusively based on historical data. For that, the historically observed changes in the market price are applied to the current portfolio, the deltas in the values are simulated in the portfolio and based on this simulated distribution the quantile and eventually the Value-at-Risk is computed (Jorion, 2007).

The second type of classification is defined through simulative and analytical approaches. Similar to the parametric approaches, the analytical methods assume also a distribution function for the aleatory

variables of the risk factors. The simulative approaches, however, generate their data based on an underlying distribution function for the risk factors by using scenarios or simulations. The simulation can either be conducted on empirical historical data (similar to non-parametric approaches) or the distribution function is determined through complex simulations by using the generated random numbers (Bonke, 2007). The resulting distribution function is forming eventually the fundament for the determination of the quantile and conclusively used for the computation of the Value-at-Risk.

The simulation is also called “*Full-Valuation-Method*”, because of a complete revaluation of the underlying portfolio based on simulated data. The advantage is the consideration of also non-linear dependencies between risk factors and portfolio value (for instance in the case of options) (Jorion, 2007).

4.2. Historical Simulation

The computation model of the Historical Simulation is often considered to be the more simple and coherent models for ascertaining the Value-at-Risk (Koller, 2005). As described in the prior chapter, the Historical Simulation belongs to the non-parametric and simulative approaches, which mean that no distribution assumptions need to be made for the risk factors and that the underlying parameters are estimated (Koller, 2005). In fact, the observed daily changes of market variables from the past are used directly for the estimation of the probability distribution of the current portfolio value (Hull, 2011).

First of all, the risk factors such as exchange rates, share prices and interest rates need to be identified due to their impact on the portfolio’s market risk. Further, the progress of these risk factors needs to be gleaned over a historical period (Hull, 2011). The risk factors’ progress from the past is building the foundation on which numerous scenarios are generated. The current portfolio can be confronted with specific circumstances and it can be observed how it could proceed over a predefined period of time (Bonke, 2007).

For instance, one of the risk factors is describing a changing exchange rate due to the portfolio is held in a different country with its own local currency. An unforeseen change in the exchange rate might cause a loss in the portfolio value. The similar case could be constructed regarding a changing interest rate (Deutsch, 2008). However, this paper is focusing mainly on the market risk. For this reason, the following risk factors represent exclusively possible changes of the market price for financial instruments in the portfolio. Hence, the risk factor is determined as the market price of the respective

financial instrument. Conclusively, a portfolio, which exists out of two financial instruments is subject to two risk factors.

For ascertaining the Value-at-Risk for the following trading day, the observed daily returns are considered. Instead of the daily returns, the daily changes regarding the risk factors can be used as well. The historical data for the target horizon $m - T \leq t \leq m$ is serving as the base. $m - T$ is describing the value, which is the furthest in the past regarding the target horizon T and m is the present value. Based for N risk factors and T historical data, the value of the risk factor j at the past day t is defined as $S_{j,t}$ with $1 \leq j \leq N$ and $m - T \leq t \leq m$. The computation for the changing risk factors $\Delta S_{j,t}$ is based on the daily movement of the risk factors from the prior day $t - 1$ to day t (e.g. Alexander, 1998, S. 92; Jorion, 2007, p. 263):

$$\Delta S_{j,t} = \frac{S_{j,t}}{S_{j,t-1}} \text{ for } 1 \leq t \leq T \text{ and } m - T \leq t \leq m$$

Equation 12: Daily Return of risk factors for historical point of time t

Hence, as a result there are T simulated scenarios $S_t^{simulated}$ caused through multiplication of the empirical historical changes $\Delta S_{j,t}$ with the at this point of time m current value of the market variable $S_{j,m}$ (Alexander, 1998, p. 92):

$$S_t^{simulated} = \sum_{i=1}^N (S_{j,m} * \Delta S_{j,t})$$

Equation 13: Simulated scenarios of risk factors

The simulated changes of the risk factors are transferred afterwards to changes in the portfolio, which predict possible prices of the portfolio value for the next trading day. Eventually, the simulated values yield a simulated empirical probability distribution of daily portfolio changes. These serve as a fundament to determine the quantile for the requested confidence interval and finally the Value-at-Risk (Neumann, 2000).

Alternatively, the quantile for the Value-at-Risk results also from reading the value $(T * (1 - c))$ after the simulated values are sorted in an ascending order. For instance, the Value-at-Risk of $T =$

200 simulated values and with a confidence interval of $c = 0,95$ is resulting out of the tenth value of the sorted values.³

In case of computing the Value-at-Risk for a period, which lasts longer than one day, the requested Value-at-Risk can be converted with the root function⁴ for the holding period. Another way is to compute the returns for the requested holding period (e.g. monthly return) ab initio. The further computations are made with the assumption of calculating the Value-at-Risk for one day (Jorion, 2007).

It has a crucial importance that the risk factors are not regarded individually, their individual changes are applied to the portfolio and that an isolated frequency distribution of the returns is compiled. In fact, the time coherence of the isolated frequency distribution always needs to be considered and kept stable, because through the frequency distribution the correlation and stochastic dependencies are expressed between the risk factors (Eisele, 2004). Based on formula 13, within the defined scenario the dependencies and the correlations are considered and kept stable for the simulation.

Besides the implied consideration of correlations, another benefit of the Historical Simulation is that while computing the Value-at-Risk a complete revaluation of the portfolio is processed (*“Full-Evaluation-Method”*). Thus, the computation is not restricted by linear coherences between the risk factors and portfolio values, such as it is the case for the Variance-Covariance method. Instead, a random number of complex financial products can be used and be compared to non-linear dependencies of risk factors and get integrated in the simulation (Scherpereel, 2005, p. 50).

The computation of the Value-at-Risk crucially depends on the length of the chosen historical target horizon, because the predefined target horizon for the simulation of the portfolio changes serves as the starting point for the actual computation (Eisele, 2004). At first view, the target horizon should be chosen as long as possible to use all available historical data as a fundament and to eliminate the stochastic error. Based on the big amount of data, the ascertained frequency distribution is flattened, and the quantile can be determined more precisely (Eisele, 2004). However, this decision needs to be thoroughly considered and is not always supported due to the fact that the prospective probability of occurrence for long gone events decreases. Thus, very old historical data are used to create a prediction for current circumstances, although they might lack the purpose of being representative for

³ With $T=200$ and $c=0,95$ resulting $200*0,05=10$

⁴ The computation of the 1-day Value-at-Risk for a random holding period T can be made through following function: $T - \text{Daily VaR} = 1 - \text{Day} - \text{VaR} * \sqrt{T}$, cf. Dowd (1998), p. 65

prospective developments (Jorion, 2007). The choice of the target horizon's length can cause a decision conflict. A chosen target horizon, which is too small is not resulting in a reliable frequency distribution, whereas a longer target horizon is not computing the Value-at-Risk on a representative way (Eisele, 2004). The choice of the historical period as well as its length have significant impact on the outcome of the Value-at-Risk and consequently, on its risk prediction power. If irrelevant historical data is considered, while computing the Value-at-Risk, it will cause a decrease of the measurement's prediction reliability.

Another critiqued aspect of the Historical Simulation is that future events are exclusively described with passed data. The computation model supposes that events from the past will happen again in an analogous way in the future (Bizjak, 2011). This also includes also extreme situations such as the financial crisis in 2008 or the "*Dotcom-Bubble*" in 2000. If the target horizon includes such extreme situations, the Value-at-Risk is calculated at a higher value in Historical Simulations because of predicting a repetition of these events. This phenomenon is also called "*Ghost Feature*" or "*Echo Effect*" (Eisele, 2004, p. 72) due to an immediate change of the Value-at-Risk as soon as the defined historical target horizon does not consider these kinds of extreme events anymore. This phenomenon can also be observed in chapter 5, during our empirical investigation.

Furthermore, the Historical Simulation assumes implicitly that the volatility and the correlation of the risk factors are stable during the observation period and implies consequently a stationarity of the market variables. However, compared to practice this is not the case due to empirical proof that volatility as well as correlation change while time passes. Phases can be observed with high volatility as well as phases with low volatility (e.g. Alexander, 1998; Hull, 2011). To avoid this problem, the Historical Simulation needs to be adjusted with an additional estimation procedure for the volatility or correlation based on how the Historical Simulation is defined in this academic paper.⁵ Because of assuring comparability between the events of the researched computation models, this paper relinquishes the use of these additional procedures in the further process.

A further requirement, which needs to be fulfilled is the availability of the historical data. As appropriate, new financial instruments may not have any historical data available, which leads consequently to an expulsion of the Historical Simulation as a computation model. Moreover, with an increase in the number of financial products – such as banks using many variants of different types

⁵ Possible estimation procedures (EWMA or GARCH-Model) can be found in chapter 4.5. Please refer to the literature for a combination of these models with the Historical Simulation (e.g. Hull (2011), pp. 305)

of instruments and derivatives in large amounts – there is also an increase in demand for a better data administration as well as for a higher computational power to handle the numerous simulations (e.g. Alexander, 1998; Eisele, 2004).

The described disadvantages and issues of the Historical Simulation lead eventually to an inaccuracy of the quantification of the existing market price risks, whereby an impairment of the risk prediction power for the computation model becomes visible. This will be analyzed in the empirical investigation, chapter 5.3.1.

To sum up, the Historical Simulation and its computation method is, however, easy to implement and comprehend. Among others, this is one of the reasons why the Historical Simulation is frequently used in companies and has become popular throughout the entire industry (Christoffersen, 2009)

4.3. Variance-Covariance Simulation

Besides the Historical Simulation, the Variance-Covariance Simulation is also widespread and used by several banks and companies, which utilize the Value-at-Risk as a risk figure. Since 1996 the Variance-Covariance method is implemented and marketed in the database and software *RiskMetrics* from J.P. Morgan (e.g. J.P. Morgan/Reuters, 1996; Schroeck, 2002).

The Variance-Covariance method belongs to the parametric and analytical approaches to ascertain the Value-at-Risk. Thus, it is essential for the computation to make a distribution assumption for the risk factors' returns – here it means the risk of a change in the market price. In the most cases the normal distribution is chosen, and it is supposed that the risk factors are normally distributed. This assumption is made due to its simplicity. The normal distribution is determined with only two parameters, the mean μ (expected value) and the standard deviation σ (Bonke, 2007, p. 97). Based in this assumption, the returns of the total portfolio ensure the normal distribution accordingly (Jockusch, 2001). Consequently, the result is the Value-at-Risk of the portfolio P – under the assumption of a normal distribution for the return distribution of the risk factors – with the inverse of a cumulated normal distribution N^{-1} , the defined confidence interval c , the expected mean μ and the standard deviation σ_P of the portfolio's returns within the target horizon (Jorion, 2007, p. 111; Penza & Bansal, 2001, p. 69):

$$VaR_c = [\sigma_P * N^{-1}(c) + \mu] * P$$

Equation 14: Computation of Value-at-Risk assuming a normal distribution

Generally, in practice it is assumed that the expected daily return is virtually zero because the value change of the risk factors is very small compared to the standard deviation in a short target horizon (Hull, 2011). Moreover, equation 14 can be simplified by replacing the defined confidence interval $N^{-1}(C)$ through a factor z . The results for a confidence interval of $c = 0.95$ is a factor of $z_{0.95} = 1.64$ pr for a confidence interval of $c = 0.99$ a factor of $z_{0.99} = 2.576$ (Alexander, 1998, p. 80; Jorion, 2007, p. 111). The adjusted formula is expressed as followed:

$$VaR_c = z_c * \sigma_P * P$$

Equation 15: Computation of Value-at-Risk assuming a normal distribution (simplified)

The determined Value-at-Risk of this simplified formula is the exclusive result of the actual portfolio value, the standard deviation of the portfolio returns and the multiplier of the chosen confidence interval. Illustration two is showing a normal distribution, which is depicting the frequency of expected portfolio returns. By using the multiplier z_c the value-at-Risk can be calculated and measured.

A brief explanation of the z multiplier is also necessary to understand how this model works. The z multiplier is known as a standard score (Siegle, 2010), or more popularly, a z -score. This ‘score’ is a decimal number which is based on the given confidence interval. In the previous paragraph, the confidence interval of 95% provided a z -score of 1.96. This can be thought of how many standard deviations away from the mean the figure is. To continue with the previous example, using 1.96 can be interpreted as being just under two standard deviations away from the mean assuming a normal distribution, and is how the Value-at-Risk integrates the chosen confidence interval.

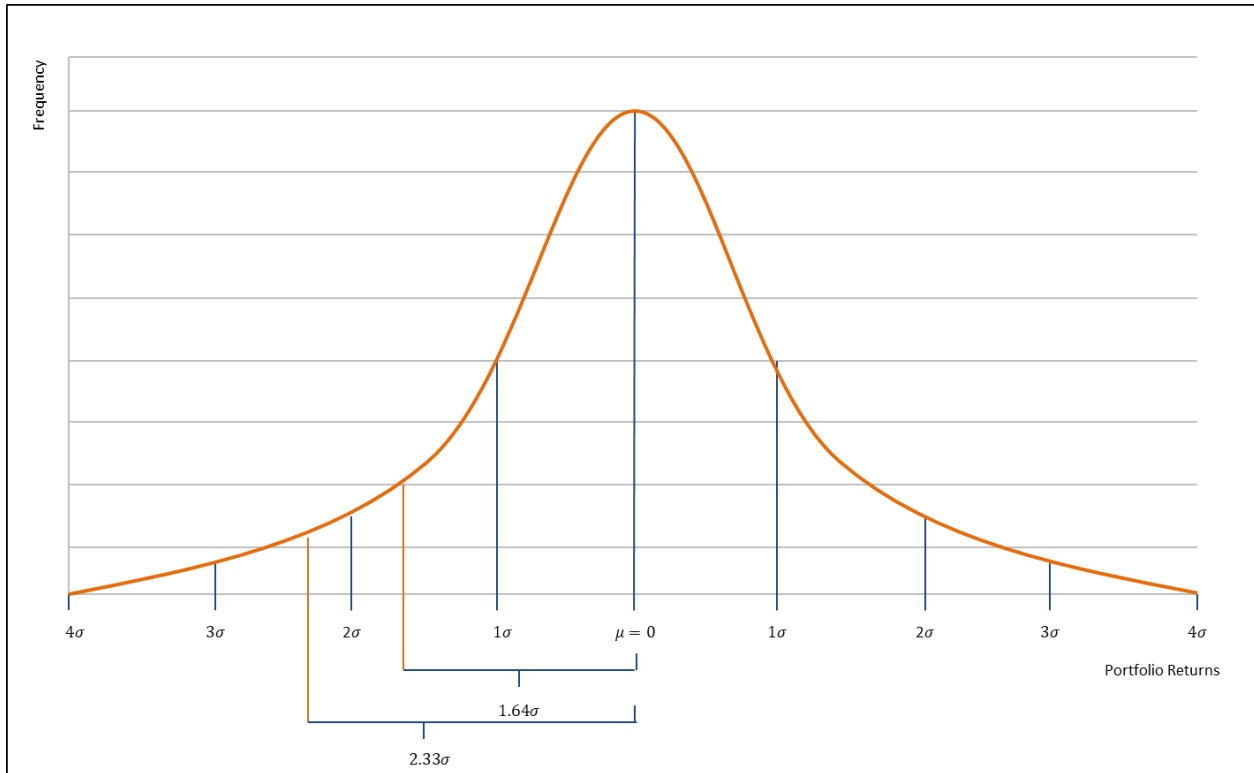


Figure 4: Schematic Normal Distribution

The standard deviation of the individual financial instruments within the portfolio as well as the correlation among these instruments is essential to ascertain the standard deviation and variance of the portfolio returns. Hence, contingent existing diversification effects can be considered while computing the Value-at-Risk (Deutsch, 2008). Applying formula 18 and formula 19 in connection with formula 15, the result for a portfolio with N values is the matrix notation for computing the Value-at-Risk. At that the covariance matrix, the weighting w of the single financial instruments in the portfolio and the chosen confidence interval c is taken under consideration (e.g. Alexander, 1998, p. 81; Dowd, 2002, p. 47; J.P. Morgan/Reuters, 1996, p. 11; Jockusch, 2001, p. 47; Jorion, 2007, p. 162,261).

$$VaR_c = z_c * P * \sqrt{(w_1 \Lambda w_N * \begin{bmatrix} \sigma_{X_1}^2 & \text{cov}_{X_1, X_2} & \Lambda & \text{cov}_{X_1, X_N} \\ \text{cov}_{X_2, X_1} & \sigma_{X_2}^2 & \Lambda & \text{cov}_{X_2, X_N} \\ \text{cov}_{X_3, X_1} & \text{cov}_{X_3, X_2} & \sigma_{X_3}^2 & \text{cov}_{X_3, X_N} \\ M & M & O & M \\ \text{cov}_{X_N, X_1} & \text{cov}_{X_N, X_2} & \Lambda & \sigma_{X_N}^2 \end{bmatrix} * \begin{pmatrix} w_1 \\ M \\ w_N \end{pmatrix})}$$

Equation 16: Computation of Value-at-Risk in Matrix Notation

By using this matrix function, the Value-at-Risk can be recalculated with small effort and does not require a complete new generation and simulation as long as the weighting, financial instruments within the portfolio and the volatility remain the same. The variance and covariance for the underlying financial instruments need to be estimated at best for the computation. Normally, this is done with historical data and estimation procedure, which are explained in more detail in chapter 5 (Alexander, 1998).

The issue with respect to the Variance-Covariance method is that it is not eligible for financial instruments with non-linear dependencies between risk factors and the portfolio value. The result would only be an estimation. The variance-covariance approach is based on the linear dependency of the changes of the risk factors and corresponds with Taylor's Theorem first order. To also depict also financial instruments with non-linear dependencies (e.g. Options), Taylor's Theorem second order needs to be chosen and the convexity should be considered in the computation, which is part of the "*Delta-Gamma approach*" approach (Bonke, 2007,). For that, vastly more complex computations are needed and the mathematical as well as the stochastic knowledge is essential for implementing this type of model in companies. Because the variance-covariance approach is sufficient for the investigations in this academic paper, the "*Delta-Gamma approach*" is not furthered researched.⁶

Due to the fact that the normal distribution is assumed for the distribution of returns for the risk factors, the Value-at-Risk is defined with only a few parameters. The simplicity of the computation model might speak for itself, however, this also leads to being stretched to its limits in practice. The assumption of a normal distribution is not always the best choice to approach to the actual return distribution of the risk factors (Jockusch, 2001). Besides other deviations between the normal distribution and the actual empirical observed distribution, the normal distribution is also lacking to consider so called "*fat tails*" of the real return distribution, whereby the Value-at-Risk is estimated too low (Bonke, 2007). This has direct impact on the risk prediction power because of a too low estimated Value-at-Risk, the probability increases that the predicted loss exceeds the Value-at-Risk measurement more often than the calculation would dictate.

To sum up, the Variance-Covariance Simulation is easy to apply in practice for ascertain the Value-at-Risk and provides an acceptable estimation for the Value-at-Risk, as far as the actual returns do

⁶ Further descriptions of the Delta-Gamma Approach can be found in the literature, cf. Dowd (1998), pp. 68; J.P. Morgan/Reuters (1996), pp. 13; Alexander (1998), pp. 85; Bizjak (2011), pp. 32

not differ too much from the normal distribution. The application under practical circumstances is the objective of investigation in chapter 5.3.2.

4.4. Monte-Carlo Simulation

As well as the Historical Simulation, the Monte-Carlo simulation belongs to the simulative approaches and thus, also to the “*Full-Valuation-Methods*”. However, the difference is that the simulation is mainly not based on historical data, but the Monte-Carlo simulation is assuming a probability distribution of the returns of the underlying risk factors, such as the Variance-Covariance Simulation. Due to this distribution several random numbers are generated for different scenarios. Therefore, the Monte-Carlo simulation is also classified as a parametric approach (Jorion, 2007). As already elucidated, the risk factors are defined as the returns, which are caused through changes of the underlying market prices.

Due to a free chosen probability distribution the application of the Monte-Carlo simulation is very flexible and non-linear dependencies are depicted. As a rule, the return distribution of the risk factors is based on a normal distribution, similar to the Variance-Covariance method (e.g. Bonke, 2007; Hull, 2011).

The procedure to ascertain the Value-at-Risk with the help of the Monte-Carlo simulation is explained as followed (e.g. Eisele, 2004, p. 127; Hull, 2011, p. 337; J.P. Morgan/Reuters, 1996, p. 26; Jorion, 2007, p. 321).

1. First of all, depending on the number of N risk factors and the number X of conducted simulations the independent random numbers $Z_{j,i}$ with $1 \leq j \leq N$ and $1 \leq i \leq X$ are generated. The random numbers comprise the codomain $[0;1]$.
2. To obtain an order of random numbers $\varepsilon_{j,i}$, which align with the normal distribution, the consistent distributed random numbers between 0 and 1 are assumed as random numbers of function values of the normal distribution N (Jorion P. , 2007, p. 312).

$$\varepsilon_{j,i} = N^{-1}(Z_{j,i}) \text{ for } 1 \leq j \leq N \text{ and } 1 \leq i \leq X$$

Equation 17: Transformation into normal distributed random numbers

3. The normal distributed and correlated random numbers $\varepsilon_{j,i}^{correlated}$ as well as the respective standard deviation σ_j and the current value $S_{j,o}$ of the risk factor j are used to simulate the change of the risk factor $S_i^{simulated}$.⁷

$$S_i^{simulated} = \sum_{j=1}^N (S_{j,o} * \sigma_j * \varepsilon_{j,i}^{correlated}) \text{ for } 1 \leq i \leq X$$

Equation 18: Simulation of risk factor changes

4. For each simulation i there will be individual risk factor changes, which result in a cumulated value change $S_i^{simulated}$ in the total portfolio. Based on the numerous simulations and the resulting probability distribution of the simulated portfolio values, the Value-at-Risk is determined by choosing eventually an appropriate quantile based on the confidence interval selected.

Compared to the Historical Simulation or the Variance-Covariance method the Monte-Carlo simulation is more flexible in its application. In addition to the normal distribution, the Monte-Carlo simulation is making a random distribution assumption, which is taken as a hypothetical model (Deutsch, 2008). Even if the returns of the risk factors differ significantly from the normal distribution and follow another known distribution function, the Value-at-Risk can still be estimated to be trustworthy. Moreover, non-linear dependencies are easier to implement and can be appropriately considered (e.g. Alexander, 1998; Bizjak, 2011; Scherpereel, 2005).

Also, the Monte-Carlo simulation is less volatile and less affected by the historical developments like the Historical Simulation. The changes of the risk factors are generated fictional due to random numbers. Extreme events can thus, seen as possible but have not occurred yet in the historical data. Another difference between the Historical Simulation and the Monte-Carlo simulation is that the Monte-Carlo simulation does not need any empirical, historical data sets, because the needed data are generated stochastically. Especially for new financial instruments this difference is an advantage. New financial instruments are often times lacking sufficient historical data, whereas this issue is avoided by using the Monte-Carlo simulation (Bonke, 2007).

Besides the big flexibility of the Monte-Carlo simulation, the model comes also with disadvantages and issues. First, the big amount of simulations requests a substantial processing power and data

⁷ The simulation is based on the assumption that the risk factors are following the stochastic process of a geometric Brownian motion. Because of observing daily returns, the expected value of the returns can be neglected

retention, which causes a certain amount of time to compute the Value-at-Risk. The conflict comprises the trade-off whether the quality of computing the Value-at-Risk is worth the large effort required for calculating this model in the form of processing power (e.g. Eisele, 2004; Jorion, 2007; Scherpereel, 2005).

Second, while repeating the execution of the Monte-Carlo simulation the results will vary. Consequently, the results will differ when computing the Value-at-Risk several times, even if the Monte-Carlo model and the applied parameters are identical. This is caused through the randomized numbers, which influence the simulation replicability. This problem can, however, be reduced by increasing the executions of simulations because the simulated distribution function is flattened, and the result is a more reliable estimation of the Value-at-Risk (Eisele, 2004). Yet small discrepancies will still exist, and users should be aware of this when using the Monte-Carlo Method.

Moreover, by assuming the normal distribution as the problem is the same as with the Variance-Covariance method. The actual returns of the risk factors might as the circumstances require not align with the normal distribution. Thus, the Monte-Carlo simulation has a comparable prediction power to the Variance-Covariance Simulation (Neumann, 2000). As a consequence, it can be recorded that the Monte-Carlo simulation, due to its required processing power and effort, should only be used in practice if the two other computation models (Historical Simulation and Variance-Covariance method) do not depict the underlying risk factors in the portfolio in a suitable way (Bonke, 2007).

In the same way as the Variance-Covariance method the Monte-Carlo simulation requests an estimation of the parameters for the underlying distribution function of the risk factors. This means that the computation approach and the Value-at-Risk are essentially dependent on the fundamental determination and estimation of the parameters. The procedure of this chapter alleges a constant volatility for a risk factor within the target horizon. However, this assumption is barely found in practice, which is why the model can be extended by parameters which change by time. A further explanation follows in the next chapter and in the empirical investigations, which applies for the estimation of the parameters. The amount which risk prediction power can be improved by applying time changing parameters will also be shown in the empirical investigation.

4.5. Estimation of the Volatility & Correlation

The parametric-based calculation methods for the Value-at-Risk require a determination of specific parameters, such as volatility or correlation to receive a result. The parameters are the most influential part for the Value-at-Risk measurement. Therefore, it is essential to estimate highly practically-oriented parameters, which are assured to receive a significant prediction capability and thus, achieve a higher risk prediction of the calculation models. Both in theory and in practice there are several estimation models for the parameters, which differ mainly regarding the calculation complexity. The simpler models assume a stable volatility and correlation for the target horizon. This target horizon is serving as the fundament for the data source in the estimation process. However, this procedure implies that the risk remains unchanged. This assumption seems to be very theoretical and is hard to project to reality, especially in more turbulent market phases (Jorion P. , 2007). Due to this fact, flexible models, which include a dimension of adjusting parameters on an ongoing basis increase in their use and accuracy.

Besides numerous models, there are three models with similar fundamental procedure. The first one is called *Moving-Average Model* (MA), the second one is called *Exponentially-Weighted-Moving-Average* (EWMA) and the third one is known as the *Generalized-Autoregressive-Conditional-Heteroskedastic* (GARCH) model (e.g. Hull, 2011; Jorion P. , 2007).

4.5.1. Moving-Average

The actual idea of the Moving-Average model is to observe the empirical return and ascertain the volatility estimation for prospective investigations, based on a historical target horizon. The basic procedure is comparable to the Historical Simulation, which is also using a specific target horizon as the data source. Premised on daily computations of the Value-at-Risk at time t and the daily return observations, this target horizon considers T last daily returns in its consideration. Consequently, a new estimation of the parameters would result in a shift of the target horizon on the next day $t+1$, meaning that the oldest return will drop out of the observation horizon. The latest return will be added to the contemplation horizon.

Thus, the variance is expressed as σ_t^2 at time t based on daily returns u_t at time T (e.g. J.P. Morgan/Reuters, 1996, p. 78; Jorion, 2007, p. 222; Saita, 2007, p. 39):

$$\sigma_t^2 = \frac{1}{T} * \sum_{t=1}^T (u_t - i)^2$$

Equation 19: Moving-Average Estimation for Variance

Analogically, the covariance between two financial instruments X and Y in a portfolio is expressed as (J.P. Morgan/Reuters, 1996, p. 83):

$$cov_t = \frac{1}{T} * \sum_{t=1}^T (u_{t-i,X} * u_{t-i,Y})$$

Equation 20: Moving-Average Estimation for Covariance

The daily returns, which are observed within the target horizon, are weighted equally in the calculation by the use of sigma. However, it needs to be criticized that there is no differentiation in the daily returns' relevance. Returns, which are sampled a long time ago, are still weighted equally and have the same relevance as the latest return data. This is a problem because of the latest returns may have a closer linkage to the current market developments (Alexander, 1998). Moreover, a risk which may occur is of experiencing the so called “*Ghost Feature*” (Jorion, 2007, p. 222), which describes erratic changes in the estimations due to single extreme results in the applied target horizon.

The Moving-Average Model is assuming that the volatility as well as the correlation remains constant during the target horizon and even more stay constant for the future. In addition, it is also assumed that the daily returns are completely independent and show no correlation between each other. Both assumptions are not sustainable in practice due to frequent volatility clusters. The volatility as an indicator for risk is versatile during time. This issue is captured in the *EWMA*.

4.5.2. Exponentially-Weighted-Moving-Average (EWMA)

With respect to the weaknesses of the Moving-Average Model, the EWMA Model distributes the daily returns on an exponentially descending way. Hence, older returns are counted as less important, whereas the latest returns are classified as more important due to their proximity to the date Value-at-Risk is being measured for. This weight on recent returns improves the reaction time of revaluating the estimated parameters in case of unforeseen or extreme market developments and enables the ability to create a by time changeable volatility (Alexander, 1998).

The variance σ_t^2 at time t based on daily returns u_t in the target horizon T in the EWMA Model is calculated with a weighting factor λ (e.g. J.P. Morgan/Reuters, 1996, p. 78; Saita, 2007, p. 41):

$$\sigma_t^2 = (1 - \lambda) * \sum_{i=1}^T \lambda^{i-1} * (u_{t-i})^2 \text{ with } 0 < \lambda < 1$$

Equation 21: General Estimation of Variance in EWMA Model

Through repetitive application the general function can be transformed into a recursive function. Therefore, the recursive parameter bases simply on the estimated variance and daily return from the prior day (e.g. Dowd, 2002, p. 95; Jorion, 2007, p. 230; Saita, 2007, p. 41).

$$\sigma_t^2 = \lambda * \sigma_{t-1}^2 + (1 - \lambda) * u_{t-1}^2$$

Equation 22: Recursive Estimation of Variance in EWMA Model

Analogically, the covariance in the EWMA Model between two financial instruments X and Y in a portfolio is expressed as (e.g. Dowd, 2002, p. 96; Jorion, 2007, p. 234):

$$cov_t = \lambda * cov_{t-1} + (1 - \lambda) * u_{t-1,X} * u_{t-1,Y}$$

Equation 23: Recursive Estimation of Covariance in EWMA Model

A very significant fact of the recursive functions, both for the variance and the covariance, is the simplicity of the parameter computation. It is sufficient to use the already known data for the parameter and add the newly observed daily returns, instead of storing huge amounts of historical

data. Repatriating to the simplicity and easy way of implementation this model is also used in J.P. Morgan's *RiskMetrics* (J.P. Morgan/Reuters, 1996).

Regarding formula 22 and formula 23, the weight for both is allocated by the λ -times the prior weight (Hull, 2011). Hence, the choice of the right weighting factor λ – known as *Decay-Factor* – is an essential key figure for reliable parameter estimation. A low λ admeasures a high emphasis for the newest daily returns. Consequently, this model reacts on the latest market events and considers non-stationarity of the volatility. However, a *Decay-Factor* close to 1 weights the latest daily returns as less important and reacts more sluggish to new market events due to a larger amount of historical data is taken into consideration. Through this procedure single extreme events are moderated (Hull, 2011; Neumann, 2000). For instance, J.P. Morgan chooses a *Decay-Factor* of $\lambda=0.94$ because it is delivering the most suitable result for a portfolio existing out of several different financial assets (J.P. Morgan/Reuters, 1996).

The EWMA Model generates good results for estimating prospective volatility and correlation, while keeping close to practice. Moreover, it can be used with a good comprehensive calculation method. All these facts are leading to a reliable prediction power of the EWMA Model. Due to this, there EWMA Model is a special case of the GARCH Model, which we will have a closer look at in chapter 4.5.3.

4.5.3. Generalized-Autoregressive-Conditional-Heteroskedastic (GARCH)

As the EWMA Model, the Generalized-Autoregressive-Conditional-Heteroskedastic (GARCH) Model can be described with recursive functions to estimate the volatility and correlation. However, the difference is seen in an additional parameter to determine the long-term average volatility V_L or correlation C_L (Hull, 2011, p. 226).

Conclusively, the recursive function is formed similar into the EWMA Model, based on the estimate variance from the prior day, the new observed daily return and the long-term average volatility, each weighted with a factor γ, α and β (e.g. Dowd, 2002, p. 95; Jorion, 2007, p. 223; Tsay, 2002, p. 94):

$$\sigma_t^2 = \gamma * V_L + \alpha * u_{t-1}^2 + \beta * \sigma_{t-1}^2 \text{ with } \gamma + \alpha + \beta = 1$$

Equation 24: Recursive Estimation of Variance in GARCH Model

Analogically, the covariance in the GARCH Model between two financial instruments X and Y in a portfolio with long-term average covariance C_L is expressed as (Hull, 2011, p. 246):

$$cov_t = \gamma * C_L + \alpha * u_{t-1,X} * u_{t-1,Y} + \beta * cov_{t-1} \text{ with } \gamma + \alpha + \beta = 1$$

Equation 25: Recursive Estimation of Covariance in GARCH Model

Several variations exist in the literature of the GARCH Model with respect to their application to time series analyses. (Alexander, 1998). However, in practice these variations are rarely used due to the additional calculation step. Before calculating the variance or covariance, the weighting factor needs to be determined upfront with e.g. the *Maximum-Likelihood-Estimation* (e.g. Hull, 2011, p. 227; Pollock, 1999, p. 667; Tsay, 2002, p. 14). This process contains a high computation complexity, because of solving the optimization issue of several variables (Jorion, 2007).

The improvement of the GARCH Model, compared to the EWMA Model, is the utilization of the long-term average volatility or correlation. This key figure is adjusted in its respective relevance, depending on the choice of the weighting factor. However, the ratio of implementation and improvement does not justify the higher prediction power of the model. (Jorion P. , 2007) The EWMA Model is often as capable to consider the changing volatility or correlation and therefore, to figure the common volatility cluster. With respect to this academic paper, the Moving-Average and the EWMA Model are exclusively used to estimate the time changing parameters.

5. Empirical Investigation of the Value-at-Risk

The following sections will provide some high-level insights on our data used in the analysis, in depth explanation of how we will evaluate our results, and most importantly, a full thorough presentation of the results from our research.

5.1. Data

In order to compare the computation models, it is necessary to determine a sample portfolio as well as an observation period. The choice of both is further explained in the upcoming paragraphs.

The data for the empirical analysis is composed of two different positions. On the one side, the share index DAX Performance Index (DAX)⁸ is observed and on the other side the Dow Jones Industrial Average Index⁹ (DowJones). The representative role of both indexes covers the German market as well as the American market while mirroring a suitable picture of the respective equity markets.¹⁰ The daily changes of the indexes are used to calculate the daily returns for the portfolio, which is further used to compute the Value-at-Risk figure. Further, we assume a similar investment in both positions of €10.000 at the same time, for a total investment of €20.000. Both positions will be shown later in the same currency (for simplicity reasons). In this context, it makes sense to keep the positions on a constant level. The investment amount, for which the Value-at-Risk is computed for every day stays at €20.000 (independently of the performance from the day before). This procedure is essential to allow the comparability of the Value-at-Risk results.

The historical time horizon is based on the premise of having the larger time horizon available for our analysis to be sure any of the trends we are noting are not anecdotal, or just an outcome of the fixed period of which we take into consideration. In this sense, we went back in time as far as we could and used a period of almost 30 years (12/30/1987 – 06/02/2017) with the DAX being established in the early 1980's, but officially published in 1988 (Andlil, 2016). By utilizing this time horizon, we will also be able to highlight specific smaller time periods within which draw attention. When using Value-at-Risk in practice, a shorter time horizon from 1-4 years is more common among financial institutions and traders when they are using Value-at-Risk measurement, so we will provide focus to this also. Additionally, only trading days are considered and weekends as well as common

⁸ DAX Performance Index (ISIN: DE0008469008)

⁹ DowJones Industrial Average Index (ISIN: US2605661048)

¹⁰ Further investigations between performance & price index will be neglected in this paper

holidays (on German and American exchange) are excluded. Differences in trading days due to various holidays in both countries are avoided by using the last closing price from the prior trading day for the non-trading day. Hence, it is also possible to compute a Value-at-Risk for days on which the exchange in the respective country is closed. However, this procedure will cause minimum differences regarding the quantification of the actual market price risk (Jorion P. , 1996), which can be neglected. The temporal course of the DAX and DowJones can be seen below for the chosen observation period in a graphical illustration.

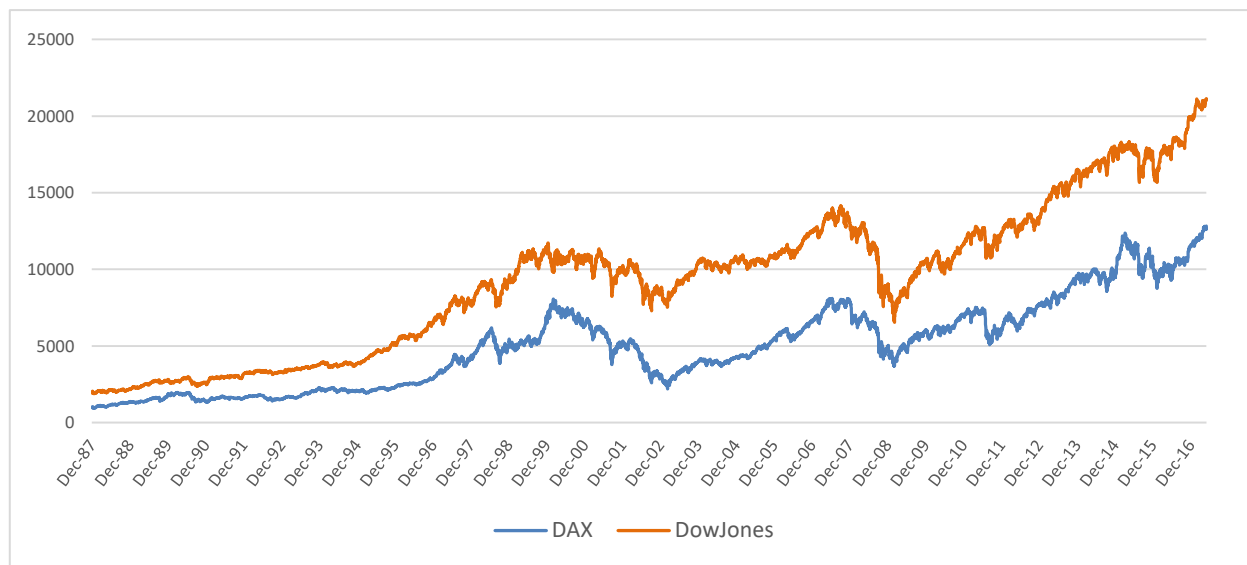


Figure 5: Development of DAX and DowJones Index

Shown above are 250 data points¹¹ of the DAX and the DowJones (means respectively from the 06/01/2007 until 05/20/2008) are used as an example snapshot of our dataset. This is obligatory due to having a solid and reliable historical data setting for the first computed Value-at-Risk. Hence, the first Value-at-Risk is computed for the 03/09/1998 and subsequently on a daily basis until the end of the observation period (06/01/2017). For each model, the number of Value-at-Risk measurements which will be included varies. This is further discussed in the section below, 5.2. But to provide some summary figures, there will be a minimum of 7347 daily Value-at-Risk measures, and a maximum of 7547 Value-at-Risk measures.

The applied historical data sets¹² of DAX and DowJones are based on the daily closing prices of the indexes. The prices during the trading days are not observed and will not be considered in the Value-

¹¹ Data point means the price of DAX and DowJones on the same trading day

¹² The applied data of DAX and DowJones were downloaded from Yahoo! Finance

at-Risk computation. Further, a continuous compounded return is used for the assessment of the daily return. The application of the continuous logarithmic return results in the following function to compute the daily return for one of the two indexes (Jorion P. , 2007, p. 93):

$$\text{Daily Logarithmic Return}_t = \frac{\text{Log}(\text{ClosingPrice})_t}{\text{Log}(\text{ClosingPrice})_{t-1}}$$

Equation 26: Logarithmic Daily Return

5.2. Objectives & Procedure of Investigation

To begin with our analysis, it was important to get a general sense of the data that we are working with. As written above, we used logarithmic continuous returns for calculating the returns of our daily stock data. We began by plotting the price and daily returns during our time horizon for both of the indexes, DowJones and DAX, in order to get an overview of the performance of these indexes during our tested horizon and also pulled some of the key figures of each index.

Time	DowJones Return	DAX Return
10/30/1991	-0.00873	-0.003942
10/31/1991	-0.00426	-0.004562
11/01/1991	-0.00351	-0.002859
...		
5/30/2017	-0.00010	0.001299
5/31/2017	0.00643	0.003944
06/01/2017	0.00293	0.012399

Table 2: Daily Returns for DAX and DowJones (Exemplary Extract)

Sample Size	Mean	Median	Std. Dev.	Minimum	Maximum	Skewness
7597	0.0003137	0.0002686	0.01051468	-0.0820051	0.1050835	-0.227763

Table 3: Summary of DowJones Statistics for Daily Returns

Sample Size	Mean	Median	Std. Dev.	Minimum	Maximum	Skewness
7597	0.0003351	0.0005748	0.01406121	-0.1409117	0.1079747	-0.242749

Table 4: Summary of DAX Statistics for Daily Returns¹³

With our initial analysis of the data completed, we next moved to assessing the predictive capabilities of each of the models. The paper “Value-at-Risk Prediction: A Comparison of Alternative Strategies,” (Kuester, Mittnik, & Paolella, 2006) was able to provide important guidance in performing an academically valuable way of conducting the predictive ability of our models chosen. It was not possible for us to concretely say one model ‘performed better’ than the other without this type of underlying statistical approach to evaluating each model.

While conducting the analysis and calculating the Value-at-Risk for dates in the early years, we ran into the challenge of the data being heavily impacted by single dates. This hurt the accuracy of our Value-at-Risk measures, and this also did not represent a real-world use of the model, since any trader or banker with some sort of industry knowledge would likely have a better sense if the appropriate parameters for a market rather than solely relying on a month, or even a year of historical data. This is also backed by the previously mentioned article (Kuester, Mittnik, & Paolella, 2006) and the author was able to handle this challenge by introducing a “window”. Instead of beginning our analysis on day one, we had a moving window, which would start at the 50th trade date¹⁴, and from then on this would be a ‘moving window.’ This would mean each Value-at-Risk calculation for a given day has 50 daily returns that make up the Value-at-Risk on a given day. By using a window, which includes 50 trade dates, it encompasses approximately two months of trade data which is a reasonable basis to use to for evaluating the effectiveness of the model (e.g. Hull, 2011; Jorion P. , 2007) This procedure is also applied by the previously cited article in their analysis.

¹³ Skewness represents a measure of the asymmetry of a probability distribution. In this case, the negative skewness can be generalized to represent that the tail on the left side of the function (negative) is longer or fatter than the tail on the right (positive) side of our function. [Joanes, 1997]

¹⁴ Respectively for 120 & 250 trade days; this will be more explained later in the procedure

	Historical Simulation	Variance-Covariance Simulation		Monte-Carlo Simulation
		Moving-Average	EWMA	
Confidence Interval	99%	99%	99%	99%
Holding Period	1 Day	1 Day	1 Day	1 Day
Assumption for Return Distribution	-	Normal Distribution	Normal Distribution	Normal Distribution
Historical Time window	50 Days 120 Days 250 Days	50 Days 120 Days 250 Days	$\lambda=0.94$ $\lambda=0.97$ $\lambda=0.99$	50 Days 120 Days 250 Days
Weighting of Historical Observations	Equally Weighted	Equally Weighted	Exponentially Weighted	Equally Weighted
Number of Simulations	-	-	-	10,000

Table 5: Framework of the Empirical Analysis¹⁵

By using this moving window, it also provides the added benefit of replicating the situation that a risk manager or trader would be in. As another data point is added to the window, the Value-at-Risk measure has a unique pool of data to be computed from. However, the only data that is included are dates that have occurred, and no dates in the future.

For instance, the Value-at-Risk measure for the 56th trading day under the Historical Simulation method is € 931, - (assuming an investment of 20,000). What this calculation encompasses at this point is all the daily returns from the prior 50-trade days. In short, as the window is rolling further to include another return amount, the Value-at-Risk calculation internalizes this amount and it is reflected in the calculation, while the oldest amount is falling out of the window. In our analysis, by

¹⁵ Respectively for the Confidence Interval of 95%; not shown in the following due to redundancy; please find further information in the Appendix

having the initial window as 50, we are left with 7,547 ‘one-step’ Value-at-Risk calculations for our analysis.

To assess and compare the accuracy of a Value-at-Risk model; the model must satisfy two properties which are defined by (Kupiec, 1995) and (Christoffersen, 1998) respectively. These two properties are the test of standard coverage and a test for independence. Glyn Holton’s book, “Value-at-Risk” provides additional guidance on how to test for these properties for a Value-at-Risk model (Holton, 2003).

To begin, a standard coverage can be modeled as follows:

$$I_t \begin{cases} 0 & \text{if loss } p_{t-1} - p_t \text{ is less than or equal to VaR} \\ 1 & \text{if loss } p_{t-1} - p_t \text{ exceeds VaR} \end{cases}$$

Equation 27: Standard Coverage

I_t is the variable which indicates if a violation occurred on a given date, meaning that the actual loss on a given date exceeded the forecasted Value-at-Risk on the date.

Next can define coverage, q^* by the frequency in which the Value-at-Risk model experience actual exceedances (Holton, 2003), more specifically when $I_t = 1$ from the model above, and N is the number of trade days Value-at-Risk is being evaluated for. This is modeled below by stating the standard expectation.

$$q^* = E(I_t)$$

Equation 28: Exceedance Percentage

This provides us with our null hypothesis, where q is the confidence interval being tested for.

$$H_o \rightarrow q = q^*$$

Equation 29: Null Hypothesis – Standard Coverage

A Value-at-Risk model is considered to be underestimating the risk and value that can be lost if the number of violations exceed the number of expected losses at a given confidence level, or too conservative if the number of dates the actual loss exceeds the forecasted Value-at-Risk is below the given confidence level expectation.

The test can be simulated and shown as follows, where x is the number of times which the data exceeds the calculated Value-at-Risk measurement:

$$X = \sum_{t=0}^{\alpha} i^{[t]}$$

Equation 30: Sum of Exceedances

The next property is (Christoffersen's 1998) Exceedance Independence Test of violations which occur. When violations occur in groups instead of spreading across a normal distribution, it may cause a risk manager to not be able to prepare and handle the violations. The violations should be independently distributed with probability α . The independence property can be shown as follows, referencing our notation from above for the confidence level, q^* :

$$\begin{cases} q_0^* = \Pr(I^t = 0 \mid i^{t-1} = 0) \\ q_1^* = \Pr(I^t = 0 \mid i^{t-1} = 1) \end{cases}$$

Equation 31: Null / Alternative Likelihood Ratio

This model can be understood as the q_0^* being the probability that Value-at-Risk was not exceeded in the prior period, and q_1^* that the Value-at-Risk was exceeded in the prior period.

5.3. Presentation of Results

As stated in chapter 3, the different methods provide unique benefits in the way of timeliness, understandability, and resources required being a few of the attributes that were expanded upon earlier. To get an initial sense of the data, we graphed the Value-at-Risk for each day, showing value-at-risk as a percentage of the investment. The value-at-risk is shown for each of the three models, at both 95% and 99% confidence intervals, and on both index portfolios during the specified time horizon. Each graph originates at the data point after the length of the relevant window, i.e., for the 250-trade day window, the first Value-at-Risk measure is on trade day 251. Each Value-at-Risk calculation contains a moving data set of the agreed upon trade day windows, as described in the previous section.

5.3.1 Historical Simulation

The Historical Simulation method is directly based on the historical daily returns of the two indexes. We are able to note the movement of returns with this Value-at-Risk, but as can be expected because of this model's dependency on historical returns, there is a 'lag' between when returns occur and when they are reflected in Value-at-Risk measure. Lag in this sense meaning, the large loss which exceeds the Value-at-Risk measure is only included the measurement day after the loss is realized. Since the historical method is making no assumptions regarding the distribution of the returns, and instead focuses on the idea that history will repeat itself (FAN, 2011) there can be points where Value-at-Risk 'flattens' in a way that can be noticed when analyzing the graph below. This is during periods of low variance in returns, so the Value-at-Risk measure is more static, as can also visualized in the following figure.

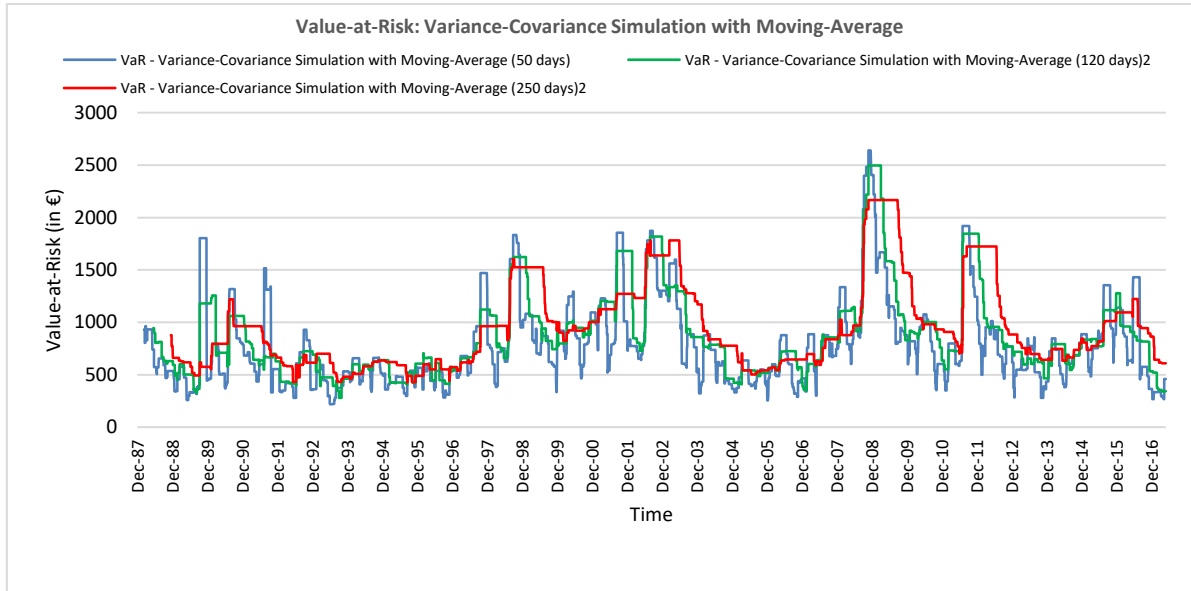


Figure 6: Value-at-Risk based on Historical Simulation

As the daily logarithmic returns explicitly enter the Value-at-Risk data on a for all prior trade days of the relevant window, as seen in the formula below, where α is the confidence interval, and q is the Value-at-Risk quantile, meaning the placement in the distribution of the daily loss value which is being returned from the historical returns.

$$VaR_t = q(1 - \alpha, \text{Distribution of Returns}[t - 1, \text{Window}])$$

Equation 32: Historical Simulation Value-at-Risk with Window

Since historic data and a window is being used in the calculation of Value-at-Risk, it is clear to see how a daily loss may exceed the Value-at-Risk using this method. A volatile spike in returns which leads to a large loss one day will itself exceed the Value-at-Risk threshold. Going forward this loss amount will be included in the historical returns that are part of the next day's Value-at-Risk measurement. Yet, if for example the window being used is 250 days, at a 99% confidence interval, the Value-at-Risk the following day would actually be the 2nd largest loss from the window. If the volatility that created the initial loss still exists, this could create another day in which the actual loss exceeds the threshold provided by Value-at-Risk. However, this deficiency is not unique to the Historical Simulation method compared to the other two models, since they also use a value placement return method in the form of a z-score, and instead the spike in volatility impacts the assumed normal distribution of returns, rather than the returns themselves being the Value-at-Risk measure (Jorion P. , 2007)

The below table provides an example output from our analysis. The data being shown is from the Historical Simulation Value-at-Risk measurement using a confidence interval of 99 and a window size of 50.

Time t	Portfolio Return r_t	Value-at-Risk (in €) VaR_t
6/01/2017	€ 248, -	€ 458, -
5/31/2017	€ 187, -	€ 458, -
...	...	...
12/19/1988	€ 269, -	€ 535, -
12/16/1988	€ 356, -	€ 535, -
12/15/1988	€ 387, -	€ 535, -
12/14/1988	€ -200, -	€ 535, -

Table 6: Value-at-Risk with historical time horizon of 50 trade days, confidence level of 99 (exemplary extract)

As we can see from the table, the impact of a return is not initially seen with a likewise change in the Value-at-Risk measurement. This also provides a clearer picture of the 'plateaus' we see in the graphs below. Since the only way that the historic simulation Value-at-Risk will change is because either a new loss occurs which decreases what the lowest 1% or 5% return is, or since a window of returns is being used instead of all historic returns, a loss in the relevant quantile 'falls' out of the window in a sense when a new return is included, also leading to a new Value-at-Risk measurement.

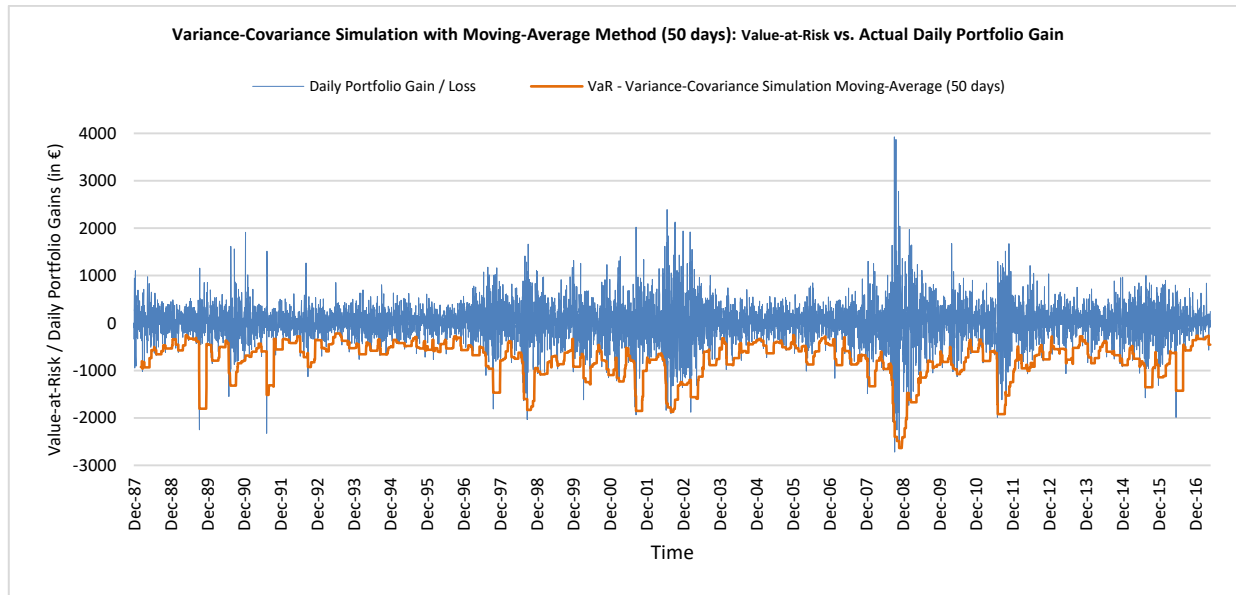


Figure 7: Historical Simulation (50) Backtesting

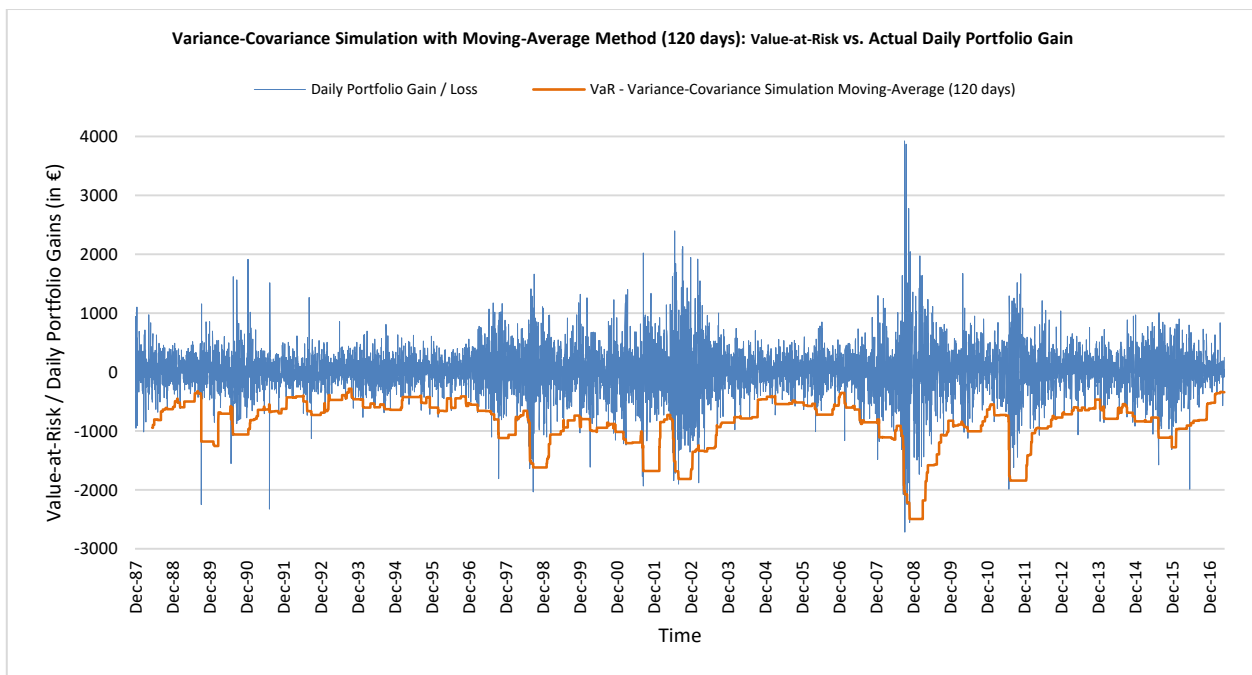


Figure 8: Historical Simulation (120) Backtesting

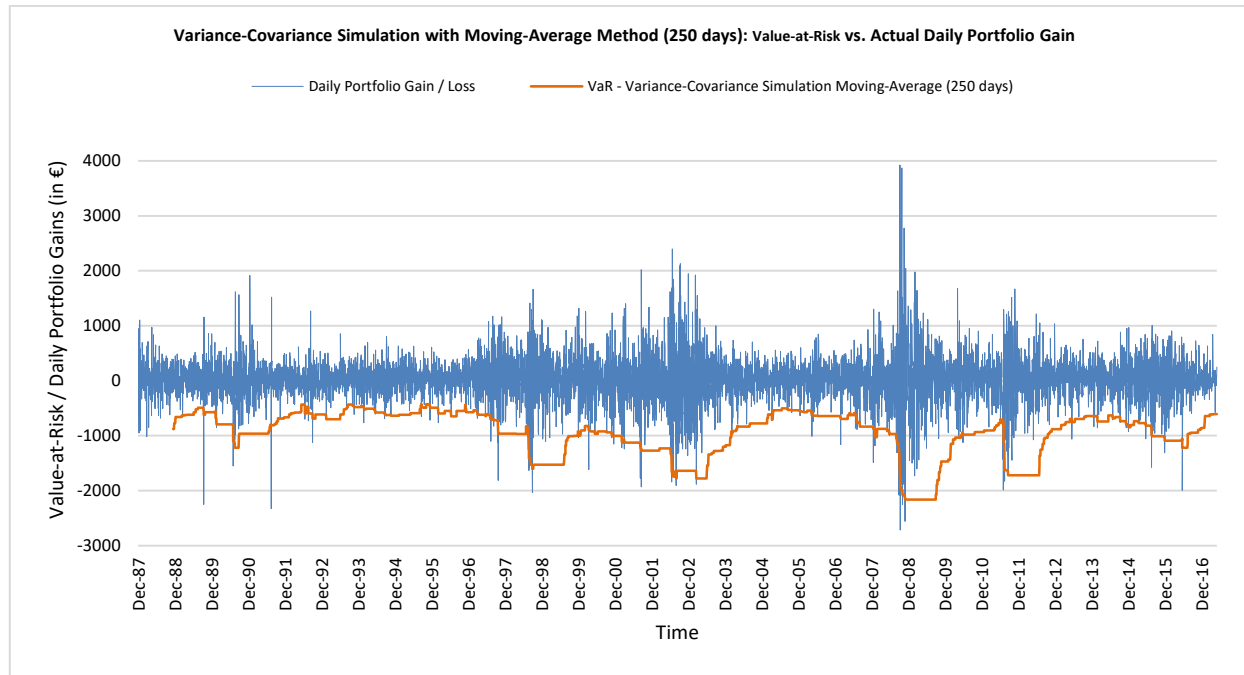


Figure 9: Historical Simulation (250) Backtesting

The graphs share many of the same tendencies since they are all calculated using the same confidence interval. However, a key difference that is impacted by the window in use, and is most notable between the 50 and 250 window graphs is the length of the ‘plateaus’ in the results. As touched upon earlier, because of the larger data set which is being referenced to with a longer window, the time at which value at risk measure stabilizes around one amount tends to last longer. This results from an instance of a large loss. If we look closer at the graphs, particularly the 250 days window graph where it is most pronounced, we can see almost ever outlier type of loss is followed by a period of constant Value-at-Risk measurement. Because of the longer time horizon, the uncommon loss spike is able to stay and be reflected in the Value-at-Risk measurement for a longer period of time.

	50 Trade Days	120 Trade Days	250 Trade Days
Minimum Value-at-Risk	€ 219, -	€ 280, -	€ 425, -
Maximum Value-at-Risk	€ 2,639, -	€ 2497, -	€ 2166, -
Average Value-at-Risk	€ 785, -	€ 868, -	€ 942, -
Number of Exceedances	170	125	94
Percentage of Exceedances	2.25%	1.67%	1.28 %

Table 7: Variance-Covariance Simulation with Moving-Average Method: Backtesting Results

The table above quantifies what we are able to note from graphing the outcomes. It should come as no surprise that the average Value-at-Risk is highest by using the 250-trade day window (€ 942, -), which allows for a large loss to exist in the Value-at-Risk calculation for the longest period of time. We can also notice a fairly linear relationship between the three measurement methods, with minimum Value-at-Risk increasing and maximum Value-at-Risk decreasing respectively as the window is further expanded. The 50-trade day's window is able to quickly spike and react to a sudden loss, yet this amount is soon left out of the calculation, and as we can tell from our results, it seems this creates many occurrences of actual loss exceeding the Value-at-Risk estimate. Meanwhile, the 250-trade days window is able to perform superiorly of the three measurements by a fairly wide margin by keeping a consistently high Value-at-Risk amount, which is particularly important in this occurrence since each of the three measurements estimate Value-at-Risk are not a high enough value that the confidence interval applied is being met.

The simplicity of adding more historical trade dates to this model to achieve substantially superior results provides some questions on what purpose having the 50 days trade window could have. The 50 days trade window was used by JP Morgan in the original *RiskMetrics* (J.P. Morgan/Reuters, 1996), so it appears the measure with these parameters would have value. The benefit of using a small window would be to capture short-term movements (Hendricks, 1996) in the underlying return data. Possibly in a more volatile environment than an index portfolio of a major market, where the mean returns over a long horizon have historically generally been positive would be a more appropriate place to use a shorter window such as 50 trade days. However, for our analysis the performance was poor when compared to its alternatives, but this could be an area for further research.

5.3.2 Variance-Covariance

The Variance-Covariance methods share many similarities because they both assume a normal distribution of returns. However, as it will be shown in the following, the impact between using a simple Moving-Average and exponential weighted Moving-Average method provides materially different results.

The mean and variance of the underlying numbers were computed by using the historical data. The mean and variance were computed for each window, so each Value-at-Risk measure has unique attributes. This action was the same for both the simple Moving-Average and the exponential

weighted Moving-Average. However, the specifics of these two models will be addressed separately in the sections below.

5.3.2.1 Variance-Covariance Simulation under Consideration of Constant Volatility (Moving-Average)

The following is devoted to the research of the Variance-Covariance Simulation based on the Moving-Average method for the parameters, which comprise the return distribution of the market price risk factors of the normal distribution. Thereby, the parameters of the normal distribution are ascertained based on historical data such as for the Historical Simulation. The rolling time window is chosen with a length of 50, 120 and 250 trade days for the empirical analysis. First, the application of the Variance-Covariance method, based on the Moving-Average method, is described for the chosen target horizon and afterwards the calculated Value-at-Risk results are analyzed with respect to their preciseness resp. risk prediction power, through backtesting procedures.

The daily changes of the market prices u_t (respectively of DAX and DowJones as risk factors) incorporate in the estimation of the parameters such as variance (formula 19) and covariance (formula 20). The variance of DAX and DowJones as well as the correlation between the two indexes are produced based on the chosen historical time window T on every day t in the target horizon.

$$\sigma_{t,DAX}^2 = \frac{1}{T} * \sum_{i=1}^T (u_{t-i,DAX})^2$$

$$\sigma_{t,DJ}^2 = \frac{1}{T} * \sum_{i=1}^T (u_{t-i,DJ})^2$$

Equation 33: Moving-Average Method for the Variance of DAX and DowJones

$$cov_t = \frac{1}{T} * \sum_{i=1}^T (u_{t-i,DAX} * u_{t-i,DJ})$$

Equation 34: Moving-Average Method for the Covariance of DAX and DowJones

Therefore, at one observation point the Value-at-Risk is calculated assuming the normal distribution, the usage of the estimated Moving-Average parameters for covariance, variance and correlation as

well as the factor of the confidence interval of 99% $z_{0,99} = 2.576$ and the weighting factors of the portfolio $w_{DAX} = 10,000$ and $w_{DJ} = 10,000$.

$$VaR_t = z_{0,99} * \sqrt{w_{DAX}^2 * \sigma_{DAX}^2 + w_{DJ}^2 * \sigma_{DJ}^2 + 2 * w_{DAX} * w_{DJ} * \rho_{DAX,DJ} * \sigma_{DAX} * \sigma_{DJ}}$$

Equation 35: Value-at-Risk for the portfolio existing out of DAX and DowJones

The parameter will be ascertained for every trade day within the observation period and the Value-at-Risk is calculated on this basis. For instance, in the below table it is displayed with a historical time window of 50 trade days.

Time t	DAX Variance $\sigma_{DAX,t}^2$	DowJones Variance $\sigma_{DowJones,t}^2$	DAX / DowJones Covariance cov_t	Value-at-Risk (in €) VaR_t
6/01/2017	0.000046	0.000023	0.000017	€ -441, -
5/31/2017	0.000044	0.000023	0.000017	€ -437, -
...	...	...	...	...
12/19/1988	0.000079	0.000065	0.000010	€ -553, -
12/16/1988	0.000078	0.000065	0.000012	€ -558, -
12/15/1988	0.000077	0.000071	0.000012	€ -566, -
12/14/1988	0.000075	0.000070	0.000010	€ -553, -

Table 8: Value-at-Risk with historical time window of 50 trade days (exemplary extract)

Based on the introduced computations, the result is respectively the overtime measurement of the Value-at-Risk of either 50, 120 or 250 trade days, shown in the below graph.

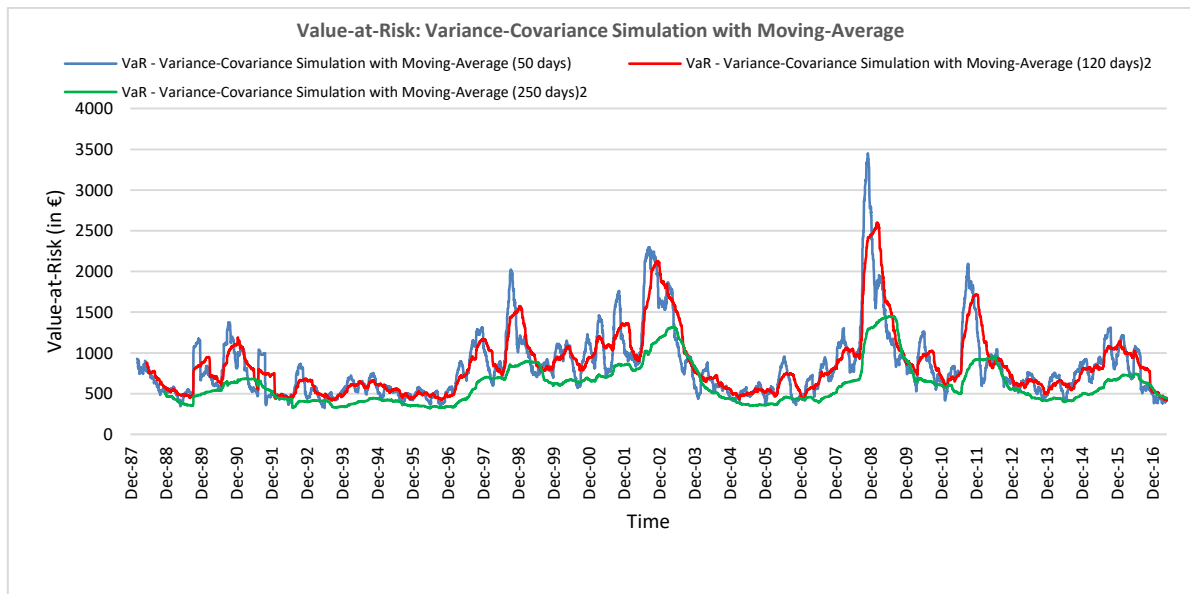


Figure 10: Value-at-Risk based on Variance-Covariance Simulation (Moving-Average)

The simple Moving-Average method for calculating Value-at-Risk provides a measurement that follows many of the same trends as the Historical Simulation before. Once again, the largest spikes we see come from our shortest trade day window, 50 days. The 250-trade day window can be described as being ‘smoother’ than the other trend lines. This should come as no surprise, because with a longer time horizon for which the Value-at-Risk measure is based on, the less volatility that a single return can add. Additionally, since the Variance-Covariance method assumes a normal distribution of returns, we do not note the same ‘plateaus’ that were prevalent in the Historical Simulation method, even though these two methods share many of the same characteristics, such as being based on a fixed number of past returns.

The analysis of the risk prediction power of the Variance-Covariance Simulation with the Moving-Average method will be executed identically to the Historical Simulation through backtesting procedure. The actual occurred losses are contrasted with the daily calculated Value-at-Risk and hence, the compliance of the predicted maximum portfolio loss for the chosen confidence interval is verified. The three below graphs show the comparison of actual occurred daily portfolio loss with the predicted Value-at-Risks in the historical time buckets. The overlapping portions of the trend lines within the graphs showing the realized exceedances.

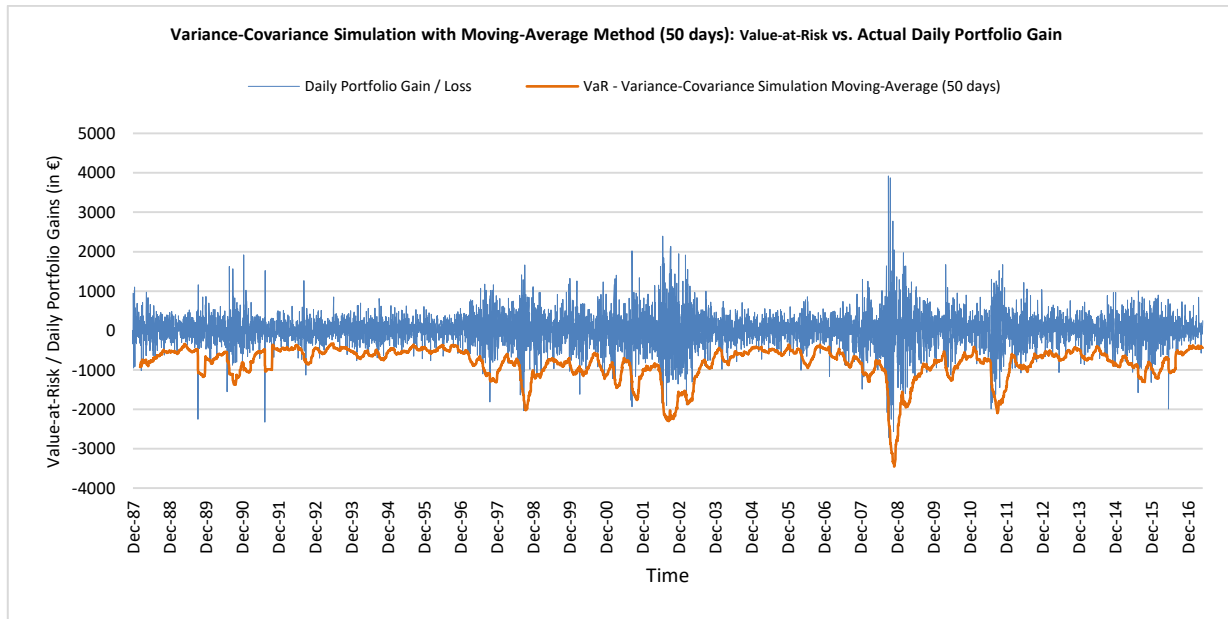


Figure 11: Variance-Covariance Simulation with Moving-Average Method (50): Backtesting

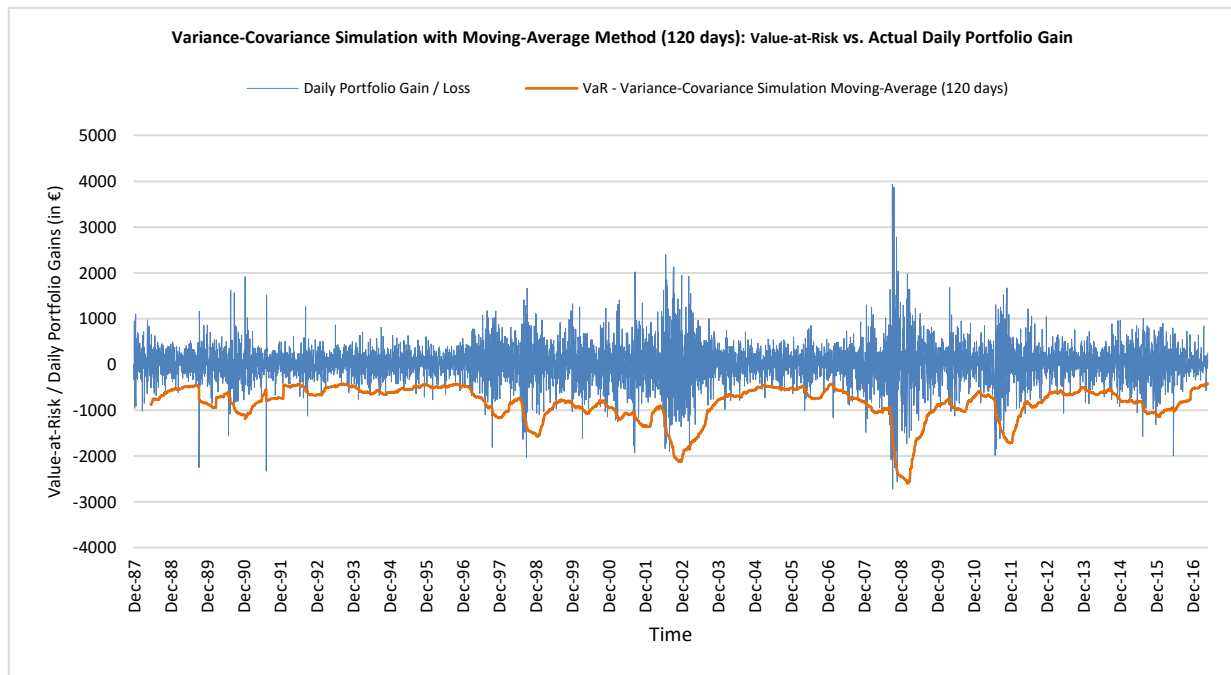


Figure 12: Variance-Covariance Simulation with Moving-Average Method (120): Backtesting

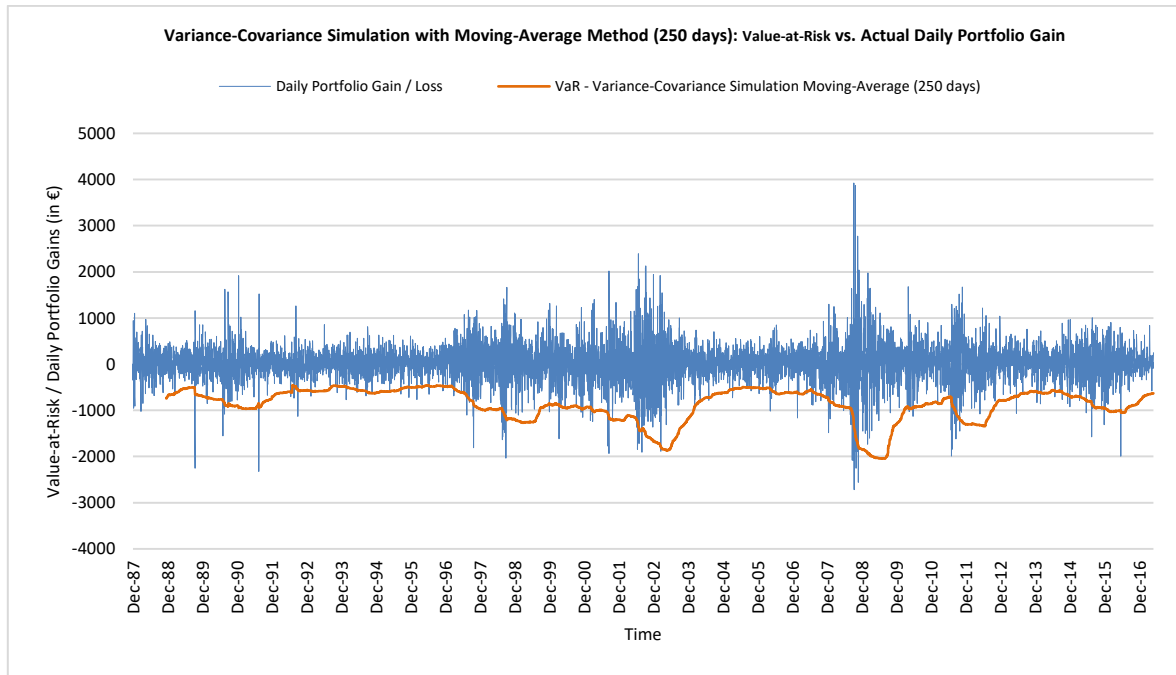


Figure 13: Variance-Covariance Simulation with Moving-Average Method (250): Backtesting

In a similar way to the previous model, Moving-Average, we can note that the longer horizon provides for more ‘smooth’ increases and decreases in the Value-at-Risk compared to the steeper jumps when using a shorter trade day horizon. Once again, the 120-trade day window falls between the two other measurements.

	50 Trade Days	120 Trade Days	250 Trade Days
Minimum Value-at-Risk	€ 324, -	€ 412, -	€ 455, -
Maximum Value-at-Risk	€ 3,450, -	€ 2,601, -	€ 2,050, -
Average Value-at-Risk	€ 833, -	€ 850, -	€ 869, -
Number of Exceedances	130	132	139
Percentage of Exceedances	1.72%	1.77%	1.89%

Table 9: Variance-Covariance Simulation with Moving-Average Method: Backtesting Results

The Variance-Covariance model with the Moving-Average method has an initial contrast to our previously analyzed model, the Historical Simulation, where the performance of the model correlated with the size of the trading window used in the measurement. For this model, the trend moves in the opposite direction, with the smaller trade windows performing the best on the metric of losses not exceeding Value-at-Risk. Yet, it should be noticed that all losses still exceed the Value-at-Risk more than the confidence level would imply. The quicker, more flexible reaction to returns appears to be

more effective when using this method. Related to this insight, is the lack of a trend of a higher average Value-at-Risk a lower exceedance rate. It would be expected since at all the window sizes, the amount in which losses exceed the Value-at-Risk is larger than the calculated confidence level, that having a higher Value-at-Risk would be a sure indicator of a superior measure.

A notable area that appears to lead to the 250 trade windows poor performance is the maximum Value-at-Risk calculated during the time horizon. The Value-at-Risk estimation of €2,050 is approximately 68% of the maximum Value-at-Risk using a 50-day trade window. Analyzing these figures and the graphs above makes it appear that the lack of a timely response rate while using the 250 window is what leads to this measure poorer performance. With that considered, the outcome of the Variance-Covariance Moving-Average does not see the same level of impact depending on the trade window being used. With the difference in exceedance percent between the 50 and 250 trade day window being 17%, it is an important figure for a trader or financial institution, but not the same type of difference seen from the Historical Simulation method.

5.3.2.2 Variance-Covariance Simulation under Consideration of time changing Volatility (EWMA)

After we had researched the Variance-Covariance Simulation based on the Moving-Average method, the same Value-at-Risk computation model is used with the EWMA method (see also chapter 3.5.2.).

On every trade day within the observation period the Value-at-Risk is computed identically (see previous formula) as in the prior investigation. Only the parameters variance and covariance, are calculated using recursion, using the measure itself to calculate the new measure which is used in the next day's Value-at-Risk, when using EWMA method. The daily determination of the parameters is based on the computation formulas, introduced in chapter 3.5.2.

$$\sigma_{DAX,t}^2 = \lambda * \sigma_{DAX,t-1}^2 + (1 - \lambda) * u_{DAX,t-1}^2$$

$$\sigma_{DowJones,t}^2 = \lambda * \sigma_{DowJones,t-1}^2 + (1 - \lambda) * u_{DowJones,t-1}^2$$

Equation 36: EWMA Method for Variance of DAX and DowJones (recursive)

$$cov_t = \lambda * cov_{t-1} + (1 - \lambda) * u_{DAX,t-1} * u_{DowJones,t-1}$$

Equation 37: EWMA Method for Covariance of DAX and DowJones (recursive)

The weighting- / Decay-Factor λ (Holton, 2003) determines the weighting of the latest market price changes. A λ factor of 0 heavily weighs the most recent returns, while a λ factor of 1 heavily weighs the returns further away. Hence, the parameters react faster and more agile to latest market events when the Decay-Factor is smaller, whereby variance and covariance could adjust to the market conditions during the time. This takes the non-stationarity of the distribution of market variables for DAX and DowJones into account. Based on the choice of which Decay-Factor is chosen, the length of the historical time window is determined due to the daily returns being exponentially descendant weighted. Hence, it incorporates the values, which are further in the past, less heavily. The Decay-Factor is usually estimated to be between 0.95 and 0.99, with 0.97 being a common value (Bollen, 2015) used in practice and also in our analysis. To guarantee a comparison between the Variance-Covariance Simulation with the EWMA method and the other researched computation models for the same historical time window, three Decay-Factors are used ($\lambda=0.94$, $\lambda=0.97$, and $\lambda=0.99$), which are comparable with the trade day windows of 50, 120 and 250 trade days (J.P. Morgan/Reuters, 1996, p. 100).

For the recursive computation of the estimated parameters an initial value is needed for the first observation point on 10/30/1991. Therefore, the first calculated variance for the DAX and DowJones as well as the covariance based on the Moving-Average method is ascertained through the first 250 trade days when using a Decay-Factor of 0.99 as an example. The following computations are done recursively so that the new measure is explicitly a product of the previous one. In the beginning this procedure could cause smaller deviations compared to the EWMA estimation due to the starting values being generated based on the Moving-Average method. However, this will have only small impact on the analysis because the estimated parameters adjust shortly to the current development of the market prices (Holton, 2003).

The parameters as well as the Value-at-Risk are computed for every trade day within the observation period. The below table shows the Value-at-Risk with the Decay-Factor of $\lambda=0.94$ exemplarily.

Time t	DAX Variance $\sigma_{DAX,t}^2$	DowJones Variance $\sigma_{DowJones,t}^2$	DAX / DowJones Covariance cov_t	Value-at-Risk (in €) VaR_t
06/01/2017	0.0000301	0.0004359	0.000013	€ -393, -
05/31/2017	0.0000311	0.0004295	0.000012	€ -391, -
...	...	...	...	...
12/19/1988	0.0000646	0.0002992	0.000020	€ -560, -
12/16/1988	0.0000633	0.0003022	0.000016	€ -538, -
12/15/1988	0.0000557	0.0002823	0.000012	€ -506, -
12/14/1988	0.0000554	0.0002768	0.000012	€ -509, -

Table 10: Value-at-Risk for Decay-Factor $\lambda=0.94$ (exemplary extract)

The following ascertains the Value-at-Risk daily with the Variance-Covariance Simulation with the EWMA method. The graph below illustrates the timely process of the Value-at-Risk in the observation period, respectively for each of the three Decay-Factors.

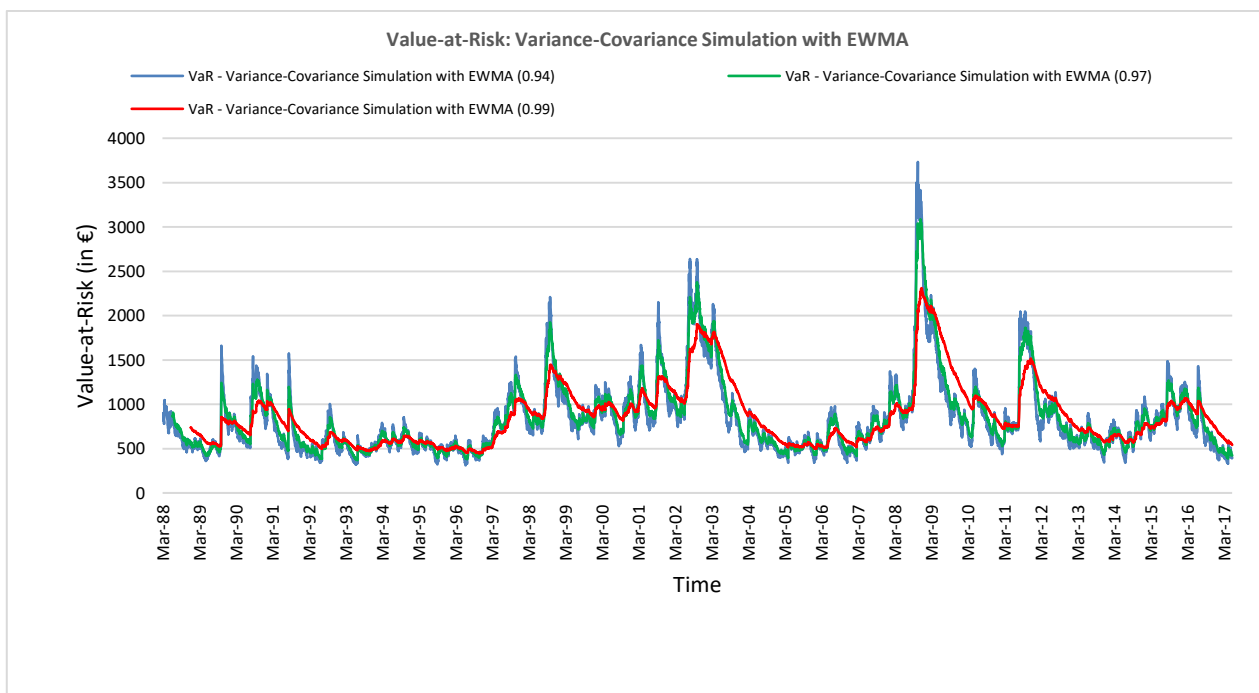


Figure 14: Value-at-Risk based on Variance-Covariance Method (EWMA)

What we can note from this graph by comparing the performance of unique Decay-Factors is how they impact the Value-at-Risk measurement. The trends in the graph are very similar, which is as expected since the measurements are being compared at the same confidence level. The key

difference to note is how each different Decay-Factor impacts the rate at which the Value-at-Risk ‘spikes.’ This can be seen most clearly by contrasting the Lambda 0.94 with 0.99. As described earlier, the higher the Decay-Factor, the more the older returns are weighted. Because of this, the occurrences of Value-at-Risk increasing quickly, will occur using the 0.94 Decay-Factor, as the recent returns are more weighted using 0.94 than 0.99. The seemingly small differences between Lambdas have a large impact as seen in the illustration 14, with a Lambda of 0.97 falling between the other two measurements.

Analogous regarding the empirical analysis of the risk prediction power, the Variance-Covariance model with the EWMA method calculates the predicted Value-at-Risk results with the actual realized portfolio gains and losses in a backtesting procedure. The illustrations 15-17 show the contrast of the daily ascertained Value-at-Risk results and the actual exceedances for the different Decay-Factors. The exceedances can be seen where the two graphs show overlaps.

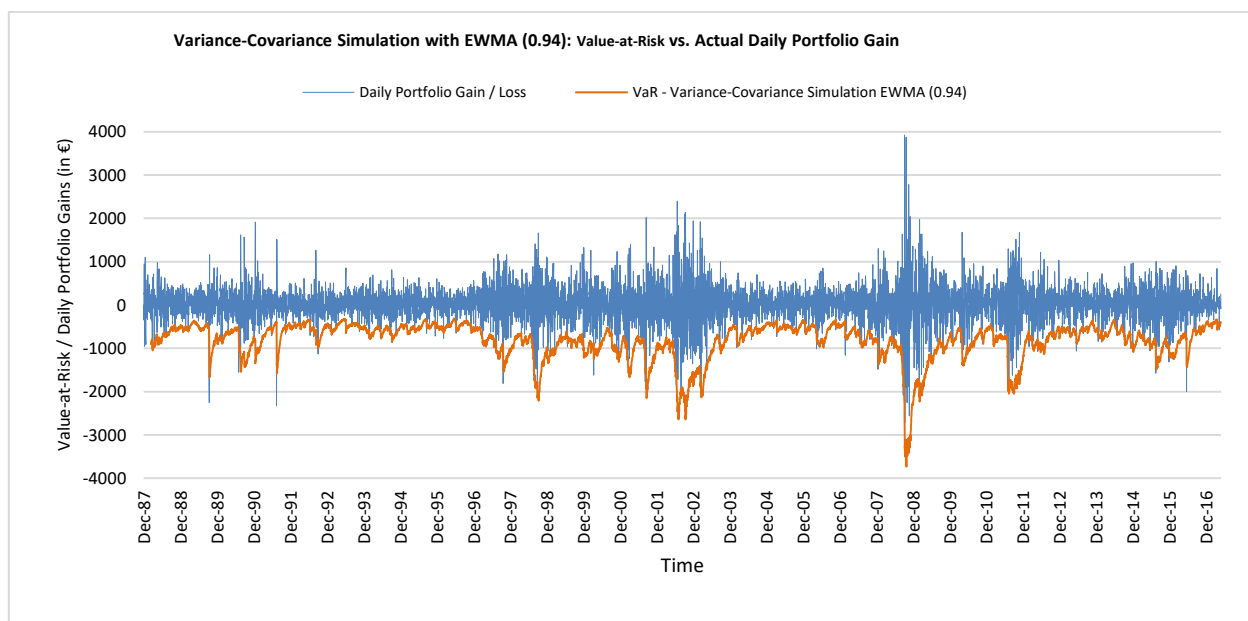


Figure 15: Variance-Covariance Simulation with EWMA Method ($\lambda=0.94$): Backtesting

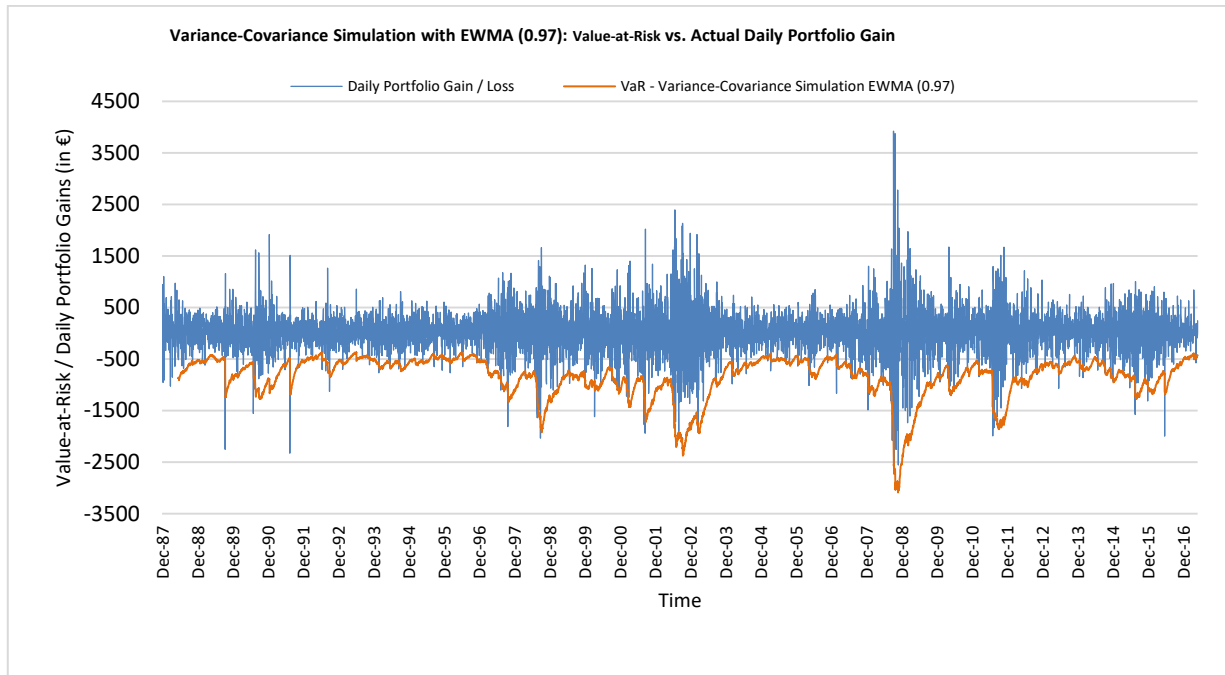


Figure 16: Variance-Covariance Simulation with EWMA Method ($\lambda=0.97$): Backtesting

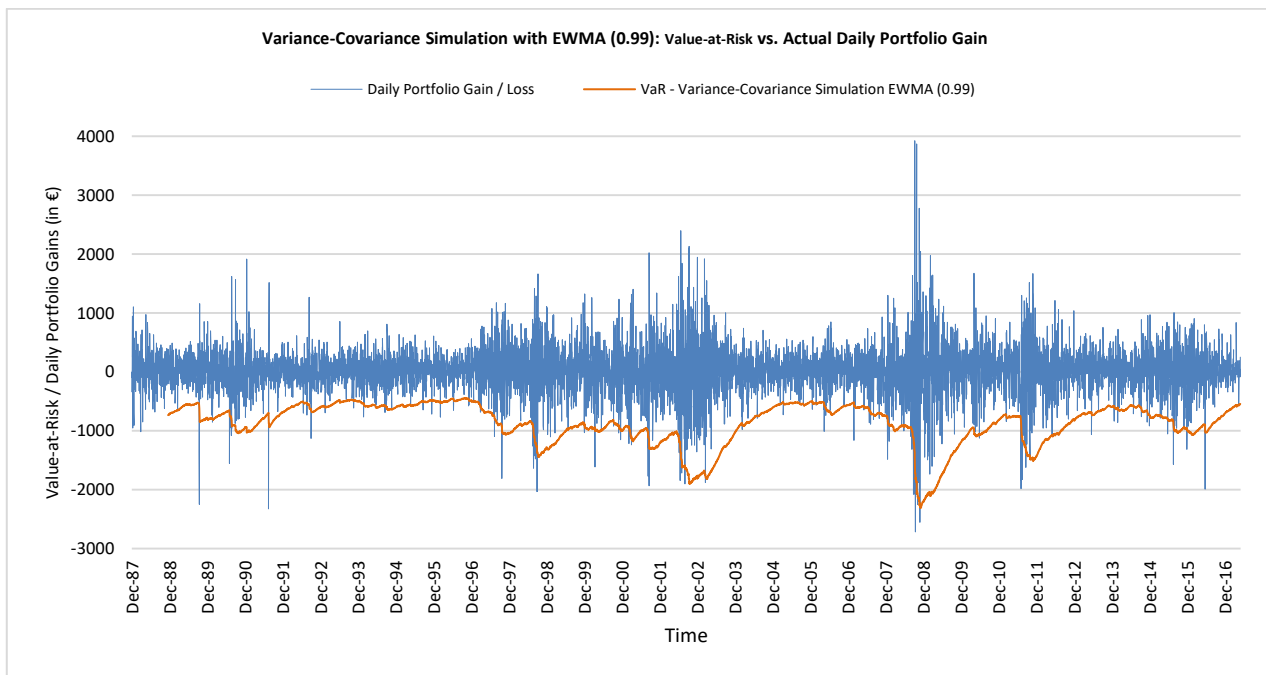


Figure 17: Variance-Covariance Simulation with EWMA Method ($\lambda=0.99$): Backtesting

With the graphs above, we have a solid picture of the performance of this method for evaluating Value-at-Risk. Many of the same trends highlighted under Illustration 14 are seen here again, with the additional context, which the actual returns accompanying the Value-at-Risk measure provide.

We can note anecdotally from the graphs that it seems the models often tend to fail at the same data points, which are the days in which particularly large losses take place. Especially between the years 2001-2002 and 2007-2008, noticeable swings can be marked due to high volatility in the markets. Focusing on the points, where the daily portfolio gain/loss (blue) is crossing the Value-at-Risk results (orange), it could be assumed looking only at these graphs that the fewest exceedances are noted with the a Decay-Factor of 0.94. However, it is also impossible to say for sure which model performed the best under the coverage test from only looking at the graphs. Nevertheless, looking more closely at the underlying data may provide another conclusion.

	$\lambda=0.94$	$\Lambda=0.97$	$\Lambda=0.99$
Minimum Value-at-Risk	€ 312, -	€ 370, -	€ 445, -
Maximum Value-at-Risk	€ 3,734, -	€ 3,087, -	€ 2,313, -
Average Value-at-Risk	€ 829, -	€ 843, -	€ 871, -
Number of Exceedances	141	129	121
Percentage of Exceedances	1.87%	1.73%	1.65%

Table 11: Variance-Covariance Simulation with EWMA Method: Backtesting Results

As alluded to above, by comparing the day by day Value-at-Risk against the daily gain / loss from the portfolio return provides the results of Decay-Factor 0.99 performing the best, and Decay-Factor 0.97 and 0.94 performing worse respectively at the 99% confidence interval. It appears that having the longer view, which the Decay-Factor 0.99 is giving, by providing more weight on data points which took place further back in time strengthens the Value-at-Risk measure produced in this case. From the graphs and table above, we can note that the maximum value at risk is greater using the Decay-Factor 0.94 method. But just as important as this is that the 0.94 Decay-Factor also has the minimum value at risk amount of the three models evaluated, which leads to this model having the lowest average Value-at-Risk. This should not be considered an inherent problem, however in the comparison of these three models it is, since each is understating the Value-at-Risk measurement that is needed to achieve the 1% confidence level. The 0.99 Decay-Factor amount can be considered more ‘stable,’ with the lowest range between its maximum and minimum Value-at-Risk measurement during the time horizon, while also having the highest average Value-at-Risk.

5.3.3 Monte-Carlo Simulations

Because the Monte-Carlo Simulation belongs to the parametric approaches, such as the Variance-Covariance model, the parameters for the assumed distribution of the returns need to be estimated.

For this purpose, the Moving-Average method is used with a historical time window of 120 and 250 trade days. Moreover, the normal distribution is assumed for the probability distribution of the returns.

The Monte-Carlo Simulation is based on the simulation of a variety of possible daily market price changes for the financial instruments within the portfolio. Hence, the Value-at-Risk is calculated on the base of simulated hypothetical portfolio changes for every day during the observation period. Therefore, the parameters like the variance (for DAX and DowJones) and their dependent correlation resp. correlation coefficient are estimated through the Moving-Average method.¹⁶ Based on the estimation of the parameters, the normal distribution of the daily returns is determined on which the simulation can be founded. First of all, independent random numbers are generated for all simulation scenarios of the Monte-Carlo Simulation. The random numbers are transformed into independent normal distributive random numbers with the support of formula 17. Subsequently, they are converted into normal distributive correlated random numbers based on the daily estimated correlation coefficient p_t with the support of the Cholesky Decomposition (Jorion, 2007, p. 321).

$$\varepsilon_{DAX,t,i}^{correlated} = \varepsilon_{DAX,t,i} \text{ for } 1 \leq i \leq 10,000$$

$$\varepsilon_{DowJones,t,i}^{correlated} = (\varepsilon_{DAX,t,i} * p_t) + (\varepsilon_{DowJones,t,i} * \sqrt{1 - p_t^2})$$

Equation 38: Computing normal distributed and correlated random numbers

The consideration of the correlation is necessary to constitute the dependency between DAX and DowJones as well as the diversification effect.

Based on the generated random numbers the changes of the portfolio are computed, whereas, in combination with the estimated volatility σ_t on day t , the random numbers determine the level of change in the financial instrument for DAX and DowJones. Based on formula 18 the 10,000 simulation scenarios show up in the observation period on day t :

$$S_{t,i}^{simulated} = p_{DAX} * \sigma_{DAX,t} * \varepsilon_{DAX,t,i}^{correlated} + p_{DowJones} * \sigma_{DowJones,t} * \varepsilon_{DowJones,t,i}^{correlated}$$

$$\text{for } 1 \leq i \leq 10,000$$

Equation 39: Simulated Changes of Risk Factors

¹⁶ The parameters can also be estimated with different methods, such as the EWMA method or the GARCH method. For this case, these methods are not considered in this academic paper.

The actual determination of the Value-at-Risk is resulting out of the quantile of the probability distribution, which shows in the simulation scenario on day t . Similarities can be detected when considering the Historical Simulation. However, the scenarios are ascertained based on historical data and not solely on a simulated normal distribution. Nevertheless, the advantage of the Monte-Carlo Simulation is the amount of simulations, which can be run to generate numerous hypothetical market price changes.

The graph below illustrates the computation of the Value-at-Risk in the observation period, while applying the Monte-Carlo Simulation with daily simulations of 10,000 hypothetical scenarios. This is in combination with the Moving-Average method for 120 and 250 trade days.

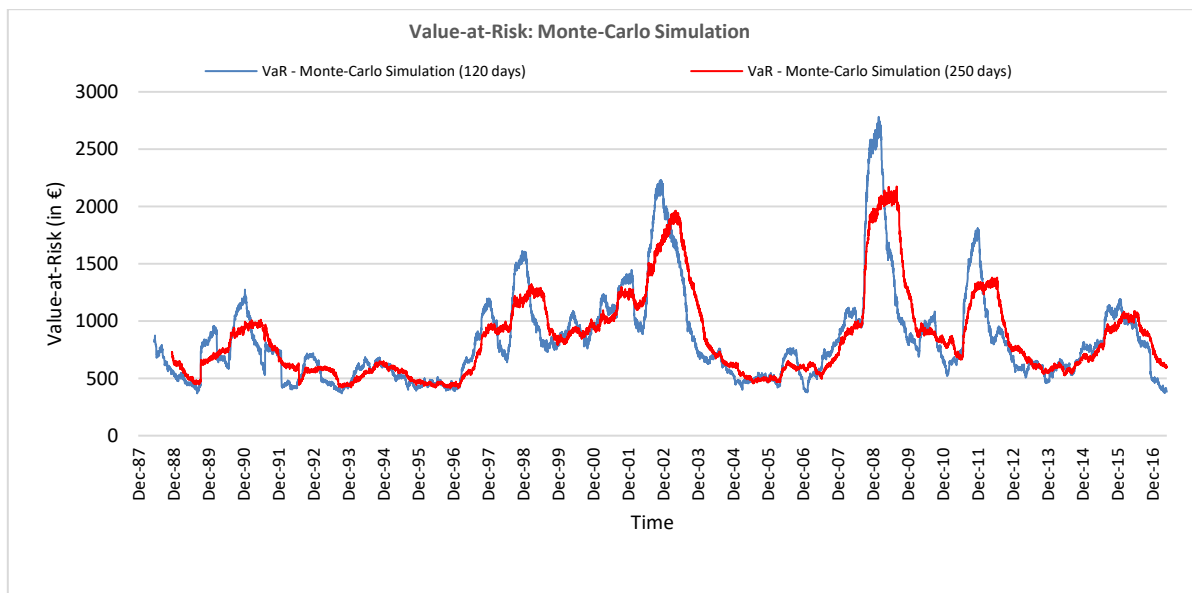


Figure 18: Value-at-Risk based on Monte-Carlo Simulation

Even though the Monte-Carlo method is simulation based on random numbers, many of the same general trends noted in the previous models hold true here. The 120-day trade window is more sporadic while the 250-trade day window does not have the same ‘peaks’ but instead flattens out more at a lower amount of Value-at-Risk. However, the path of the trend lines of these two lines is more unique than our previous models. In the prior models, the Value-at-Risk trend lines could be described as being on the same trend line with unique rates of volatility impacting the motion. For the Monte-Carlo method, because of the random number generation aspect of the model, the trend of the lines is more differentiated than the models we analyzed prior.

The backtesting procedure is used again to reveal the risk prediction power of the Monte-Carlo Simulation. The predicted Value-at-Risk results and the actual occurred portfolio gain / loss are shown in the graphs below, for 120 trade days and 250 trade days.

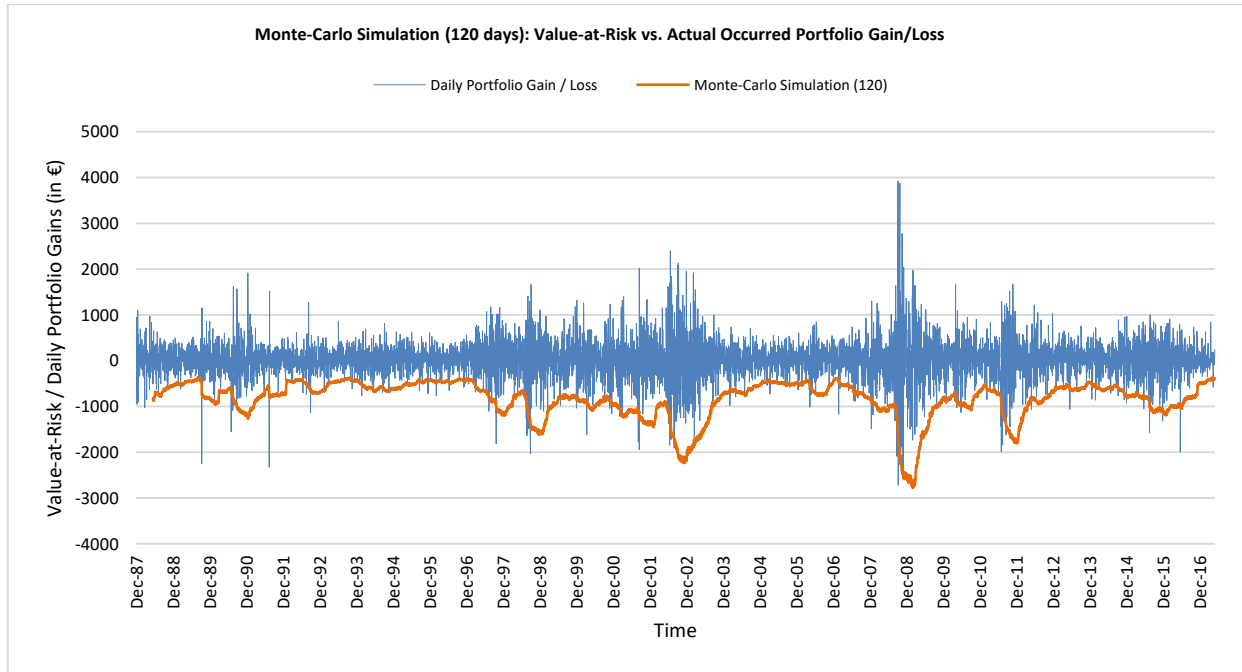


Figure 19: Monte-Carlo Simulation (120): Backtesting

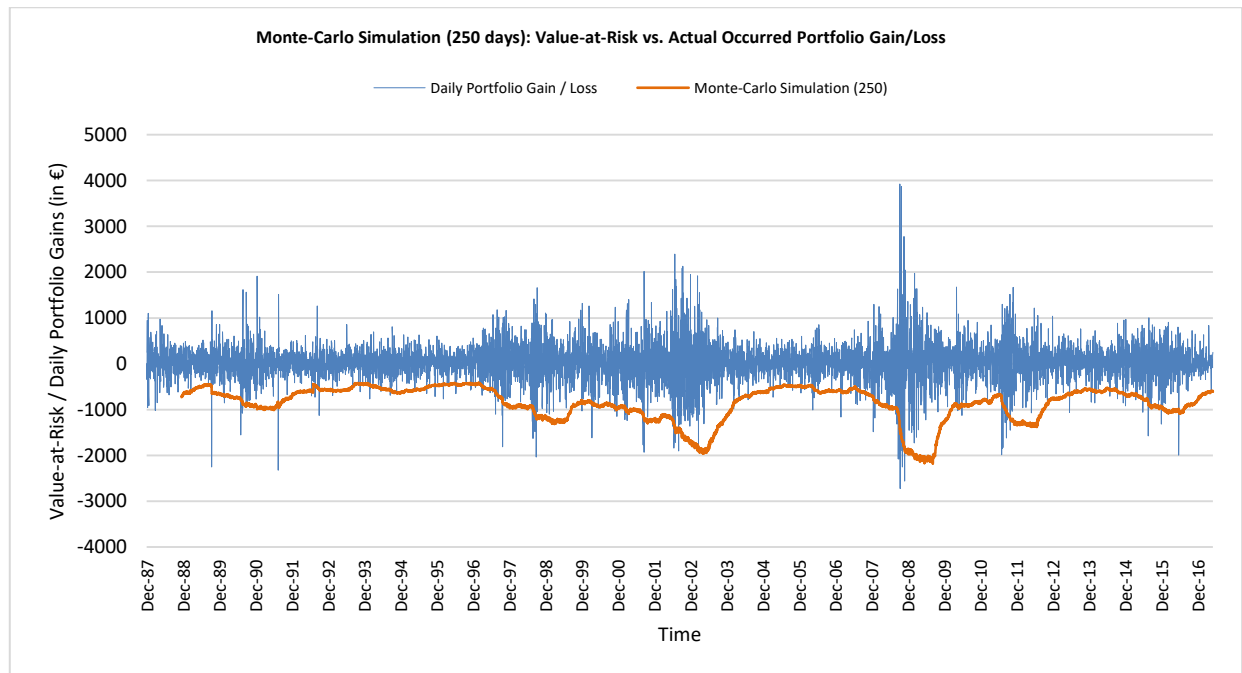


Figure 20: Monte-Carlo Simulation (250): Backtesting

Comparing the graphs to the daily returns, it is clear regarding the Monte-Carlo model that the occurrence of the outlier type of large losses tends to exceed the estimated Value-at-Risk. This can be seen in the both the 120 and 250 trade day window graphs. A note that can be made is the seemingly slower and lower spike in Value-at-Risk that accompanies the 250-trade day window. Even with the introduction of random numbers into the simulation, the shorter trade window has the impact of integrating recent returns into the measure quicker and more explicitly, as seen in the difference in the ‘peaks’ in figure 18.

	120 Trade Days	250 Trade Days
Minimum Value-at-Risk	€ 369, -	€ 415, -
Maximum Value-at-Risk	€ 2,783, -	€ 2,174, -
Average Value-at-Risk	€ 838, -	€ 857, -
Number of Exceedances	143	147
Percentage of Exceedances	1.91%	2.00%

Table 12: Monte-Carlo Simulation: Backtesting Results

In a similar vein as the Variance-Covariance Moving-Average method, the rate at which actual losses exceed the calculated Value-at-Risk is greater the longer the time horizon. In addition, the 250-trade day window also has a higher average Value-at-Risk, yet still the number of exceedances and percentage is greater.

However, it is important to note that the absolute performance of the Monte-Carlo method is particularly poor at either window size. With the model created with a 99% confidence interval applied, the expectancy would be that the percentage of exceedances would fall near 1%, but both exceeded this percentage. A hypothesis is that the utilization of random numbers is a powerful tool, but needs large amounts of domain and technical knowledge (Jorion P., 2007) to do so in a way that adds value. It is possible that a savvy investment banker or stock trader would be able to use better inputs than historical performance to generate random numbers, and thus a more accurate Value-at-Risk measurement.

5.4 Comparing Model Results

Some trends to briefly note for all models are that the Value-at-Risk response takes place after a major event. We can see this reflected around 2000, when the dot com bubble burst, and again around 2007 after the financial crisis. However, overtime after these events took place the Value-at-Risk will slowly decrease again, likely not rising until another major financial event takes place.

Using the test of standard coverage and Exceedance Independence Test explained above in section 5.2, we can showcase the relative rates of all the methods in the table below.

Model	Window Size	Violation Rate	Absolute Variance	LR Independence	Average Value-at-Risk
99th Percentile Value at Risk Measures					
Historical Simulation	50	2.25%	0.0125	0.00	3.93%
	120	1.67%	0.0067	0.00	4.34%
	250	1.28%	0.0028	0.00	4.71%
Variance-Covariance (MA)	50	1.72%	0.0072	0.00	4.17%
	120	1.77%	0.0077	0.00	4.25%
	250	1.89%	0.0089	0.00	4.35%
Variance-Covariance (EWMA)	$\lambda=0.94$	1.87%	0.0087	0.00	4.15%
	$\lambda=0.97$	1.73%	0.0073	0.00	4.22%
	$\lambda=0.99$	1.65%	0.0065	0.00	4.36%
Monte-Carlo	120	1.91%	0.0091	0.00	4.19%
	250	2.00%	0.0100	0.00	4.29%

95 th Percentile Value at Risk Measures					
Historical Simulation	50	6.15%	0.0115	0.00	2.74%
	120	5.30%	0.0030	0.00	2.92%
	250	5.68%	0.0068	0.00	3.02%
Variance-Covariance (MA)	50	5.41%	0.0041	0.00	2.94%
	120	5.11%	0.0011	0.00	3.01%
	250	5.39%	0.0039	0.00	3.07%
Variance-Covariance (EWMA)	$\lambda=0.94$	5.74%	0.0074	0.00	2.93%
	$\lambda=0.97$	5.22%	0.0022	0.00	2.98%
	$\lambda=0.99$	5.00%	0.0000	0.00	3.08%
Monte-Carlo	120	5.40%	0.0040	0.00	2.95%
	250	5.55%	0.0055	0.00	3.01%

Table 13: Predicted Results of Computation Models for Value-at-Risk

The initial takeaway is that at the 99% and 95% confidence interval, the rates at which actual returns violate the Value-at-Risk model estimates exceed the confidence interval used in all but one scenario. This is the case for the four models, under all additional variables window size and upon and different confidence intervals. However, they all fulfill the test for independence at the specified confidence interval.

This evaluation was done in R programming, using a package (Acharya, 2016) which provides the capability for this specific test of Christoffersen Independence. This result can be interpreted as being able to conclude with a probability equal to the confidence interval, in this case 99 and 95 percent, that the cases of loss exceedance being greater than Value-at-Risk are independent of one another. This is evaluated by comparing the vectors of actual daily returns to the vector of Value-at-Risk measures, and the program is able to decipher if the distribution of the excess violations leads to a conclusion that they are independent of one another. Appendix IX provides the code for running this test.

With this in mind, since none of the models are able to fulfill the null hypothesis for standard coverage, we will place emphasis on which performs superior relatively rather than on an absolute spectrum.

Evaluating the results on the criteria of difference between the violation rate and the given confidence interval provides a result for which model is superior of the three tested. At the 99% confidence interval, the Historical Simulation method has an average variance of 0.0073, while the Variance-Covariance exponential weighted Moving-Average and simple weighted Moving-Average are 0.0075 and 0.0079 respectively. Lastly, the Monte-Carlo method has an average difference of 0.0095. We can interpret this as the Historical Simulation method performing superiorly based on this metric. The lower the value deviations from the expected deviations, the more accurate a model (Christoffersen P., 2009) is according to the standard coverage theory.

Another aspect to highlight is the single most accurate occurrence, which is however, from the Variance-Covariance – weighted average method. Additionally, the 95% confidence interval fares particularly well for using the Variance-Covariance exponential weighted average method, where we calculated the one Value-at-Risk measure that has only been exceeded by losses as often as the confidence level gives latitude. However, this was the only particularly notable deviation between the 99 and 95 confidence intervals, which is why we relegated the 95 confidence intervals to the appendix to reduce redundancy, and highlighted the most used confidence interval, 99%.

The largest variances resulted from the Monte-Carlo method for calculating Value-at-Risk. Based on higher results for the variance, the model might get more inaccurate due to that in reality results do not always fall within a normal distribution. However, generally high variance results do not have to be negative because the size of the mean is essential for evaluating the results correctly. But multiple reasons can be theorized for the poor performance of this model. In our analysis, we used the historical data as additional inputs for our random generation, with each Value-at-Risk data point being the result of 10,000 simulations. Our thought was that with this number of simulations, the Value-at-Risk data points would normalize, and the curve would be ‘smoother.’ The results contrast with our expectations, resulting in the random number input having a significant impact on the Value-at-Risk outputs. With this amount of volatility in the day to day Value-at-Risk numbers, it is not surprising this model fared the worst. It is likely that using another method to generate the random numbers, such as market rates, or a shorter time horizon may be able to generate Value-at-Risk, which creates a more consistent outcome and thus a better performing Value-at-Risk measure.

5.6. Evaluation of Risk Prediction Power

This conclusive subchapter of the empirical analysis is dealing with an overarching comparison of the prediction quality and the risk prediction power of all three researched computation models, especially with respect to the market price risk. Subsequently, the results of the investigation will be assessed and evaluated with respect to the supervisory requirements. This will show whether the researched computation models could be implemented and used in banks and fulfill the regulatory requirements.

5.6.1. Comprehensive Comparison of the Risk Prediction Power

The empirical analysis has shown a different reliability and prediction quality for the analyzed Value-at-Risk computation models when applying a backtesting procedure. Hence, it is highly significant to check the quality of the computation models in practice on a continuous base and adjust the computation approaches respectively, so that the number of occurred exceedances aligns with the chosen confidence interval. Particularly the banking sector is highly monitored by the supervisory authorities regarding the utilization of the Value-at-Risk. The focus lies on the results of backtesting analysis to guarantee the control of the internal risk quantification procedures (Basel Committee on Banking Supervision, 2006).

While contemplating the quality of the risk prediction of a Value-at-Risk computation model, the focus is first and foremost on the backtesting procedure results, which give information about the number of occurred exceedances of the predicted Value-at-Risk compared to the actual occurred portfolio loss. However, an essential element of the assessment of a computation model is also the consideration of the Value-at-Risk's size. For instance, if a computation model calculates too high results for the Value-at-Risk in average, the exceedances would be theoretically less due to external losses occur with a lower probability. A model with this average higher Value-at-Risk would hence, show a better risk prediction power, but is only limited deployable in practice, especially for banks. The size of the Value-at-Risk is essential for banks to calculate the equity rate, which needs to be deposit (Schroeck, 2002). Conclusively, if the model computes a too high and not needed Value-at-Risk, expensive and unnecessary equity is bonded (Fricke, 2006). Therefore, it is important to ponder between the level of the average Value-at-Risk and the resulting risk prediction power of the computation model. An analysis of this relation can be found in illustration 21 for the empirical investigation of the observed target horizon. It shows the size of the average Value-at-Risk compared to the number of occurred exceedances for each of the researched computation models.

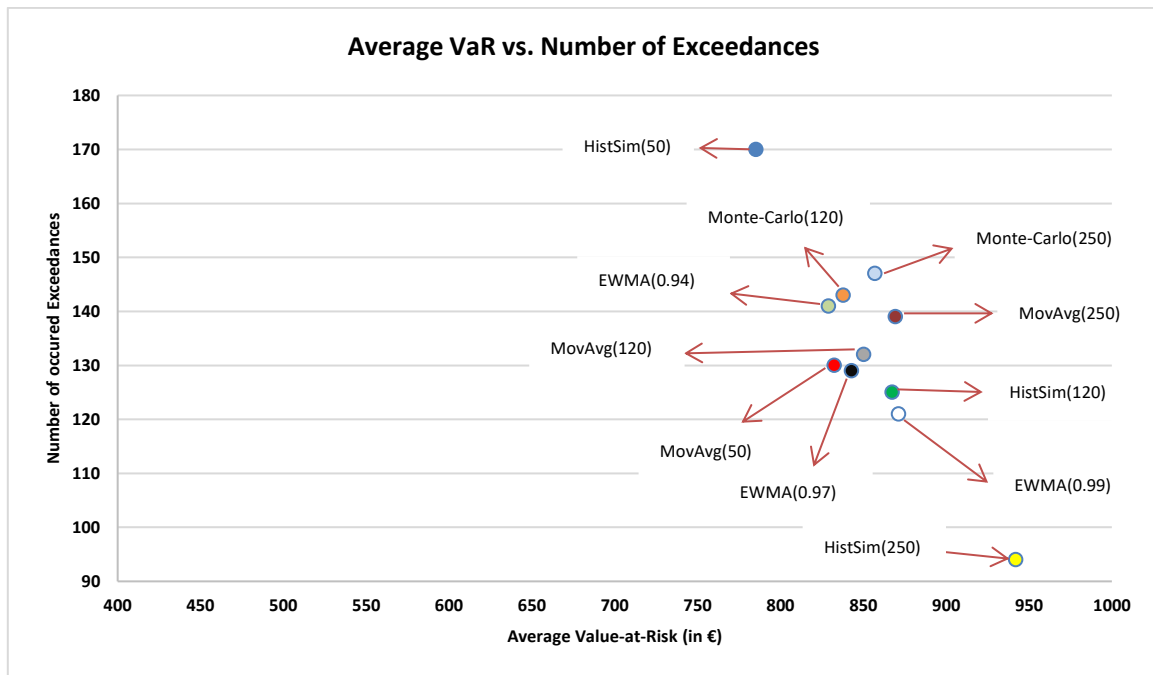


Figure 21: Comparison of Risk Prediction Quality & Average Value-at-Risk

The comparison of the computation models shows clear patterns, which might give a better enlightenment and understanding.

On the one side, the Historical Simulation, which is based on 50 trade days is showing the highest number of exceedances (170), while also having the second lowest average Value-at-Risk (€ 785, -). With respect to the evaluation of the risk prediction power, this model seems to deliver the worst results due to the number of exceedances.

On the other side, the Historical Simulation, which is based on 250 trade days is giving the lowest number of exceedances (94), while also having the highest average Value-at-Risk (€ 942, -). Hence, the Historical Simulation, based on the longest time window, seems to be the most reliable procedure, however, being also the most expensive regarding the equity rate, which needs to be deposit.

Furthermore, it is noticeable that the rest of the results are heavily bundled within a field from 121¹⁷ until 147¹⁸ exceedances and an average Value-at-Risk from € 829, -¹⁹ until € 871, -²⁰. Within this field, the Variance-Covariance Simulation based on the EWMA method shows the lowest number of

¹⁷ Variance-Covariance Simulation with EWMA Method, based on a Decay-Factor of 0.99

¹⁸ Monte-Carlo Simulation, based on a historical time window of 250 trade days

¹⁹ Variance-Covariance Simulation with EWMA Method, based on a Decay-Factor of 0.94

²⁰ Variance-Covariance Simulation with EWMA Method, based on a Decay-Factor of 0.99

exceedances as well as the lowest average Value-at-Risk. Thus, this computation model is distinguished with the best results, while considering equally the risk prediction power and the level of the Value-at-Risk.

As elucidated in chapter 5.2. *Objectives & Procedure of Investigation*, the number of exceedances within a confidence interval of 99% should remain significant. While the Historical Simulation, based on a time window of 50 trade days, shows more than twice as many exceedances as accepted²¹, the rest of the computation models, which are mostly bundled, showing results between 1.3% and 2.0% of accepted exceedances. To sum up, it is shown that the Value-at-Risk models do not contain a sufficient reliability regarding the quantification of the market price risk, considering the chosen assumptions and application of the described computation models.

5.6.2. Comparison of the Risk Prediction Power considering Supervisory Requirements

In the following, the prediction quality of the researched Value-at-Risk computation models is investigated under consideration of supervisory and regulatory requirements for financial institutions. There are slight deviations compared to the empirical analysis in this academic paper with respect to the legitimacy for exceedances.

The Basel Committee of Banking Supervision, which is responsible for the publication of the framework Basel II, writes that it is necessary to conduct constant backtesting analyses to check on internal Value-at-Risk computation models, while fulfilling already the quantitative and qualitative requirements (*Basel Committee on Banking Supervision, 2006*). In case of that a bank's computation model is missing the requested requirements, the bank will be imposed with additional regulations, which contain an increase of the essential deposit equity, calculated with an own (equity-) factor that is determined with respect to the size of the Value-at-Risk.

While using the Value-at-Risk computation models certain error types exist due to statistical issues.²² When considering these error types, in chosen time horizons it is legitimate to show a higher number of exceedances of the Value-at-Risk than the confidence interval actually demands. The framework Basel II is serving this legitimacy while defining concrete exceedance probabilities, which consider already these error types. For instance, during the last 250 trade days it is conformable to law showing

²¹ 2.9% exceedances were detected of the total number of Value-at-Risk results, whereas only 1% outliers are accepted according to a confidence interval of 99%

²² Regarding the evaluation of the statistical tests (binomial distribution) and the resulting error type 1 and error type 2, please refer to the following literature; Jorion (2007), pp. 142

up to four exceedances without fearing any additional regulations for the required equity levels. This corresponds to an exceedance probability of 1.6%²³, although a maximum of 1% exceedances is set based on the required confidence interval of 99%. Dependent on the number of occurred exceedances in the last 250 trade days an equity factor is formed within the "Traffic Light Approach" (Deutsch, 2008, p. 450) based on supervisory regulations. This approach determines a color (green, yellow or red), which indicates the measure of exceedances. In table 14, the mentioned colors and their subdivisions are sorted with respect to the occurred exceedances as well as the resulting equity factor, according to Basel II regulations.

Basel II			
	Number of Exceedances in last 250 trade days	(Equity-) Factor	Number of Exceedances in Target Horizon (Horizon = 7,347 days)
Green Zone	0-4	3.00	0-117
Yellow Zone	5	3.40	118-146
	6	3.50	147-176
	7	3.65	177-205
	8	3.75	206-235
	9	3.85	236-264
Red Zone	10+	4.00	265+

Table 14: Supervisory Regulations for Backtesting in Financial Institutions²⁴

The occurred exceedances relate to a historical target horizon of 250 trade days respectively, which means that four exceedances within the last 250 trade days cause an increase in the equity multiplied by factor three of the Value-at-Risk, which needs to be deposited. This illustrates the lowest equity amount, which needs to be deposit. In case of repetitive occurrence of exceedances, the equity factor increases and conclusively also the bonded equity. The highest equity factor is four times the Value-at-Risk, which is applied if more than ten exceedances are measured.

²³ $\frac{4}{250} = 0.016 = 1.6\%$

²⁴ Own Illustration referring to Basel Committee on Banking Supervision (2006), p. 321

If the supervisory regulations of the underlying confidence interval of 99% and the time horizon of 250 trade days are applied to the empirical analysis in this academic paper with a target horizon of 7,347 trade days, the results are within the allowed exceedance probability, as shown in table 14.²⁵ For this purpose, the allowed number of exceedances was proportional projected from 250 to 7,347 trade days.

Hence, a computation model is classified as “good” if not more than 105 exceedances are measured in the actual investigation. This was only reached from the Historical Simulation based on a time horizon of 250 trade days. All other researched computation models are classified in the yellow zone with respect to their prediction quality, meaning they cause additional deposits of equity because of average higher exceedances. No computation model shows a prediction quality within the red zone.

Basically, the Value-at-Risk computation models deliver “good” to “sufficient” results regarding their utilization in banks while considering the supervisory regulations. However, there is clear potential to enhance the risk prediction of all researched computation models until the number of occurred exceedances is in the green zone and hence, no additional equity is necessary for hedging.

²⁵ Number of Exceedances was proportional calculated from 250 to 6,597 trade days

6. Critical Appreciation of the Value-at-Risk & Prospect

The Value-at-Risk constitutes a risk measure to quantify risks and allows with a certain probability a prediction of the prospective development resp. states a probability of the risks' possible occurrence. Special attention should be directed to the fact that the Value-at-Risk is not exclusively used in the market price risk but can basically be applied to all types of risk (Scherpereel, 2005), so that these diverse types of risk and risk positions are comparable through one common risk figure. The resulting consolidated Value-at-Risk is expressed in a monetary unit (Fricke, 2006) and is easy to comprehend for management and other decision makers within a company. Hence, the risk of a portfolio, existing of different financial instruments can be quantified and also be compared to other risk types within the own company or to similar risks in competing companies. Basically, the Value-at-Risk can be used across sectors because the measurement is fundamentally feasible for any arbitrarily quantifiable risk. Therewith, the Value-at-Risk constitutes a universal risk measure for users (Bonke, 2007). Therefore, the Value-at-Risk is used by companies from many different business sectors, whereby often financial institutions use this risk measure and publish it in their financial reports (Fricke, 2006). However, the focus is not only on the size and the historical data of the prior business year, but also on the applied assumptions and the used computation methods. Only this information enables a comparison of the Value-at-Risk with other companies. Our empirical investigations show that the choice of the computation model affects the results for the Value-at-Risk significantly.

Besides the interpretability and the flexible usage, the Value-at-Risk also faces critiques regarding its conceptual fundamentals. The empirical investigation of this academic paper has shown that the Value-at-Risk models offer an appropriate quantification of the market price risk and hence, a prediction of the prospective development, however, the risk prediction and the validity are not reliable enough due to too many exceedances of the predicted maximum loss in each computation model. Replicating these observations into the field of a trader in a bank (in which the traders' limits are determined through the latest accepted Value-at-Risk), repetitive exceedances of the projected maximum loss could have crucial consequences; even jeopardizing the bank's existence (Saita, 2007). Hence, the prediction quality of the applied Value-at-Risk computation procedure needs to be checked continuously through backtesting analyses and the basic conditions of the models (e.g. length of historical time horizon) need to be adjusted respectively.

The use of historical data within the Value-at-Risk calculation shows up as critical. Each of the researched computation models uses information from the past to estimate the potential future

development from that. Even when the historical data are weighted differently in each model for the computation, the data are eventually responsible for the level of the Value-at-Risk. Hence, the risk prediction power is dependent on the fact whether the historical data are actually representative for the future. In case of a significant difference between the historical and future development, the prediction quality of the computation models will consequently be lower (Dowd, 2002). An exception is only the Monte-Carlo simulation, which can build and simulate a random number of scenarios in a way that the historical data constitute a starting basis, the actual expected future return distribution will be, however, simulated. In this case, the Value-at-Risk is less dependent on the historical data.

Moreover, the availability of the historical data is not always given, especially if no historical data exist for newly introduced financial instruments (e.g. Christoffersen, 2009; Dowd, 2002). But especially for new financial instruments a risk assessment should be conducted because no empirical values of possible developments are known, and the financial instruments might contain hidden risks.

As already shown in the empirical investigation of this academic paper the Value-at-Risk can be ascertained for a portfolio, which exists out of a random number of financial instruments. The portfolio used in this academic paper exists of two positions, which correlates with each other and shows consequently a result of diversification effect. The consideration and inclusion of the diversification effect is an enormous advantage of the Value-at-Risk concept, but exacerbates the computation while the number of financial instruments in the portfolio increases. Furthermore, while the calculation of the Value-at-Risk for one portfolio, existing out of a small number of positions, can be still managed within a spreadsheet program (e.g. in this investigation used program *Microsoft Excel* to store and transfer the input and output data) on a computer, the inquiry of the market price risks within a financial institution demands complex software, big datastores, and numerous interfaces to provide the necessary data. This could be a considerable challenge for big institutions, which may have risk exposures all around the world.

The complexity of the computation as well as the resulting amount of data increase more and more due to the comparability and the fact that the portfolio mix cannot be changed during the time horizon of the Value-at-Risk. However, this is not the case in the real world, which means that positions in the portfolio change several times during the day. Therefore, it is necessary to capture and archive electronically different historical data paths, for instance the date sets which are necessary for the Value-at-Risk when the portfolio has not been changed and on the other side the data sets after the portfolio had been changed.

This leads to another weak point regarding the interpretation of the Value-at-Risk. It constitutes only a snapshot, which was created at the end of the trade day for the following trade day in this empirical investigation. Thereby, the following closing prices of the respective financial instruments were considered, whereby any development during the trade day is not pictured. If the course of a financial instrument in the portfolio would be exposed to significant volatility during the trade day, however, the closing price would be mostly around the same, the Value-at-Risk would be lower if instead considering the intraday prices. Consequently, as an extension of the Value-at-Risk models, prices should be incorporated in the computation and to provide perhaps, even a Value-at-Risk in “*real-time*”²⁶, which could be integrated into a business controlling system. Thus, developments of the Value-at-Risk could be tracked in real-time for management. The opportunity would give the opportunity to react contemporaneously on an increasing risk with corrective actions. Concerning this matter, a few publications have been released (Frye, 1998), which base on the Monte-Carlo simulation while the Value-at-Risk is repetitive and is recalculated during the trade day.

In addition, the empirical investigation has proved that the Value-at-Risk results (minimum, maximum and average Value-at-Risk) differ significantly, depending on the computation method and the chosen length of the historical time horizon. It was also mentioned that additional costs are caused through consistent overestimations of Value-at-Risk because of an increasing equity rate. Hence, a goal conflict is detected between the quality of the computation models and the size of the Value-at-Risk, which should be solved as optimal as possible based from an economic perspective. Moreover, considering these aspects while observing the “Ghost Feature” issue, as seen in the Historical Simulation or the Variance-Covariance simulation, it can be concluded that the goal conflict for these models was not optimally solved. For the mentioned models the Value-at-Risk remains on a high level, although the course of the underlying financial instrument has already increased. The Value-at-Risk needs to decrease as a consequence. After a certain period of time, which normally corresponds to the length of the historical time horizon, the Value-at-Risk is adjusted to the actual market conditions as soon as the extreme historical event is not further considered in the computation procedure. In the entire run of the time horizon the Value-at-Risk is mostly estimated too low, which can be perceived through low equity costs. Eventually, the calibration of the Value-at-Risk models is

²⁶ Real-time means here the continuous update and recalculation of the Value-at-Risk as soon as an influencing factor (e.g. market price) is changing

inevitable in every business to guarantee the risk prediction quality but also to obtain precise Value-at-Risk predictions.

In order to claim that the predictions of the Value-at-Risk models are very precise, the applied assumptions for the models need to be as similar as possible to the events, which are observable in the practice. The fundamental assumption of all researched models is regarding the distribution of the returns of financial instruments within the portfolio. This academic paper used the normal distribution due to the fact that it is commonly used in practice (Jorion, 2007) based on its simplicity and its handling. Besides the advantages the normal distribution has (e.g. simple determination through mean and standard deviation), several disadvantages and inaccuracies showing up in the computation. Repetitive empirical observations showed that the distribution of the returns in practice does not match with the normal distribution but instead display “fat tails” and “thin waists” (Eisele, 2004, p. 59). Thus, the reality shows more often a greater frequency of extreme deviations in the mean and a higher amount of observations for the mean compared to the normal distribution.²⁷ As a consequence, the broader ends of the actual distributions can lead to a significant underestimation of the risk when applying the Value-at-Risk (Bonke, 2007). Therefore, before applying a Value-at-Risk computation model in practice it should be guaranteed that the underlying return distribution function is precisely analyzed and also that a fitting approximation is chosen to avoid eventually bigger estimate errors for the Value-at-Risk results. For this purpose, the Monte-Carlo model seems to be the most convenient and flexible one of the researched computation models. The simulation can base on random distribution functions (e.g. logarithmical or multivariate t-distributions) (Hull, 2011).

Another point of criticism regarding the Value-at-Risk concept is the limited validity of the actual expecting loss. While the Value-at-Risk gives a probability of a predicted maximum loss, no information is given for the case if the Value-at-Risk is actually exceeded (Boudoukh, Richardson, & Whitelaw, 1997). This problem and the estimation of the mean if the Value-at-Risk is exceeded are discussed in the “Extreme Value Theory.” It forms an extension of the Value-at-Risk approach and gives answers to questions such as “How big is the expected loss resulting from prospective natural catastrophes and what effect it has on insurance companies.” (Dowd, 2002, p. 131; Saita, 2007, p. 55)

²⁷ A distribution with the named characteristics is called leptokurtic distribution; Eisele (2004), p. 59

In addition, the question arises whether the volatility can still be put on a level with the risk, as it was presented in former publications. For instance, the volatility of a financial instrument increases if underlying price changes erratically. As a consequence, the Value-at-Risk, which is based on the volatility, increases as well. Normally, an increase in the price does not automatically constitute a risk.²⁸ This specific phenomenon showed up during our empirical investigation, especially in the variance-covariance approach, when the Value-at-Risk paradoxically increased in times of recovery (e.g. after the peak of the financial crisis) due to a bigger impact of the volatility on the computation. This brings up a question whether the Value-at-Risk models are meaningful and sufficient in times of increasing volatility on the financial markets and decoupling of volatility and risk. It is already slightly discernible on the financial markets that the Value-at-Risk is not exclusively crucial anymore for the risk of a financial instrument. Especially governmental bonds show a very low volatility compared to other financial instruments, whereas the risk increased continuously during the euro crisis and a payment default could not be excluded at this point for certain countries. This proves clearly that the volatility does not reflect the risk exclusively and thus, the Value-at-Risk as the sole measure of existing risk is not completely comprised.

Hence, the described and analyzed Value-at-Risk computation models and the results in this empirical investigation should not be contemplated and interpreted separately, but rather researched further regarding an extension of these models to include and adopt among aspects extraordinary events in the models, such as 9/11 or the fiscal crisis in 2008. For this purpose, the models need to be changed in a way that manual input of data is added, changes of the model behavior, as well as definition for parameters are possible to edit. In this way, the models would be more flexible and could react contemporarily to extreme events. Another option could be an extension or combination of the models with stress test scenarios, in which different extraordinary prospective developments are assumed and their impacts are simulated and analyzed.

Despite the mentioned points of criticism, the Value-at-Risk is a well-known and prevalent risk figure. The empirical investigation showed that the Value-at-Risk did not reflect completely the risk, while using the different computation models. Nevertheless, by introducing the Value-at-Risk in companies the level of transparency increases and creates a starting point for impactful risk analyses. Furthermore, the investigation showed how important the influencing parameters are (e.g. length of historical target horizon or estimation procedure for parameters) for the models' prediction power.

²⁸ Assuming a Long-Position. For a Short-Position the scenario would be reversed.

While approximating the risk quantification is sufficient for smaller companies, it is essential to determine the risk precisely in large financial institutions under their tighter regulations.

Consequently, already a simple table calculation program could help to implement the Value-at-Risk in a small company so that the management can use it for steering the risks. Based on this academic work, a company could adjust the Value-at-Risk concept in own internal processes and hence, implement an own key risk figure system. The implementation procedure can be basically observed from the computation model description in chapter 3.

However, bigger financial institutions face greater IT-requirements for software and hardware to implement a Value-at-Risk computation model. Moreover, it is highly essential to calibrate influencing factors and adjust them continuously with respect to the actual market situation to guarantee an optimal prediction quality. Even though the computation models showed good results during our empirical investigation regarding the risk prediction power, it is advisable to have a more differentiated view on the results and expect better results before using it in larger financial institutions for a productive environment. According to that, a pursuing research task shows great potential and might have interesting insights, whether the Value-at-Risk is for example able to comply with the requested level of prediction quality by changing the distribution assumption or using different estimation processes for the parameters.

Finally, the consideration of extraordinary or rare events as well as of complex computations provides no guarantee for an accurate prediction of the future volatility and measurement. Each of the Value-at-Risk computation models constitutes a simple approximation of the reality, which differs in its accuracy and in its resulting prediction power. Eventually, the utilization of the computation models in companies or financial institutions increases the consciousness in the entire business, especially the management's consciousness for focusing on their organization's own risks and possible losses. This is clearly conducive for the stability of decision makers, but even more for the security and robustness of the general financial system.

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Appendix

Figure I: Calculation of Daily Logarithmic Returns Both Indexes

```
n <- length(DAXData$DAXPrice)
i = 2
DAXDailyReturn <- c(0)
#Calculating Logarithmic Returns
while(i <= n)
{
  r <- log(DAXData$DAXPrice[i]/DAXData$DAXPrice[i-1],
           exp(1))
  DAXDailyReturn[i] <- r

  i = i + 1 }
```

```
n <- length(DJIData$DJIPrice)
i = 2
DJIDailyReturn <- c(0)
#Calculating Logarithmic Returns
while(i <= n)
{
  r <- log(DJIData$DJIPrice[i]/DJIData$DJIPrice[i-1],
           exp(1))
  DJIDailyReturn[i] <- r

  i = i + 1 }
```

Figure II: EWMA Portfolio Covariance with Recursion

```

n = length(DJIData$Date)
i = 51 #Initialization window
lambda = 0.94 #decay factor
#Portfolio Covariance, Recursion method for EWMA

PortfolioCovariance <- c(0)
SqrtCovariance <- c(0)

CovarianceInitial <- cov(DJIReturn[1:50], DAXReturn[1:50])
PortfolioCovariance[50] <- CovarianceInitial

while(i < n)
{
  PortfolioCovariance[i] <-
    (lambda * PortfolioCovariance[i-1])
  + ((1 - lambda) *
    DAXDailyReturn[i-1] * DJIDailyReturnEUR[i-1])
  SqrtCovariance[i] <- sqrt(PortfolioCovariance[i])

  i = i + 1 }

```


Figure III: EWMA Portfolio Variance and Volatility (Std Dev.) with Recursion

```

PortfolioVolatilityEWMA94 <- c(0)
EWMAVariance94 <- c(0)
n <- length(DJIData$Date)
i = 51 #Initializing Window
lambda = 0.94 #Decay factor
InitialVariance <- var(IndexSumEUR[1:50])
EWMAVariance94[50] <- InitialVariance
#Portfolio Variance Recursive Method for EWMA
while(i < n)
{
  EWMAVariance94[i] <-
    (lambda * EWMAVariance94[i-1]) +
    ((1 - lambda) * IndexSumEUR[i-1]^2)
  PortfolioVolatilityEWMA94[i] <- sqrt(EWMAVariance94[i])

  i = i + 1 }

```

Figure IV: Calculation Value-at-Risk Historical Simulation Method

```

n <- length(DAXData$DAXPrice)
IndexSumEUR <- c(0)

i = 1
i50 = 50

#Portfolio Returns

while(i < n)
{IndexSumEUR[i] <- (DJIReturnEUR[i] + DAXReturn[i])
i = i + 1}

Dailyhvar9950 <- c(0)
i = 1

while(i < n-50)
{
  H <- 1          # Number of holdings
  value1 <- 20000 #Amount Invested
  valuep <- value1*H # Value of portfolio
  hp <- 1         # Holding period
  a1 <- .99       # Confidence level (1%)

  # Historical VaR

  Dailyhvar9950[i+50] <-
  quantile(IndexSumEUR[i:i50],1-a1, na.rm = TRUE)*valuep

  i = i + 1
  i50 = i + 50}

#quantile function used to select measurement
#based on location in distribution

```

Figure V: Calculation of Value-at-Risk Variance-Covariance Moving-Average Method

```

n <- length(DJIData$DJIPrice)
Dailyparvar9950 <- c(0)

i = 1
i50 = 50
while(i < n-50)
{
  H <- 1          # Number of holdings
  value1 <- 20000 #Amount Invested
  valuep <- value1*H # Value of portfolio
  hp <- 1         # Holding period (in days)
  a1 <- .99       # Confidence level (1%)

  #Variance Covariance MA, normal distribution

  Dailyparvar9950[i+50] <-
  valuep*qnorm(1-a1,mean = 0,sd = 1)*
  sd(IndexSumEUR[i:i50])*sqrt(hp)

  #qnorm -> z-score,
  #returns value based on deviation from mean
  #Volatility Return Set = 50

  i = i + 1
  i50 = i + 50}

```

Figure VI: Calculation of Value-at-Risk Variance-Covariance Exponential Weighted Moving-Average Method

```

n <- length(DAXData$DAXPrice)

i = 1

DailyEWMAVaR9994 <- c(0)

while(i < n)
{
  H <- 1          # Number of holdings
  value1 <- 20000 # Amount Invested
  valuep <- value1*H # Value of portfolio
  hp <- 1         # Holding period
  a <- .99        # Confidence level (1%)

  DailyEWMAVaR9994[i] <-
    valuep*qnorm(1-a,mean = 0,sd = 1)*
    PortfolioVolatilityEWMA94[i]*sqrt(hp)

  i = i + 1}

#qnorm -> z-score ,
#returns value based on deviation from mean

```

Figure VII: Calculation of Value-at-Risk Variance Monte-Carlo Simulation Method

```

n <- length(DJIData$DJIPrice)
i = 1
i50 = 50

#list(mode = "vector",length= 100)

Dailymontevar9950 <- c(0)

H <- 1          # Holdings
value1 <- 20000 #Amount Invested
valuep <- value1*H # Value of portfolio
hp <- 1         # Holding period
a1 <- .99       # Confidence level (1%)

set.seed(27)
#Seed used for random number initiation, #
makes results more replicable

while (i < n-50)
{
  Dailymontevar9950[i+50] <-
  quantile(rnorm(10000, mean = mean(IndexSum[i:i50]),
  sd = sd(IndexSum[i:i50])), 1-a1, na.rm = TRUE)*valuep

  i = i + 1
  i50 = i + 50}

#rnorm, randomization function for normal distribution

```

Figure VIII: Standard Coverage Backtesting

```

#Kupiec Test for Standard Coverage

n <- length(DJIData$DJIPrice)

Dailyhvar9950 <- c(0)

i = 51 #First VaR Measurement Date

# Calculating Portfolio Daily Return Compared
#to measured VaR using historical simulation

while(i < n)

{

  if(IndexSum[i] <= Dailyhvar9950[i])

    DJIhit = DJIhit + 1

  i = i + 1 }

Dailyhvar9950 <- DJIhit / (n-50)

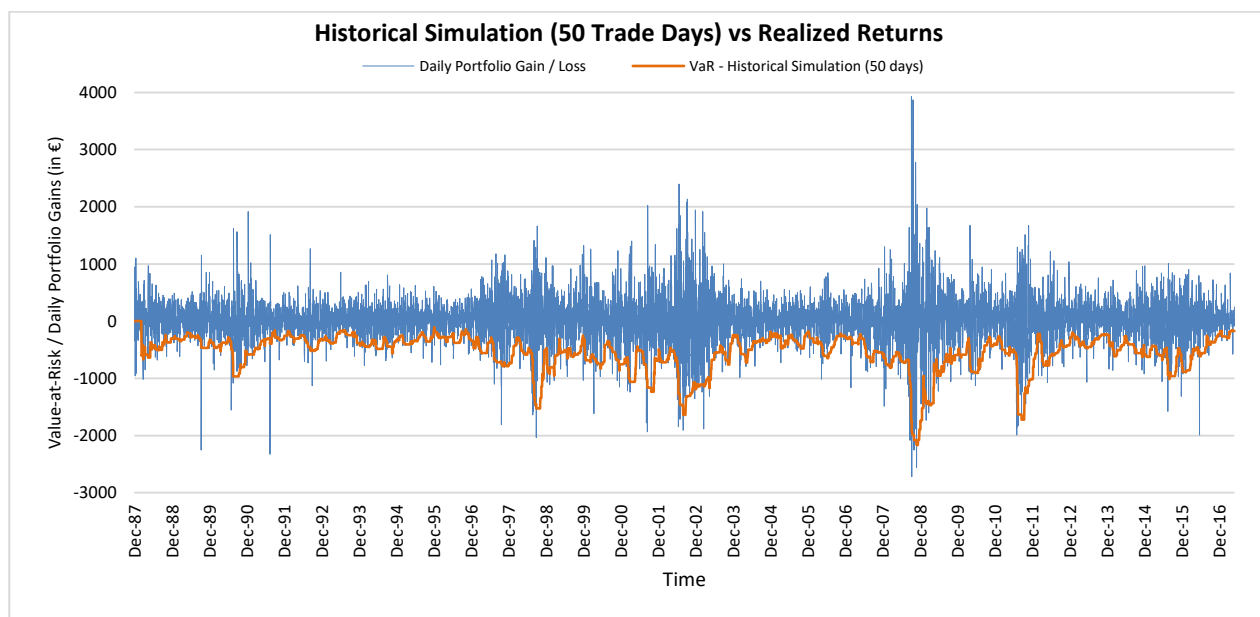
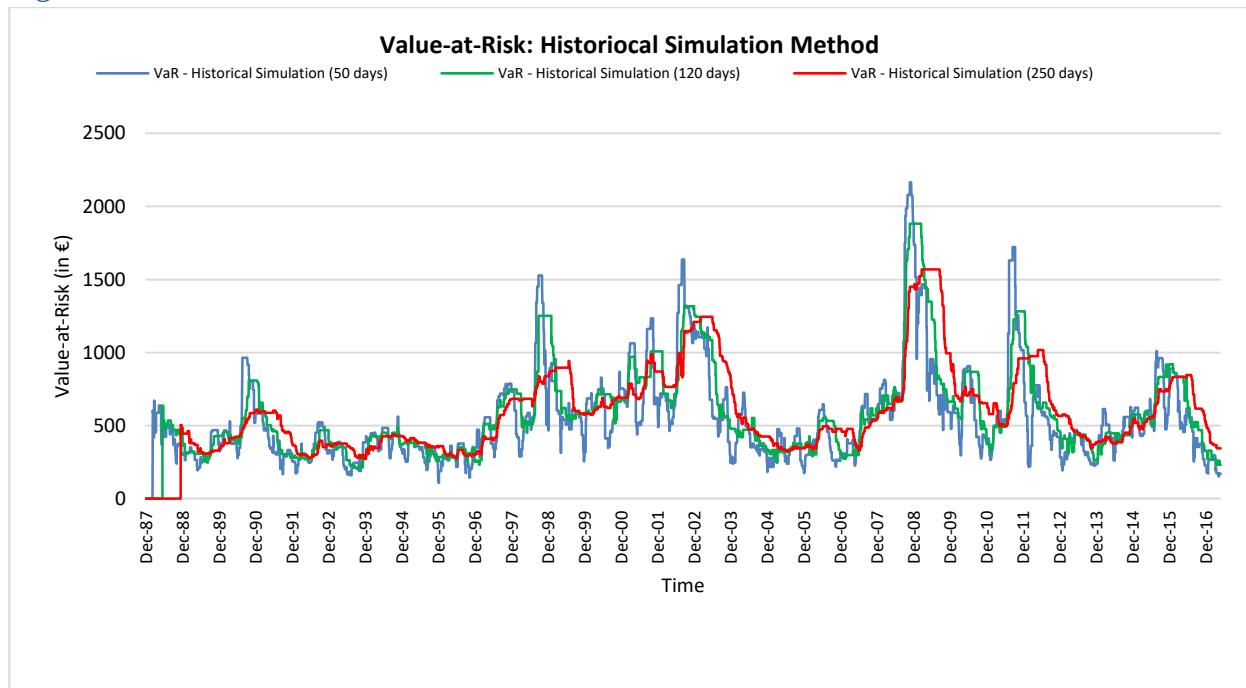
Dailyhvar9950

```

Figure IX: Christoffersen Independence Backtesting with R Package (Acharya, 2016)

```
#Chrifffsen Independence Backtest Using R Package Dowd  
  
library(Dowd)  
  
Dailyhvar9950Independence <-  
  ChristoffersenBacktestForIndependence  
  (IndexSumEUR, -Dailyhvar9950, 0.99)  
#Vector of actual returns ,  
#vector of VaR measures (positive),  
#confidence interval
```

Figure X: Historical Simulation 95 Confidence Interval



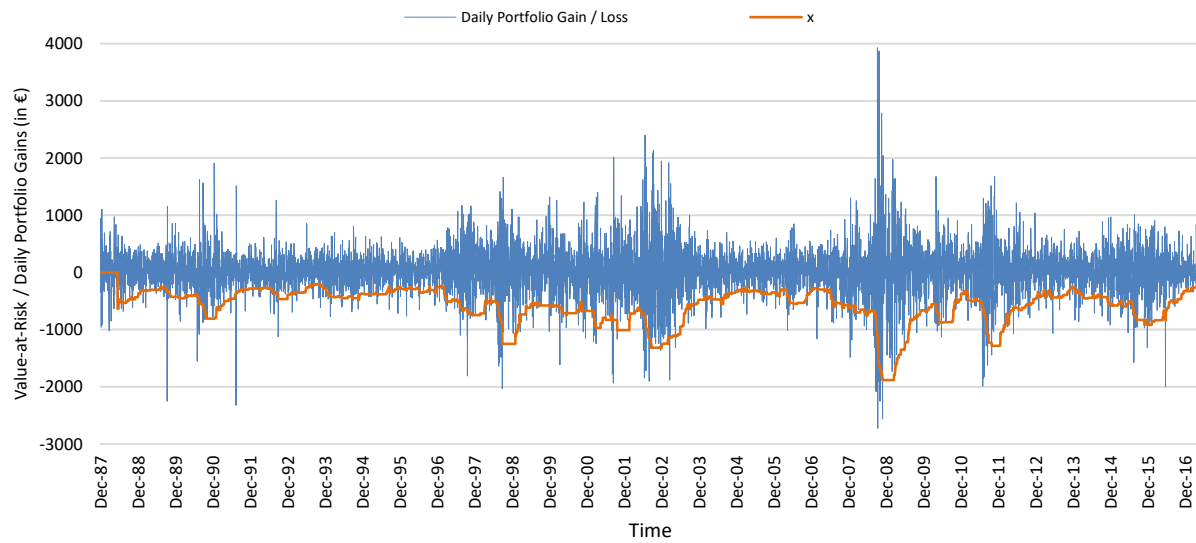
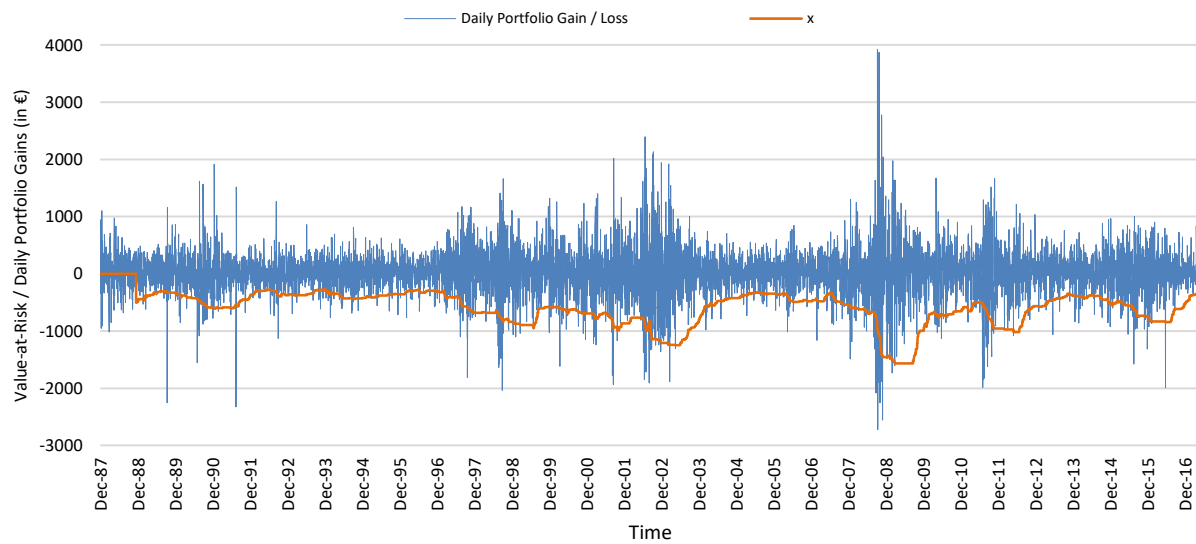
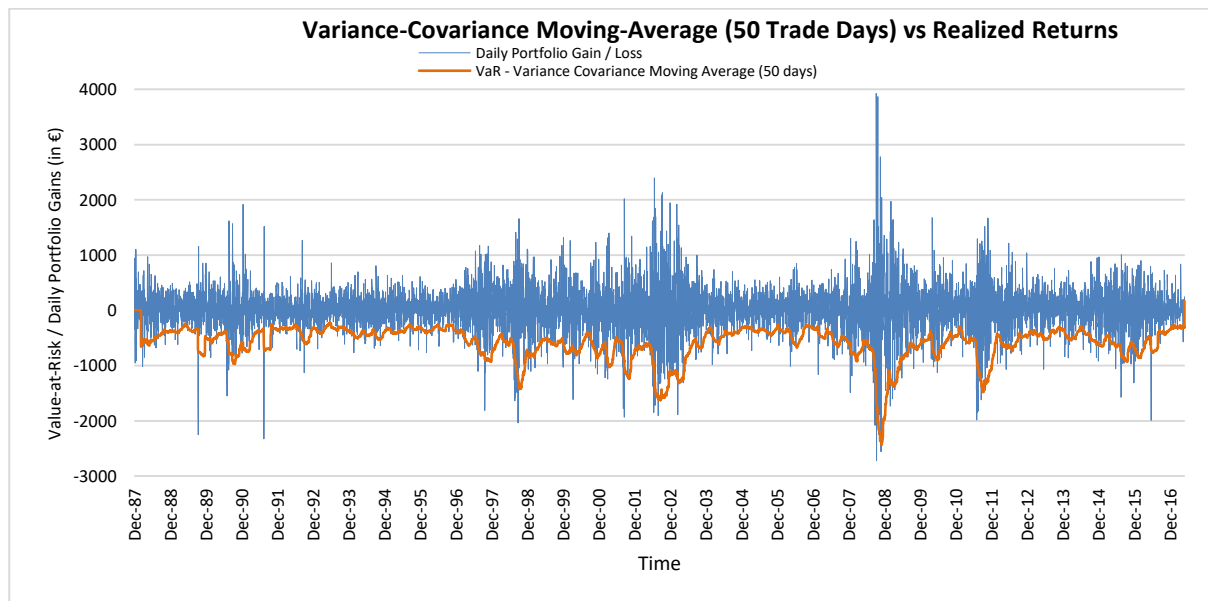
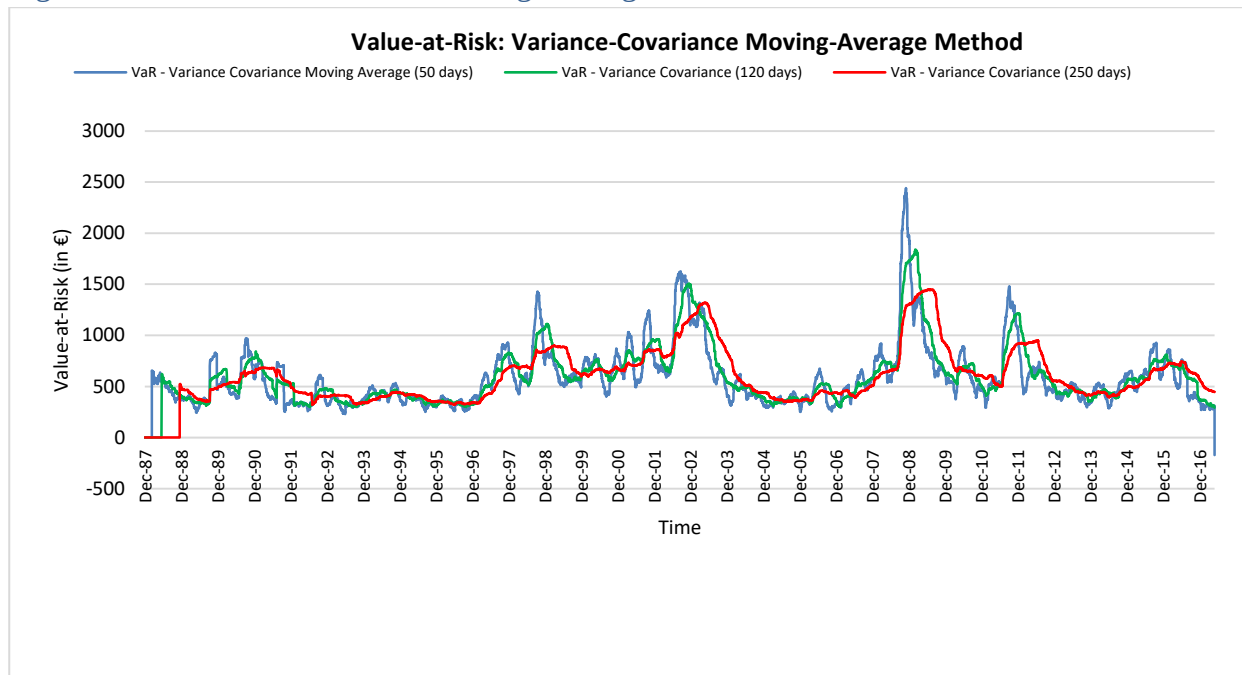
Historical Simulation (120 Trade Days) vs Realized Returns**Historical Simulation (250 Trade Days) vs Realized Returns**

Figure XI: Variance-Covariance Moving-Average 95 Confidence Interval



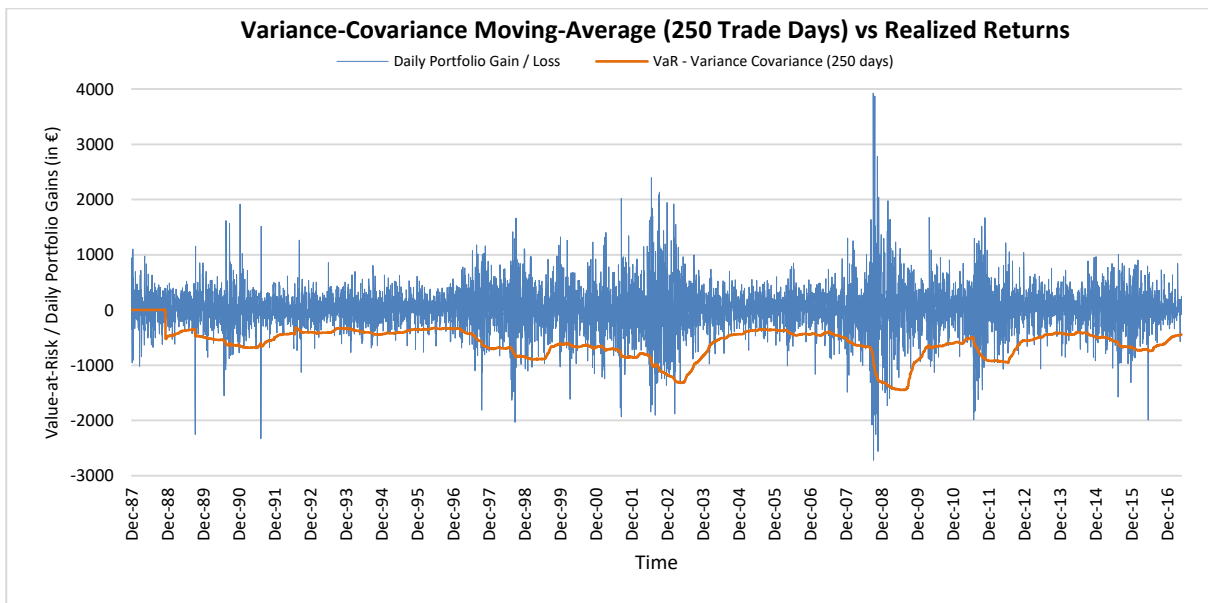
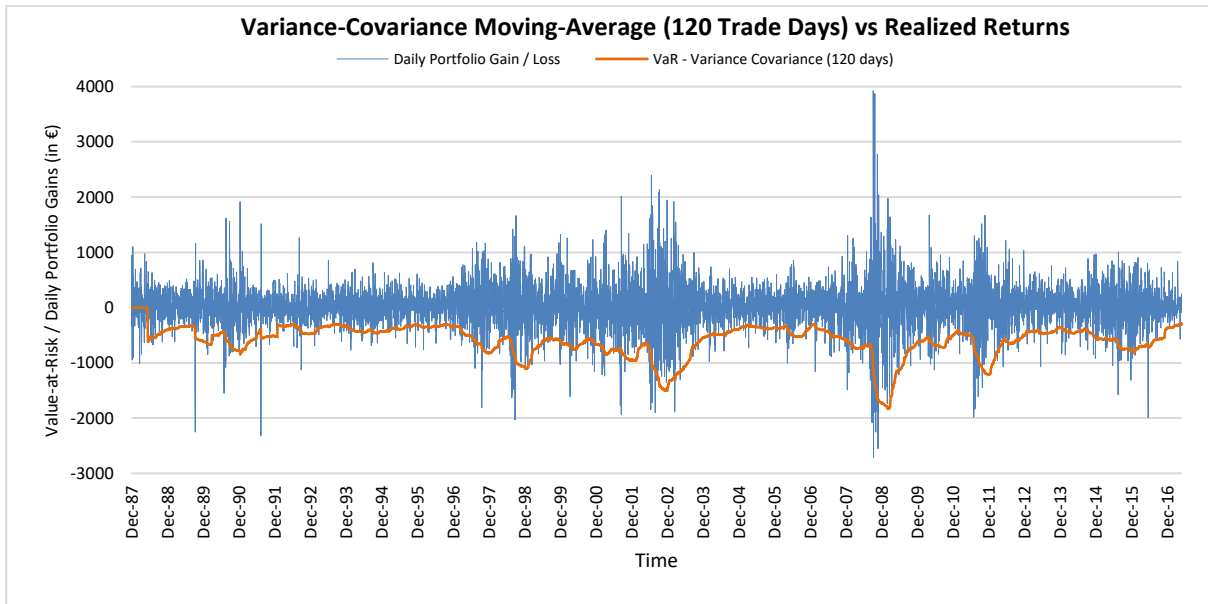
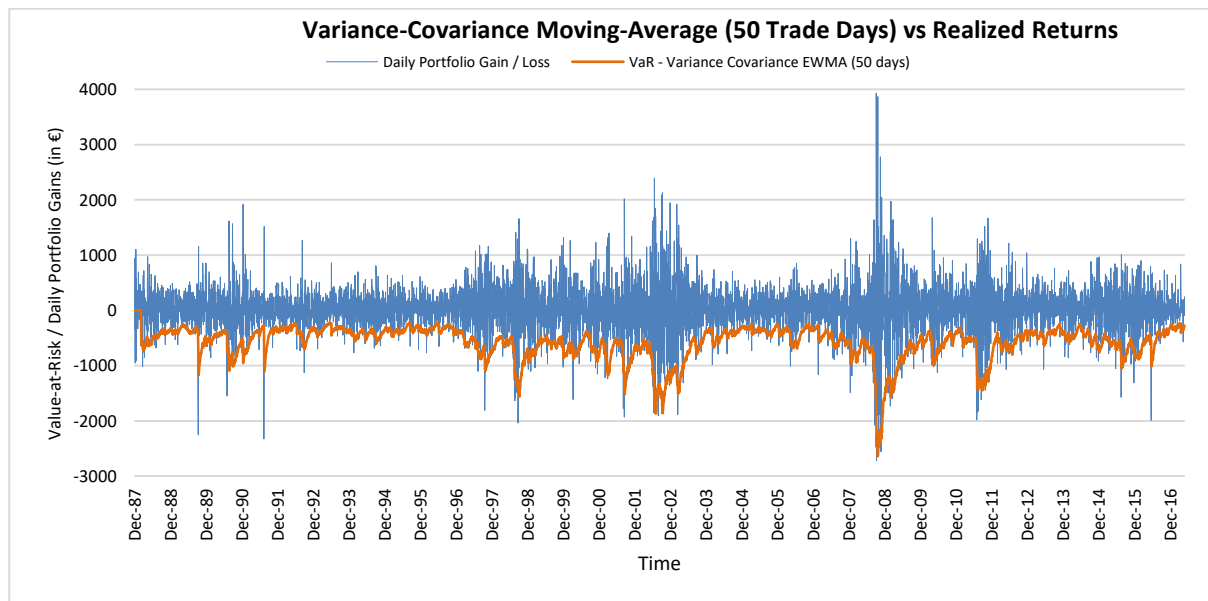
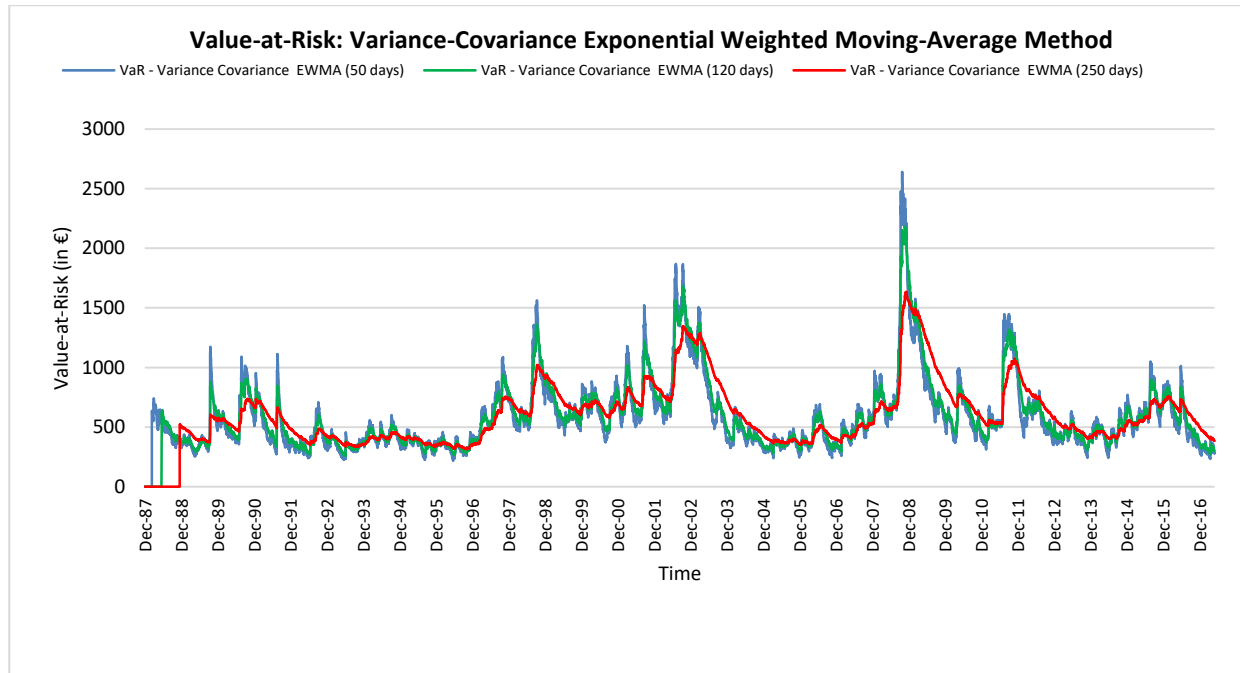


Figure XII Variance-Covariance Exponential Weighted Moving-Average 95 Confidence Interval



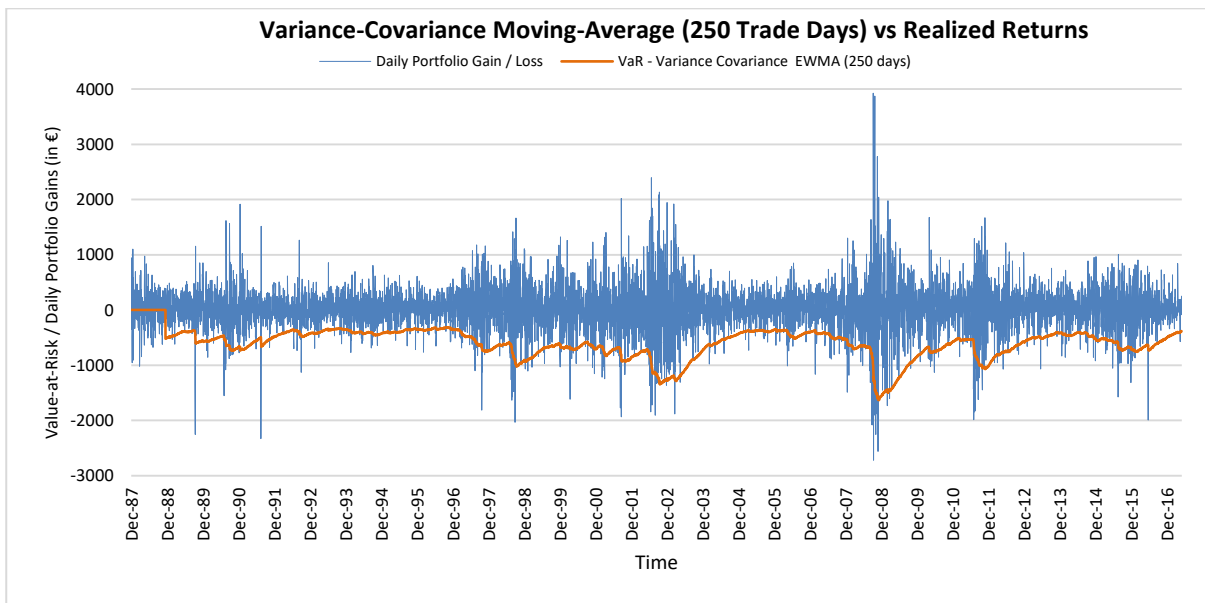
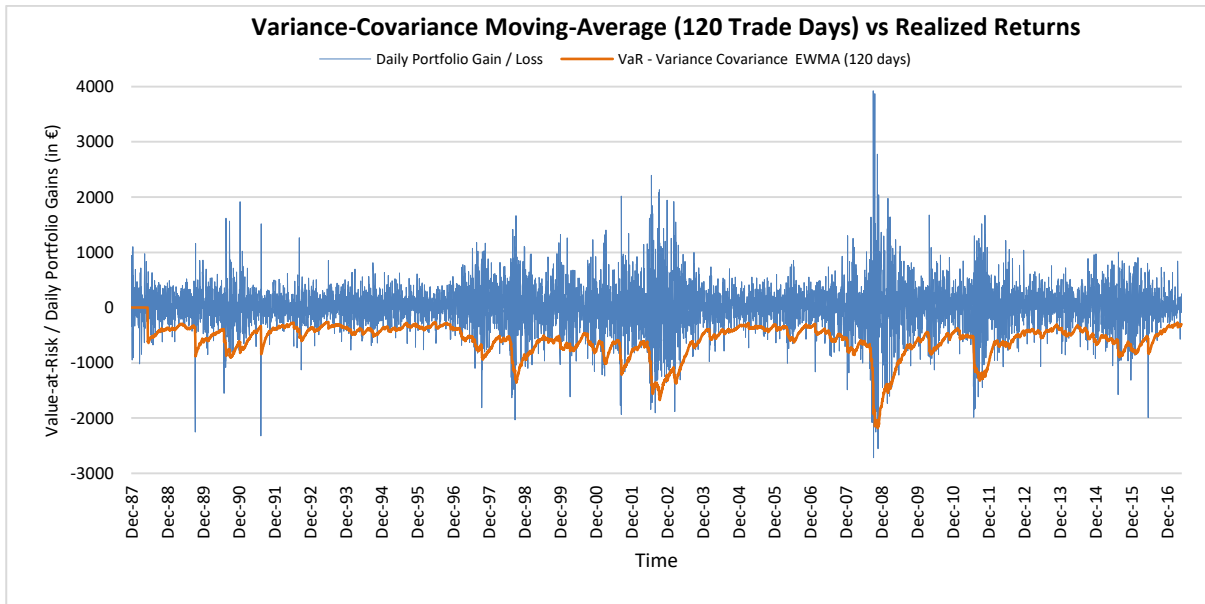


Figure XIII: Monte-Carlo Simulation Method 95 Confidence Interval

